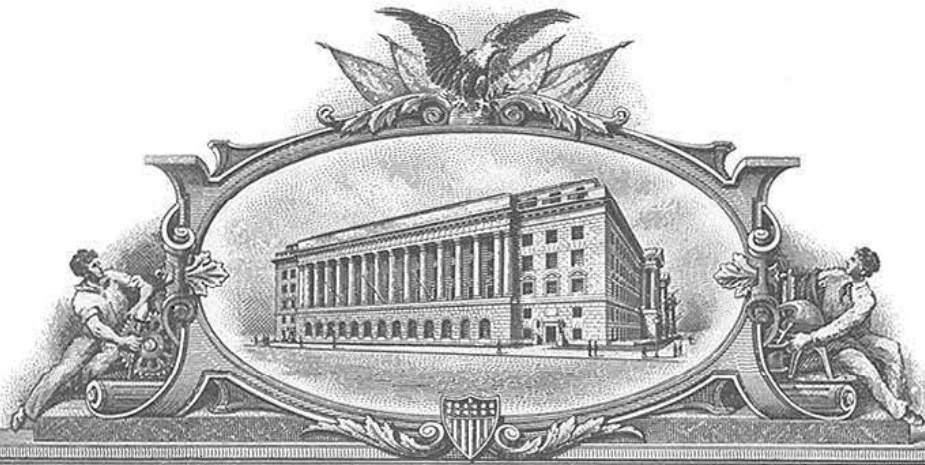


APPENDIX N

8720785



THE UNITED STATES OF AMERICA

TO ALL TO WHOM THESE PRESENTS SHALL COME:

UNITED STATES DEPARTMENT OF COMMERCE

United States Patent and Trademark Office

May 26, 2026

**THIS IS TO CERTIFY THAT ANNEXED IS A TRUE COPY FROM THE
RECORDS OF THIS OFFICE OF THE FILE WRAPPER AND CONTENTS
OF:**

APPLICATION NUMBER: 19/313,233

FILING DATE: *August 28, 2025*

PATENT NUMBER: 12646735

ISSUE DATE: *June 02, 2026*



Certified by

DIRECTOR OF THE UNITED STATES PATENT AND TRADEMARK OFFICE



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/313,233	08/28/2025 04:12:26 PM Z ET	LGES 3.3F-1654 CCCCCIICII

Title of Invention

Secondary Battery and Method for Manufacturing the Same

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	7385	FILED BY	Sharon McGee
PATENT CENTER #	72065099	FILING DATE	-
CUSTOMER #	86765	FIRST NAMED INVENTOR	Jae Woong Kim
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Kevin Kocun

Documents

TOTAL DOCUMENTS: 11

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
LGES6695 - Communication Regarding Certified Copy.pdf	1	Miscellaneous Incoming Letter	21 KB
LGES6695 - Power of Attorney by Applicant PTO AIA-82.pdf	1	Power of Attorney	17 KB
LGES6695 - Application Data Sheet Fillable PDF.pdf	9	Application Data Sheet	2202 KB
LGES6695 - Transmittal -- Utility Patent Application PTO AIA-15.pdf	1	Transmittal of New Application	33 KB
LGES6695 - Certification and Request for Prioritized Exam 37 CFR 1102 e PTO AIA- 424.pdf	1	Track One Request	34 KB

LGES6695 - Information Disclosure Statement Fillable PDF.pdf	11	Information Disclosure Statement (IDS) Form (SB08)	1317 KB
LGES6695 - IDS - Information Disclosure Statement - Reference in Previous Application LD-4.pdf	1	Transmittal Letter	21 KB
POA - LG Energy Solution Ltd.pdf	1	Power of Attorney	46 KB
Declaration.pdf	2	Oath or Declaration filed	100 KB
Specification LGES 33F-1654 CCCCIIICII-APP.TEXT.docx	22	Application body structured text document	38 KB
Warning: Document has [Balance SBCS characters and DBCS characters] option enabled and may result in reduction of pages, however, no data or text has been modified.			
Drawings LGES 33F-1654 CCCCIIICII.pdf	9	Drawings-only black and white line drawings	765 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
LGES6695 - Communication Regarding Certified Copy.pdf	8FC0603915EDA06E7251EF3DEA430338DD4C87209A312C0AF FD8E70BA4D933854A9FE61EA203FAA980508709A2DD25F7F8 EFC9479BED9899BFDE0EC5F93EA89C
LGES6695 - Power of Attorney by Applicant PTO AIA-82.pdf	8D04AF37BF1CC442AD283F9FA676DA23700F5EB2F56F3A995 CF4990B055018AD4AE0FDDD479BD6F576EB3757625F794869 1AB6F8FFDB1B18BFE9165EF5969CCA
LGES6695 - Application Data Sheet Fillable PDF.pdf	26497EA34902D52DD6CEF4C5EDF5CEFDD13FC4F2D4792EA 6A63461E8D8866B0594C45505D1BF4E3CA16A98FDE00874C0 BBD9F41CD114AF62A3CA015B5AF7D7EC
LGES6695 - Transmittal -- Utility Patent Application PTO AIA-15.pdf	54ED4CB43CDDE64EC4D550226218E7D3299C92940E17E773 8372F3CC4CA3B5C0647978D2F568CD68BAE7F5FD1C4D287F D73A55DB9D21DF4DC16C49EA91C37A22
LGES6695 - Certification and Request for Prioritized Exam 37 CFR 1102 e PTO AIA-424.pdf	B8D2CB77BB8A8D3A36DCE083DA0370329A34E8E801AEB2E3 ACB7862258E95DBD6676A92728120DCCCD837976F93EECF1 BDFDBF7CDA65406D073088F8BE92402C
LGES6695 - Information	96B34277CE3374D2F6BD89ED3F569C27B5B3380CAA970F8A8

Disclosure Statement Fillable PDF.pdf	8290C2FED9C458DE651A8F42E4D441DF0DC40EFB660232BDF2CB8FB8ACEAF7B2F89307C40649D9E
LGES6695 - IDS - Information Disclosure Statement - Reference in Previous Application LD-4.pdf	6BA567D2CD51DB7259B109A5008AF0F89A8BF0E770229E77077597BADA70D202B5010B05ECFD5DEF42ECB08B9136F7EB30A0C5D55AFD115A39E0CFB7D2683B3E
POA - LG Energy Solution Ltd.pdf	F18C66A48E1D0E420627962F7063C6E831339EDFBCB6123EE3E1ACD09BE0C007FD748A528FF87D98D94170D7E971E26A7E0AEF38D0F53940B2E4D37C2805D84F
Declaration.pdf	15E8E267EEA2282A6243812DB47489BEE5B8983900EC30C5522760E1A5FB020DD18B5BD702ECB2B185825816DD6D9A6C16A2E655660DFC39856A0F8F6E84E66C
Specification LGES 33F-1654 CCCCIIICII-APP.TEXT.docx	34F51812B3DD44BDA1ECBD20C7B5E8D09AC4CFCED4A06863727B679D984EE3D747165D94EC290B2095CA1FCC2A31A6868827892C5B16B95862F0C7AA6AF4946E
Drawings LGES 33F-1654 CCCCIIICII.pdf	F7D9599201823ECDB36B7F3197457C4A7CB48F780C5149EF5A080A88682E8058BB971B10ACFB178C4CEBA290422BACAA9A674A9E97E34C252606404D07AC119C

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES 3.3F-1654 CCCCCIICII
		Application Number	
Title of Invention	Secondary Battery and Method for Manufacturing the Same		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor 1					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jae Woong		Kim		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Daejeon	Country of Residence ⁱ	KR		
Mailing Address of Inventor:					
Address 1		LG Energy Solution Research Park			
Address 2		188, Munji-ro, Yuseong-gu			
City	Daejeon	State/Province			
Postal Code		34122	Country ⁱ	KR	
Inventor 2					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Hyoung Kwon		Kim		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Daejeon	Country of Residence ⁱ	KR		
Mailing Address of Inventor:					
Address 1		LG Energy Solution Research Park			
Address 2		188, Munji-ro, Yuseong-gu			
City	Daejeon	State/Province			
Postal Code		34122	Country ⁱ	KR	
Inventor 3					Remove
Legal Name					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII
		Application Number	
Title of Invention	Secondary Battery and Method for Manufacturing the Same		

Prefix	Given Name	Middle Name	Family Name	Suffix
	Sang Uck		Kim	
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service				
City	Daejeon	Country of Residence ⁱ	KR	

Mailing Address of Inventor:				
Address 1	LG Energy Solution Research Park			
Address 2	188, Munji-ro, Yuseong-gu			
City	Daejeon	State/Province		
Postal Code	34122	Country ⁱ	KR	
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button. Add				

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).	
☐ An Address is being provided for the correspondence information of this application.	
Customer Number	86765
Email Address	Add Email Remove Email

Application Information:

Title of the Invention	Secondary Battery and Method for Manufacturing the Same		
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	9	Suggested Figure for Publication (if any)	9

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country ⁱ

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES 3.3F-1654 CCCCCIICII
		Application Number	
Title of Invention	Secondary Battery and Method for Manufacturing the Same		

Publication Information:

☐	Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐	Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
Please Select One:			
☒	Customer Number	☐	US Patent Practitioner
☐	Limited Recognition (37 CFR 11.9)		
Customer Number	86765		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove			
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
	Continuation of	19/207758	2025-05-14		
Prior Application Status	Patented	Remove			
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
19/207758	Continuation of	18/620269	2024-03-28	12334493	2025-06-17
Prior Application Status	Patented	Remove			
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
18/620269	Continuation of	18/618609	2024-03-27	12341147	2025-06-24
Prior Application Status	Patented	Remove			
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
18/618609	Continuation of	18/132719	2023-04-10	11973177	2024-04-30

Application Data Sheet 37 CFR 1.76		Attorney Docket Number		LGES 3.3F-1654 CCCCCIICII	
		Application Number			
Title of Invention		Secondary Battery and Method for Manufacturing the Same			
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
18/132719	Continuation of	16/498751	2019-09-27	11652232	2023-05-16
Prior Application Status				Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
16/498751	a 371 of international	PCT/KR2018/004366	2018-04-13		
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.					Add

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number		Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
10-2018-0043191		KR	2018-04-13	
Application Number		Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
10-2017-0048645		KR	2017-04-14	
Additional Foreign Priority Data may be generated within this form by selecting the Add button.				

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII
		Application Number	
Title of Invention	Secondary Battery and Method for Manufacturing the Same		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII
		Application Number	
Title of Invention	Secondary Battery and Method for Manufacturing the Same		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.			
Applicant 1			Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.			
Clear			
☒ Assignee	☐ Legal Representative under 35 U.S.C. 117	☐ Joint Inventor	
☐ Person to whom the inventor is obligated to assign.		☐ Person who shows sufficient proprietary interest	
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:			
Name of the Deceased or Legally Incapacitated Inventor:			
If the Applicant is an Organization check here. ☒			
Organization Name	LG Energy Solution, Ltd.		
Mailing Address Information For Applicant:			
Address 1	Tower 1, 108, Yeoui-daero, Yeongdeungpo-gu		
Address 2			
City	Seoul	State/Province	
Country	KR	Postal Code	07335
Phone Number		Fax Number	
Email Address			
Additional Applicant Data may be generated within this form by selecting the Add button. Add			

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII
		Application Number	
Title of Invention	Secondary Battery and Method for Manufacturing the Same		

Assignee 1			
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.			
Remove			
If the Assignee or Non-Applicant Assignee is an Organization check here. ☒			
Organization Name	LG Energy Solution, Ltd.		
Mailing Address Information For Assignee including Non-Applicant Assignee:			
Address 1		Tower 1, 108, Yeoui-daero, Yeongdeungpo-gu	
Address 2			
City	Seoul	State/Province	
Country ⁱ	KR	Postal Code	07335
Phone Number		Fax Number	
Email Address			
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button. Add			

Signature:[Remove](#)

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the **INITIAL** filing of the application **and** either box A or B is **not** checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Joseph H. Karlin/		Date (YYYY-MM-DD)	2025-08-28	
First Name	Joseph	Last Name	Karlin	Registration Number	62,471
Additional Signature may be generated within this form by selecting the Add button. Add					

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES 3.3F-1654 CCCCCIICII
		Application Number	
Title of Invention	Secondary Battery and Method for Manufacturing the Same		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

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First Named Inventor	Jae Woong Kim
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SIGNATURE of Applicant or Patent Practitioner

Signature	/Joseph H. Karlin/	Date (Optional)	August 28, 2025
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SECONDARY BATTERY AND METHOD FOR MANUFACTURING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. Application No. 19/207,758, filed on May 14, 2025, which is a continuation of U.S. Patent Application No. 18/620,269, filed on March 28, 2024, now U.S. Patent No. 12,334,493, which is a continuation of U.S. Patent Application No. 18/618,609, filed on March 27, 2024, now U.S. Patent No. 12,341,147, which is a continuation of U.S. Application No. 18/132,719, filed on April 10, 2023, now U.S. Patent No. 11,973,177, which is a continuation of U.S. Application No. 16/498,751, filed on September 27, 2019, now U.S. Patent No. 11,652,232, which is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/KR2018/004366, filed on April 13, 2018, which claims priority from Korean Patent Application Nos. 10-2017-0048645, filed on April 14, 2017, and 10-2018-0043191, filed on April 13, 2018, the disclosures of all of which are hereby incorporated by reference herein in their entireties.

TECHNICAL FIELD

[0002] The present invention relates to a secondary battery and a method for manufacturing the same, and more particularly, to a secondary battery in which an electrode tab increases in mechanical strength and a method for manufacturing the same.

BACKGROUND ART

[0003] Batteries (cells) that generate electric energy through physical or chemical reaction to supply the generated electric energy to the outside are used when AC power to be supplied to the building is not obtained, or DC power is required according to the living environments surrounded by various electric and electronic devices.

[0004] Among such batteries, primary batteries and secondary batteries, which are chemical batteries using chemical reaction, are being generally used. The primary batteries

are consumable batteries which are collectively referred to as dry batteries. Also, secondary batteries are rechargeable batteries that are manufactured by using a material in a redox process between current and a substance is repeatable several times. When the reduction reaction is performed on the material by the current, power is charged, and when the oxidation reaction is performed on the material, power is discharged. Such the charging-discharging is repeatedly performed to generate electricity.

[0005] A lithium ion battery of the secondary batteries is manufactured through the following processes. An active material is applied to each of a positive electrode conductive foil and a negative electrode conductive foil at a predetermined thickness, and a separator is disposed between the positive electrode conductive foil and the negative electrode conductive foil, and then, an electrode assembly, in which the positive electrode conductive foil, the separator, and the negative electrode conductive foil are wound several times in a jelly-roll or cylindrical shape, is accommodated into a cylindrical or prismatic can, a pouch, and the like to seal the resultant product, thereby manufacturing the lithium ion battery.

[0006] An electrode assembly and a secondary battery including the same according to the related art is disclosed in Korea Patent Publication No. 10-2012-0006389.

[0007] In the secondary battery according to the related art, it is necessary to apply a technology in which a metal having low resistance is applied to a negative electrode tab, or the number of electrode tabs increases in order to develop a low resistance cell.

[0008] A negative multi-tab structure developed by this necessity was a two-tab structure in which an electrode tab is attached to each of a core and an outer portion or a three-tab structure in which an electrode tab is applied to each of a core and an outer portion, and a middle tab is attached to an intermediate portion.

[0009] However, in order to attach the middle tab on the characteristics of the jelly roll, a non-coating portion has to be formed on a predetermined portion in the middle of the electrode. Thus, there is a problem that the battery capacity is lowered due to an increase of the non-coating portion.

DISCLOSURE OF THE INVENTION

TECHNICAL PROBLEM

[0010] Therefore, the present invention has been made to solve the abovementioned problem, and an object of the present invention is to provide a secondary battery in which a multi-tab is formed while minimizing a non-coating portion and a method for manufacturing the same.

[0011] Also, an object of the present invention is to provide a secondary battery which is capable of increasing in mechanical strength while notching electrode foil to form an electrode tab and a method for manufacturing the same.

TECHNICAL SOLUTION

[0012] A secondary battery according to an embodiment of the present invention includes: an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state.

[0013] The notching tab part may continuously extend across 1/3 or more of a total length of the coating portion in the longitudinal direction.

[0014] When the electrode is in the wound state, the notching tab part may form a

plurality of notching tabs.

[0015] The foil tab may have a width of 4 mm to 120 mm.

[0016] The electrode may be accommodated in a can member, and each of the plurality of notching tabs may be in a bent state and may be connected to the can member.

[0017] A bent part formed by bending each of the plurality of notching tabs may be bonded to the can member.

[0018] The secondary battery may further include an insulation member disposed between the bent parts and an end of the electrode.

[0019] The secondary battery may further include an electrode tab attached to the non-coating portion.

[0020] When the electrode is in the wound state, the notching tab part may form a plurality of notching tabs, and the electrode may be accommodated in a can member, and each of the plurality of notching tabs may be in a bent state and may overlap the electrode tab.

[0021] The electrode tab and the plurality of notching tabs may be bonded together to the can member.

[0022] A method for manufacturing a secondary battery according to an embodiment of the present invention includes: a preparation step of preparing an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; a notching tab part formation step of notching the electrode to form a notching tab part not coated with the active material and extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part extending across 1/3 or more of a total length of the coating portion in the longitudinal direction; a winding step including stacking the

electrode with a separator and with another electrode having a polarity different from that of the electrode and winding the electrode, the separator, and the another electrode to form an electrode assembly; a notching tab formation step of cutting the notching tab part of the electrode assembly to form a plurality of notching tabs; and an installation step of accommodating the electrode assembly into a can member and bonding each of the plurality of notching tabs to the can member.

[0023] The electrode may be a negative electrode.

[0024] During the preparation step, the coating portion coated with the active material on the electrode collector and the non-coating portion on which the active material is not applied may be arranged in parallel to each other in the width direction of the electrode collector.

[0025] during the notching tab part formation step, the notching tab part on which the active material is not applied may be notched in the width direction of the coating portion while cutting the coating portion and the non-coating portion in the longitudinal direction of the electrode collector.

[0026] During the notching tab part formation step, notching may be performed so that the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the electrode.

[0027] During the notching tab formation step, the notching tab part may be cut using a laser.

[0028] During the installation step, the electrode assembly may be accommodated in the can member in a state in which the plurality of notching tabs are bent and bonded to the can member.

[0029] During the installation step, a bent portion of each of the plurality of notching tabs may be welded to the can member.

[0030] During the installation step, bent portions of each of the plurality of notching tabs may overlap each other and may be bonded to the can member together.

[0031] During the winding step, the notching tab part may be wound to form two or more overlapping layers.

ADVANTAGEOUS EFFECTS

[0032] According to the present invention, the multi-tab may be formed while minimizing the formation of the non-coating portion.

[0033] According to the present invention, the multi-tab may be formed to minimize the electric resistance.

[0034] According to the present invention, the formation of the non-coating part may be minimized to secure the battery capacity.

[0035] According to the present invention, the notching tab may overlap each other to form two or more layers while forming the notching tab that is the foil tab from the electrode foil, whereby increasing in mechanical strength.

[0036] According to the present invention, the notching tab that is the foil tab may increase in mechanical strength to prevent the notching tab from being broken in the bent state, and thus, the notching tab may be bonded to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

[0037] According to the present invention, the notching tab and the electrode tab may be bonded to the can member at the same to secure the welding processability.

[0038] According to the present invention, the plurality of notching tabs may be formed outside the electrode assembly to install the insulation member between the plurality of notching tabs and the end of the electrode assembly, and thereby to prevent the notching tabs and the electrode assembly from coming into contact with each other, thereby improving the safety.

BRIEF DESCRIPTION OF THE DRAWINGS

[0039] FIG. 1 is a development view of an electrode according to an embodiment of the present invention.

[0040] FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1 is wound.

[0041] FIG. 3 is a perspective view illustrating a state in which a notching tab part is cut to form a notching tab in FIG. 2.

[0042] FIG. 4 is a projective perspective view when seeing through a can member in a state in which an electrode assembly is installed in the can member according to an embodiment of the present invention.

[0043] FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

[0044] FIG. 6 is a flowchart successively illustrating a method for manufacturing a secondary battery according to an embodiment of the present invention.

[0045] FIG. 7 is a plan view illustrating a state in which a coating portion and a non-coating portion, which are formed in a longitudinal direction of an electrode collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

[0046] FIG. 8 is a bottom view illustrating a state in which a notching tab and an electrode tab are bonded to each other according to another embodiment of the present invention.

[0047] FIG. 9 is a side view of FIG. 8.

MODE FOR CARRYING OUT THE INVENTION

[0048] Hereinafter, a secondary battery and a method for manufacturing the same according to exemplary embodiments of the present invention will be described in detail with reference to the accompanying drawings.

[0049] Terms or words used in the specification and claims should not be construed as limited to a lexical meaning, and should be understood as appropriate notions by the inventor based on that he/she is able to define terms to describe his/her invention in the best way to be seen by others. Therefore, embodiments and drawings described herein are simply exemplary and not exhaustive, and it will be understood that various equivalents may be made to take the place of the embodiments.

[0050] In the drawings, the dimension of each of components or a specific portion constituting the component is exaggerated, omitted, or schematically illustrated for convenience in description and clarity. Thus, the dimension of each element does not entirely reflect an actual size. Moreover, detailed descriptions related to well-known functions or configurations will be ruled out in order not to unnecessarily obscure subject matters of the present invention.

[0051] FIG. 1 is a development view of an electrode according to an embodiment of the present invention, and FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1 is wound.

[0052] As illustrated in FIGS. 1 and 2, a secondary battery according to an embodiment of the present invention includes an electrode on which a coating portion 110 coated with an active material a on an electrode collector and a non-coating portion 120, on which the active material a is not applied on the electrode collector, are disposed in a longitudinal direction of the electrode collector and a notching tab part 111 extending from the coating portion 110 in a width direction of the electrode collector without being coated with the active material a and overlapping each other to form two or more layers when the electrode 100 is wound (FIG. 2 illustrates a notching tab part 111 that overlap each other to form two or more layers).

[0053] The foil tab may overlap each other to form two or more layers while forming the notching tab part that is a foil tab from electrode foil of the electrode collector, thereby increasing in mechanical strength.

[0054] The electrode 100 may be one of a positive electrode coated with a positive electrode active material and a negative electrode coated with a negative electrode active material. The electrode assembly 1 may be manufactured by stacking the positive electrode, the negative electrode, and a separator several times so that the separator is disposed between the positive electrode and the negative electrode.

[0055] Also, the electrode assembly 1 may be manufactured by winding a stack, in which the positive electrode, the separator, and the negative electrode are stacked, in a jelly-roll shape.

[0056] The positive electrode may be an aluminum electrode collector and include a positive electrode coating portion coated with the positive electrode active material and a positive electrode non-coating portion which is not coated with the positive electrode active material.

[0057] The positive electrode active material may include a lithium-containing transition metal oxide such as LiCoO_2 , LiNiO_2 , LiMnO_2 , and LiMnO_4 or a lithium chalcogenide compound.

[0058] The positive electrode coating portion may be manufactured, for example, by applying the positive electrode active material to a portion of at least one surface of the aluminum electrode collector, and the remaining portion of the aluminum electrode collector, which is not coated with the positive electrode active material, may be the positive electrode non-coating portion.

[0059] The negative electrode may be a copper electrode collector and include a negative electrode coating portion coated with the negative electrode active material and a

negative electrode non-coating portion which is not coated with the negative electrode active material.

[0060] The negative electrode active material may be a carbon material such as crystalline carbon, amorphous carbon, a carbon composite, and a carbon fiber, a lithium metal, or a lithium alloy.

[0061] The negative electrode coating portion may be manufactured, for example, by applying the negative electrode active material to a portion of at least one surface of the copper electrode collector, and the remaining portion of the copper electrode collector, which is not coated with the negative electrode active material, may be the negative electrode non-coating portion.

[0062] The separator may be manufactured by applying a polyvinylidene fluoride-hexafluoropropylene copolymer (PVDF-HFP co-polymer) to one base material selected from the group consisting of, for example, polyethylene (PE), polystyrene (PS), polypropylene (PP), and a copolymer of polyethylene (PE) and polypropylene.

[0063] As described above, the electrode 100 according to the present invention may be one of the positive electrode and the negative electrode, which constitute the electrode assembly 1. Particularly, the electrode 100 may be one, which requires further reduction in resistance, of the positive electrode and the negative electrode.

[0064] In the present invention, the negative electrode will be described as an example.

[0065] The electrode 100 has one end on which the non-coating portion 120 is formed. An electrode tab 121 may be attached to the non-coating portion 120 formed on the one end of the electrode 100.

[0066] The notching tab part 111 may be formed by notching the electrode 100 to extend from the coating portion 110 of the electrode 100 in the width direction. Only the electrode

collector may be formed to extend on an area corresponding to 1/3 or more of the total length of the coating portion 110 of the electrode 100, and the active material a may not be applied.

[0067] Only the active material a may be removed from the coating portion 110 extending from the electrode 100 to form the notching tab part 111. Alternatively, only the electrode foil may extend without applying the active material a on the area of the coating portion 110 of the electrode 100 to form the notching tab part 111.

[0068] Thus, there may be an effect of forming the multi-tab while minimizing the formation of the non-coating portion, and the formation of the non-coating portion may be minimized to secure the battery capacity.

[0069] FIG. 3 is a perspective view illustrating a state in which the notching tab part is cut to form the notching tab in FIG. 2.

[0070] As illustrated in FIG. 3, the notching tab part 111 may be cut in the wound state to form a plurality of notching tabs 111a. When the number of notching tabs 111a increases, the electrode tab may decrease in resistance.

[0071] Here, one notching tab 111a may have a width of 4 mm to 5 mm. This is done for a reason in which, if the notching tab 111a has a width that is less than 4 mm, the mechanical strength may be weakened, and if the notching tab 111a has a width exceeding 5 mm, deformation in the process of bonding for the electrical connection with the can member may be difficult.

[0072] The notching tab 111a that is the foil tab may increase in mechanical strength to prevent the notching tab 111a from being broken in the bent state, and thus, the notching tab 111a may be bonded to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

[0073] FIG. 4 is a projective perspective view when seeing through the can member in a state in which the electrode assembly is installed in the can member according to an

embodiment of the present invention, and FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

[0074] As illustrated in FIGS. 4 and 5, in the electrode assembly 1 of the secondary battery according to the present invention, a bent part of the notching tab 111a may be welded to be coupled to the can member 10 so that the notching tab 111a is electrically connected to the can member 10 in a state in which the notching tab 111a is bent while being accommodated in the can member 10.

[0075] The notching tab 111a may be bent to allow an end of the notching tab 111a to be coupled to a center of a bottom surface of the can member 10. Thus, the plurality of notching tabs 111a may be electrically connected to each other while the electrical connection is smooth to minimize the electric resistance.

[0076] FIG. 8 is a bottom view illustrating a state in which a notching tab and an electrode tab are bonded to each other according to another embodiment of the present invention, and FIG. 9 is a side view of FIG. 8.

[0077] As illustrated in FIGS. 8 and 9, in a secondary battery according to another embodiment of the present invention, a notching tab part 111 of an electrode assembly 1, in which an electrode 100 is wound, may be cut to form a plurality of notching tabs 111a. The plurality of notching tabs 111a may be bent and then be connected to overlap an electrode tab 121 formed on a central portion of an end of the electrode assembly 1.

[0078] Also, the electrode assembly 1 may be accommodated in a can member 10, and the plurality of notching tabs 111a connected to overlap the electrode tab 121 may be bonded to the can member 10 together with the electrode tab 121 through welding.

[0079] As described above, in the secondary battery according to another embodiment of the present invention, since the plurality of notching tabs 111a and the electrode tab 121 are welded together to the can member 10 in the state in which the notching tabs 111a and

the electrode tab 121 are connected to overlap each other, welding processability may be easily secured.

[0080] In the secondary battery according to another embodiment of the present invention, when an end of the electrode assembly 1 has a length ranging of 16 mm to 21 mm, a notching length of each of the plurality of notching tabs 111a may range of 8 mm to 11 mm. When the notching length of the notching tab 111a is less than 8 mm, the length of the notching tab 111a may be shortened, and the notching tab 111a may be bent. Thus, it may be difficult to allow the notching tab 111a to overlap the electrode tab 121 formed on a central portion of an end of the electrode assembly 1. Also, when the notching length of the notching tab 111a exceeds 11 mm, the length of the notching tab 111a may be unnecessarily lengthened to increase in cost due to waste of the material and in space occupied by the notching tab 111a within the can member 10. Thus, an amount of electrolyte filled into the can member 10 may be reduced to deteriorate capacity. Preferably, the notching tab 111a may have a notching length of 9 mm to 10 mm.

[0081] In the secondary battery according to another embodiment of the present invention, an insulation member 130 may be formed between a bent part of the notching tab 111a and an end of the electrode assembly 1 in which the electrode 100 is wound.

[0082] The insulation member 130 may insulate the notching tab 111a from the end of the electrode assembly 1 to prevent the notching tab 111a from coming into contact with the electrode assembly 1.

[0083] The plurality of notching tabs 111a may be formed on an outer portion of the end of the electrode assembly to prevent the insulation member 130 from interfering with the end of the electrode assembly 1.

[0084] The plurality of notching tabs 111a may be formed on the outside of the electrode assembly so that the insulation member 130 is installed between the plurality of notching

tabs 111a and the end of the electrode assembly 1. Thus, the contact between the notching tab 111a and the electrode assembly 1 may be prevented to improve safety.

[0085] Hereinafter, a method of manufacturing a secondary battery according an embodiment of the present invention will be described in detail with reference to the accompanying drawings.

[0086] FIG. 6 is a flowchart successively illustrating a method for manufacturing a secondary battery according to an embodiment of the present invention.

[0087] As illustrated in FIG. 6, a method for manufacturing a secondary battery according to an embodiment of the present invention includes a preparation step (S1), a notching tab part formation step (S2), a winding step (s3), a notching tab formation step (S4), and an installation step (S5).

[0088] The preparation step (S1) is a step of preparing an electrode 100 in which a coating portion 110 coated with an active material a on an electrode collector and a non-coating portion 120 on which the active material a is not applied are provided in a longitudinal direction of the electrode collector.

[0089] Here, the electrode 100 may be a negative electrode.

[0090] FIG. 7 is a plan view illustrating a state in which the coating portion and the non-coating portion, which are formed in the longitudinal direction of the electrode collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

[0091] As illustrated in FIG. 7, in the preparation step (S1), the coating portion 110 and the non-coating portion 120, which are formed in a longitudinal direction L of the electrode collector may be arranged in parallel to each other in the width direction W of the electrode collector.

[0092] In the notching tab part formation step (S2) is a step of notching the electrode 10 to form a notching tab part 111 extending in a state in which the active material is not formed in the width direction W of the coating portion 111 on an area of the electrode 100 corresponding to 1/3 or more of a length of the coating portion 110.

[0093] As described above, the notching of the notching tab part 111 in the electrode 100 may be a step of notching a portion of the non-coating portion on which the active material is not applied and which is formed in the width direction W of the coating portion 111 while cutting the electrode 100 in the electrode collector to form a notch tab part 111.

[0094] Also, when the coating portion 100 and the non-coating portion 120, which are formed in the longitudinal direction L of the electrode collector, are arranged in parallel to each other in the width direction W of the electrode collector, the non-coating portion on which the active material is not applied and which is formed between one coating portion 110 and the other coating portion 110a, which are arranged in parallel to each other in the width direction W may be notched while cutting the coating portion 110 and the non-coating portion 120, which are arranged in parallel to each other to form the notching tab part 111 in the width direction W of the coating portion 110.

[0095] The winding step (S3) may be a step of stacking the electrode 100, on which the notching tab part 111 is formed, on the separator and the other electrode having a polarity different from that of the electrode 100 and winding the stack to form the electrode assembly 1.

[0096] While the electrode 100 is wound, the notching tab part 111 may overlap each other to form two or more layers.

[0097] The notching tab formation step (S4) may be a step of cutting the notching tab part 111 to form a plurality of notching tabs 111a.

[0098] The notching tabs 111a formed by cutting the notching tab part 111 may overlap each other to two or more layers like the notching tab part 111.

[0099] Here, the cutting of the notching tab part 111 may be maximized in work efficiency by using laser.

[0100] The insulation step (S5) may be a step of accommodating the electrode assembly 1 into the can member 10 to bond the notching tabs 111a to the can member 10.

[0101] In the installation step (S5), each of the notching tab 111a may be bent so that the welding is easily performed while the notching tab 111a is electrically connected to the can member 10 to allow the electrode assembly 1 to be accommodated in the can member 10 in the state in which the notching tab 111a is bent.

[0102] Here, since the notching tab 111a has the two or more layers, even though the notching tab 111a is bent, mechanical strength may be maintained, and thus, the notching tab may not be easily broken.

[0103] The bent notching tab 111a may have an end that faces a center of a bottom surface of the can member 10.

[0104] The coupling between the bottom surface of the can member 10 and the notching tab 111a may be performed by welding the bent part of the notching tab 111a to the bottom surface of the can member 10.

[0105] In addition, according to another embodiment of the present invention, the bent portions of the plurality of notching tabs 111a may overlap the central portion of the end of the electrode assembly 1 so as to be welded and bonded to the can member 10 together with the electrode tab 121 formed on the central portion of the end of the electrode 1 at the same time.

[0106] Although the secondary battery and the method for manufacturing the same according to the present invention has been described above with reference to the exemplary

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drawings, various changes and modifications may be made thereto by one skilled in the art without departing from the scope and spirit of the invention as set forth in the appended claims.

CLAIMS

1. An electrode assembly in a jelly-roll form, the electrode assembly having a separator disposed between a first electrode and a second electrode, the electrode assembly being wound about a winding axis along a longitudinal direction, one of the first or second electrodes comprising:

an electrode collector; and

an active material coated on a coating portion of the electrode collector,

wherein the electrode collector comprises a notching tab part on which the active material is not coated, the notching tab part extending from one side of the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part only extending from a first end region of the coating portion in the longitudinal direction located at a radially outer portion of the jelly-roll form remote from the winding axis of the jelly-roll form in a radial direction,

wherein the notching tab part has a plurality of notching tabs defined by cut portions where the notching tab part is cut,

the plurality of notching tabs are arranged along the radially outer portion of the jelly-roll form such that they are spaced apart from one another in the longitudinal direction of the electrode collector by the cut portions but overlap one another in a radial direction of the jelly-roll form, and

each of the plurality of notching tabs is bent radially inward to define bent parts that overlap one another in an axial direction of the jelly-roll form perpendicular to the radial direction and the circumferential direction to provide welding portions.

2. The electrode assembly of claim 1, further comprising an insulating member disposed radially inward of the plurality of notching tabs and axially inward of the bent parts of the plurality of notching tabs.

3. The electrode assembly of claim 1, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap

one another in the radial direction are together distributed about the winding axis of the jelly-roll form.

4. The electrode assembly of claim 1, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap one another in the radial direction are spaced apart from the winding axis of the jelly-roll form.

5. The electrode assembly of claim 1, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap one another in the radial direction do not overlap one another in the axial direction.

6. The electrode assembly of claim 1, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap one another in the radial direction overlap one another in the axial direction at the winding axis of the jelly-roll form.

7. The electrode assembly of claim 1, wherein the electrode collector further comprises a non-coating portion disposed at a second end region of the coating portion opposite the first end region in the longitudinal direction, the non-coating portion being uncoated with the active material.

8. The electrode assembly of claim 7, wherein the second end region of the coating portion is located at an inner core of the jelly-roll form, and an electrode tab is attached to the non-coating portion, the electrode tab extending further in the width direction beyond the coating portion.

9. The electrode assembly of claim 8, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap one another in the radial direction overlap one another and the electrode tab in the axial direction at a central portion of the jelly-roll form.

10. A secondary battery comprising:
a can member; and
the electrode assembly of claim 1 accommodated in the can member,
wherein the bent parts are welded to the can member.

11. The secondary battery of claim 10, further comprising an insulating member disposed radially inward of the plurality of notching tabs and axially inward of the bent parts of the plurality of notching tabs.

12. The secondary battery of claim 10, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap one another in the radial direction are together distributed about the winding axis of the jelly-roll form.

13. The secondary battery of claim 10, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap one another in the radial direction are spaced apart from the winding axis of the jelly-roll form.

14. The secondary battery of claim 10, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap one another in the radial direction do not overlap one another in the axial direction.

15. The secondary battery of claim 10, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap one another in the radial direction overlap one another in the axial direction at the winding axis of the jelly-roll form.

16. The secondary battery of claim 10, wherein the electrode collector further comprises a non-coating portion disposed at a second end region of the coating portion

opposite the first end region in the longitudinal direction, the non-coating portion being uncoated with the active material.

17. The secondary battery of claim 16, wherein the second end region of the coating portion is located at an inner core of the jelly-roll form, and an electrode tab is attached to the non-coating portion, the electrode tab extending further in the width direction beyond the coating portion.

18. The secondary battery of claim 17, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap one another in the radial direction overlap one another and the electrode tab in the axial direction at a central portion of the jelly-roll form.

ABSTRACT

The present invention relates to a secondary battery in which an electrode tab is improved in mechanical strength and a method for manufacturing the same. Also, the secondary battery according to the present invention includes: an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector and a notching tab part extending from the coating portion in a width direction of the electrode collector without coating with the active material and overlapping each other to form two or more layers when the electrode is wound.

CERTIFICATION AND REQUEST FOR PRIORITIZED EXAMINATION UNDER 37 CFR 1.102(e) (Page 1 of 1)			
First Named Inventor:	Jae Woong Kim	Nonprovisional Application Number (if known):	Not Yet Assigned
Title of Invention:	Secondary Battery and Method for Manufacturing the Same		
APPLICANT HEREBY CERTIFIES THE FOLLOWING AND REQUESTS PRIORITIZED EXAMINATION FOR THE ABOVE-IDENTIFIED APPLICATION.			
1. The processing fee set forth in 37 CFR 1.17(i)(1) and the prioritized examination fee set forth in 37 CFR 1.17(c) have been filed with the request. The publication fee requirement is met because that fee, set forth in 37 CFR 1.18(d), is currently \$0. The basic filing fee, search fee, and examination fee are filed with the request or have been already been paid. I understand that any required excess claims fees or application size fee must be paid for the application. 2. I understand that the application may not contain, or be amended to contain, more than four independent claims, more than thirty total claims, or any multiple dependent claims, and that any request for an extension of time will cause an outstanding Track I request to be dismissed. 3. The applicable box is checked below: I. ☒ Original Application (Track One) - Prioritized Examination under § 1.102(e)(1) i. (a) The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a). This certification and request is being filed with the utility application via USPTO patent electronic filing system. ---OR--- (b) The application is an original nonprovisional plant application filed under 35 U.S.C. 111(a). This certification and request is being filed with the plant application in paper. ii. An executed inventor's oath or declaration under 37 CFR 1.63 or 37 CFR 1.64 for each inventor, or the application data sheet meeting the conditions specified in 37 CFR 1.53(f)(3)(i) is filed with the application. II. ☐ Request for Continued Examination - Prioritized Examination under § 1.102(e)(2) i. A request for continued examination has been filed with, or prior to, this form. ii. If the application is a utility application, this certification and request is being filed via USPTO patent electronic filing system. iii. The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a), or is a national stage entry under 35 U.S.C. 371. iv. This certification and request is being filed prior to the mailing of a first Office action responsive to the request for continued examination. v. No prior request for continued examination has been granted prioritized examination status under 37 CFR 1.102(e)(2).			

Signature /Joseph H. Karlin/	Date August 28, 2025
Name (Print/Typed) Joseph H. Karlin	Practitioner Registration Number 62,471
Note: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. Submit multiple forms if more than one signature is required. *	
I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4). Dated: August 28, 2025 Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/	

UTILITY PATENT APPLICATION TRANSMITTAL (Only for new nonprovisional applications under 37 CFR 1.53(b))	Attorney Docket No. LGES 3.3F-1654 CCCIIICII	
	First Named Inventor Jae Woong Kim	
	Title	Secondary Battery and Method for Manufacturing the Same
	Priority Mail Express® Label No.	

APPLICATION ELEMENTS See MPEP chapter 600 concerning utility patent application contents. 1. ☐ Fee Transmittal Form (PTO/SB/17 or equivalent) 2. ☐ Applicant asserts small entity status. See 37 CFR 1.27 3. ☐ Applicant certifies micro entity status. See 37 CFR 1.29. Applicant must attach form PTO/SB/15A or B or equivalent. 4. ☒ Specification [Total Pages <u>22</u>] Both the claims and abstract must start on a new page. (See MPEP § 608.01(a) for information on the preferred arrangement) 5. ☒ Drawing(s) (35 U.S.C. 113) [Total Sheets <u>9</u>] 6. Inventor's Oath or Declaration [Total Pages <u>2</u>] (including substitute statements under 37 CFR 1.64 and assignments serving as an oath or declaration under 37 CFR 1.63(e)) a. ☐ Newly executed (original or copy) b. ☒ A copy from a prior application (37 CFR 1.63(d)) 7. ☒ Application Data Sheet * See note below. See 37 CFR 1.76 (PTO/AIA/14 or equivalent) 8. CD-ROM or CD-R in duplicate, large table, or Computer Program (Appendix) ☐ Landscape Table on CD 9. Nucleotide and/or Amino Acid Sequence Submission (if applicable, items a. – c. are required) a. ☐ Computer Readable Form (CRF) b. ☐ Specification Sequence Listing on: i. ☐ CD-ROM or CD-R (2 copies); or ii. ☐ Paper c. ☐ Statements verifying identity of above copies	ADDRESS TO: Commissioner for Patents P.O. Box 1450 Alexandria, VA 22313-1450 ACCOMPANYING APPLICATION PAPERS 10. ☐ Assignment Papers (cover sheet & document(s)) Name of Assignee 11. ☐ 37 CFR 3.73(c) Statement (when there is an assignee) ☒ Power of Attorney 12. ☐ English Translation Document (if applicable) 13. ☒ Information Disclosure Statement (PTO/SB/08 or PTO-1449) ☐ Copies of citations attached 14. ☐ Preliminary Amendment 15. ☐ Return Receipt Postcard (MPEP § 503) (Should be specifically itemized) 16. ☐ Certified Copy of Priority Document(s) (if foreign priority is claimed) 17. ☐ Nonpublication Request Under 35 U.S.C. 122(b)(2)(B)(i). Applicant must attach form PTO/SB/35 or equivalent. 18. ☒ Other: Certification and Request for Prioritized Examination Communication Regarding Certified Copy
--	---

***Note:** (1) Benefit claims under 37 CFR 1.78 and foreign priority claims under 1.55 **must** be included in an Application Data Sheet (ADS).
(2) For applications filed under 35 U.S.C. 111, the application must contain an ADS specifying the applicant if the applicant is an assignee, person to whom the inventor is under an obligation to assign, or person who otherwise shows sufficient proprietary interest in the matter. See 37 CFR 1.46(b).

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	Registration No. (Attorney/Agent)		62,471

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Dated: August 28, 2025

Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/



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ELECTRONIC PAYMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/313,233	08/28/2025 04:12:26 PM Z ET	LGES 3.3F-1654 CCCCCIICII

Title of Invention

Secondary Battery and Method for Manufacturing the Same

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	7385	FILED BY	Sharon McGee
PATENT CENTER #	72065099	AUTHORIZED BY	Kevin Kocun
CUSTOMER #	86765	FILING DATE	-
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Jae Woong Kim

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
CARD / 1011	E20258RG12595901	Sharon McGee

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1011	BASIC FILING FEE - UTILITY (PAPER FILING ALSO REQUIRES NON-ELECTRONIC FILING FEE UNDER 1.16(T))	350.00	1	350.00
1111	UTILITY PATENT APPL. SEARCH FEE	770.00	1	770.00
1311	PATENT APPL. EXAMINATION FEE	880.00	1	880.00
1817	REQUEST FOR PRIORITIZED EXAMINATION	4515.00	1	4515.00
1830	PROCESSING FEE, EXCEPT IN PROVISIONAL APPLICATIONS	150.00	1	150.00
1832	FILING AN INFORMATION DISCLOSURE STATEMENT THAT	200.00	1	200.00

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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Dated: August 28, 2025

Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGES 3.3F-1654 CCCCIIICII
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: Not Yet Assigned

Confirmation No.: N/A

Filed: Concurrently Herewith

Art Unit: N/A

For: Secondary Battery and Method for Manufacturing
the Same

Examiner: Not Yet Assigned

COMMUNICATION REGARDING CERTIFIED COPY

Dear Commissioner:

The present application claims priority to one or more prior foreign applications under 35 U.S.C. § 119(b). Applicant hereby identifies prior-filed nonprovisional Application No. 16/498,751, for which a benefit is claimed under 35 U.S.C. §§ 120, 121, 365(c), or 386(c), as containing a certified copy of such foreign application(s). Accordingly, pursuant to 37 C.F.R. § 1.55(h), the requirement for timely filing a certified copy of the foreign application(s) is considered satisfied in this application.

All required fees are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments to our Deposit Account No. 12-1095.

Dated: August 28, 2025

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin

Registration No.: 62,471
LERNER DAVID LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number

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LG Energy Solution, Ltd.

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
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SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature

Han Sun Lee

Date (Optional)

June 10, 2021

Name

LEE, Han-Sun

Title

IP Department Leader

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☒ Total of 1 forms are submitted.

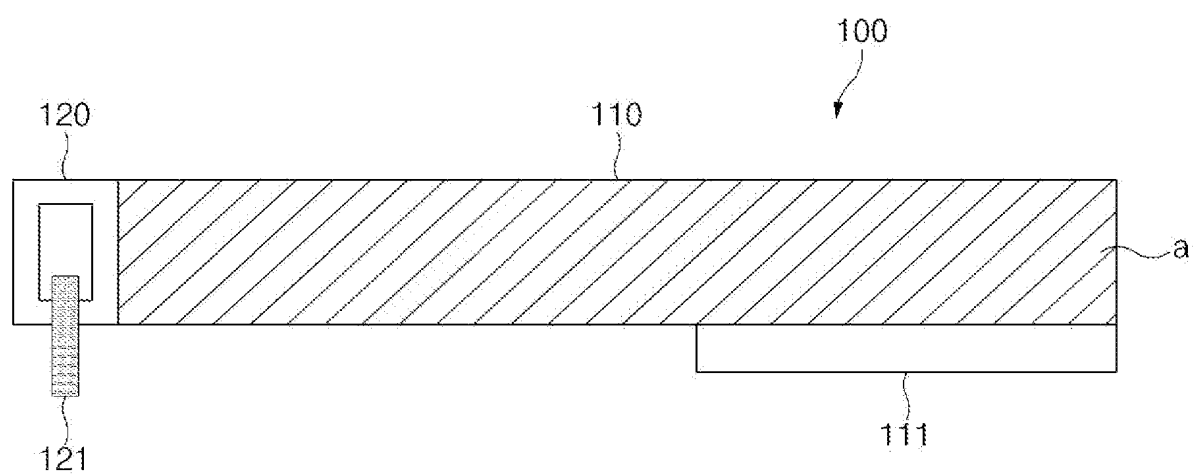


FIG. 1

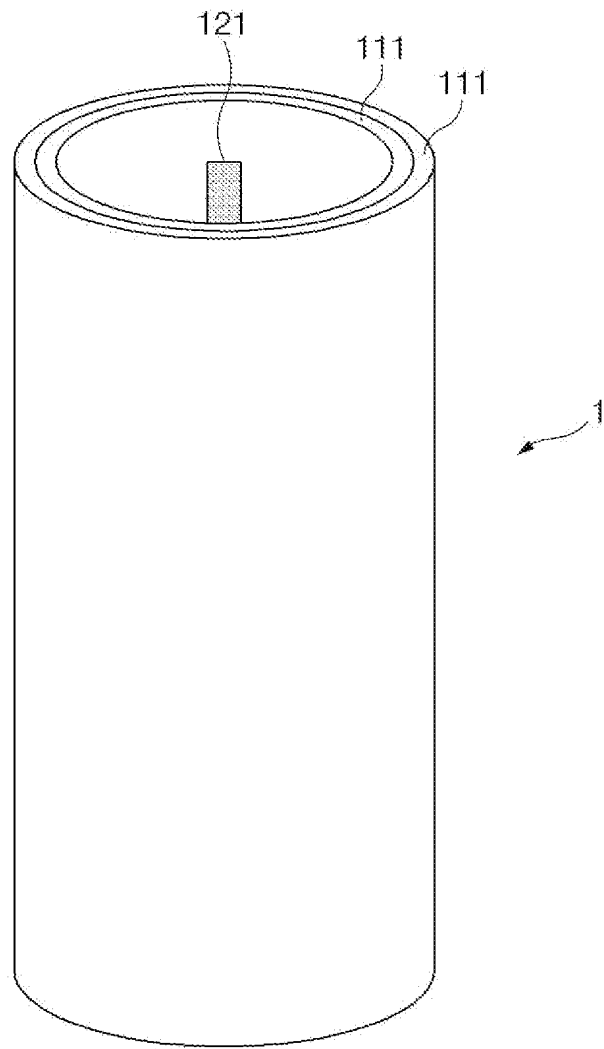


FIG. 2

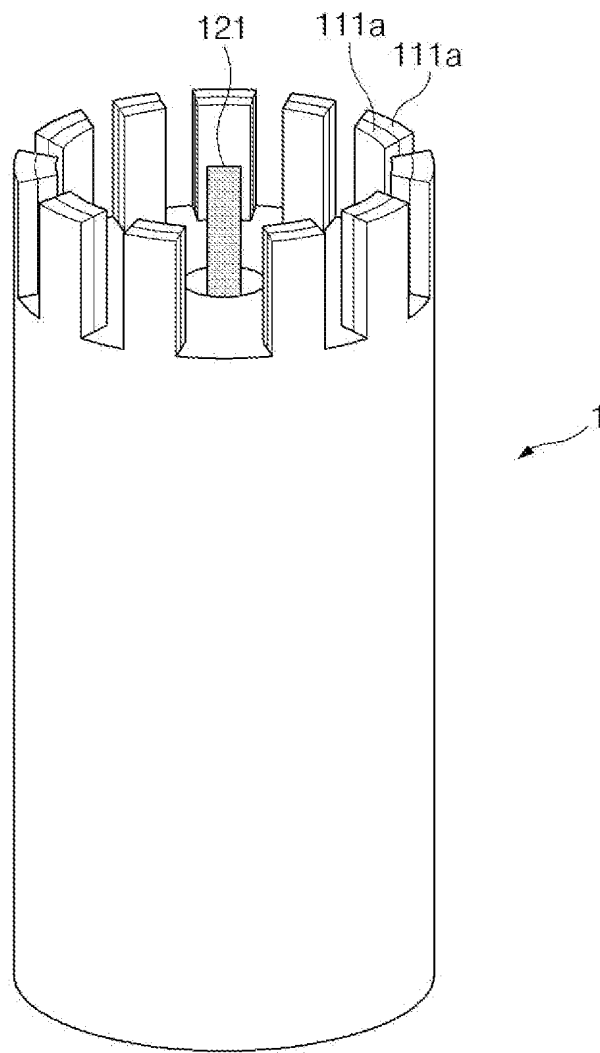


FIG. 3

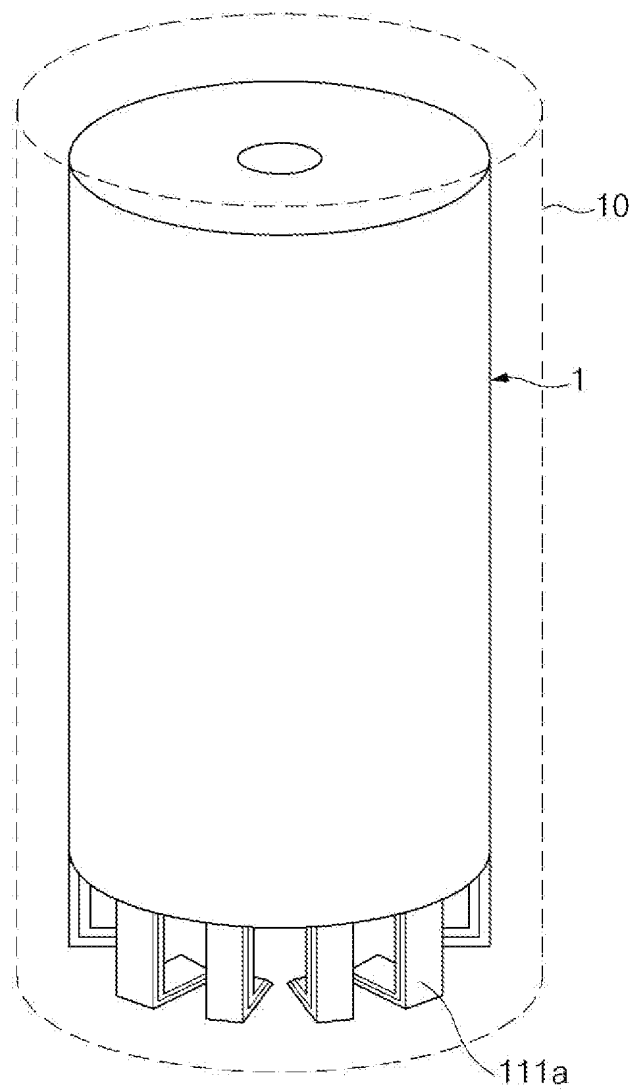


FIG. 4

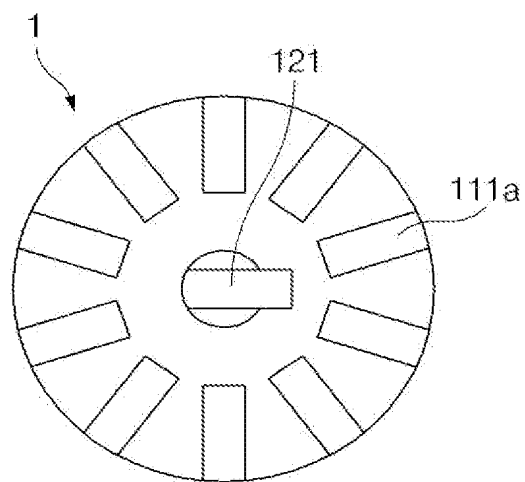


FIG. 5

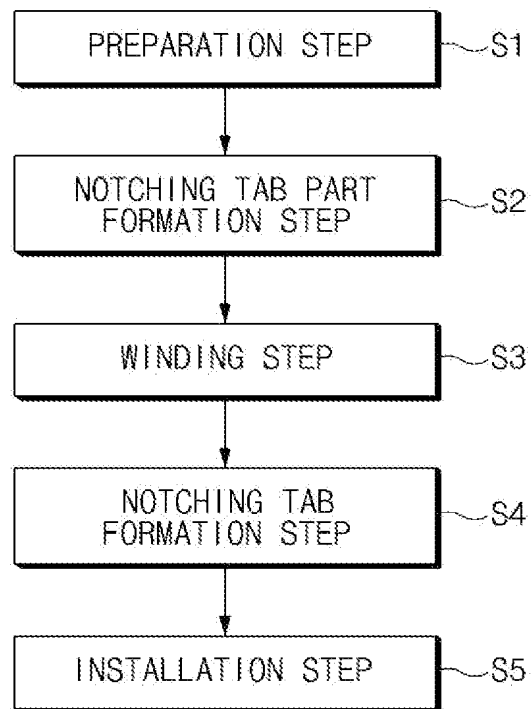


FIG.6

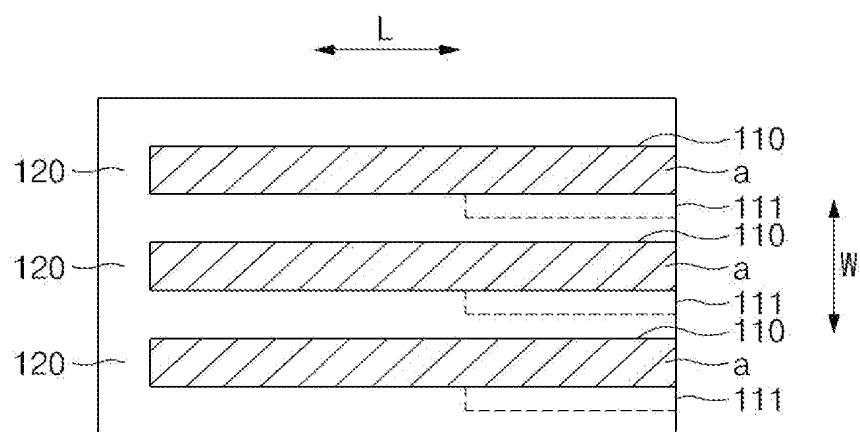


FIG. 7

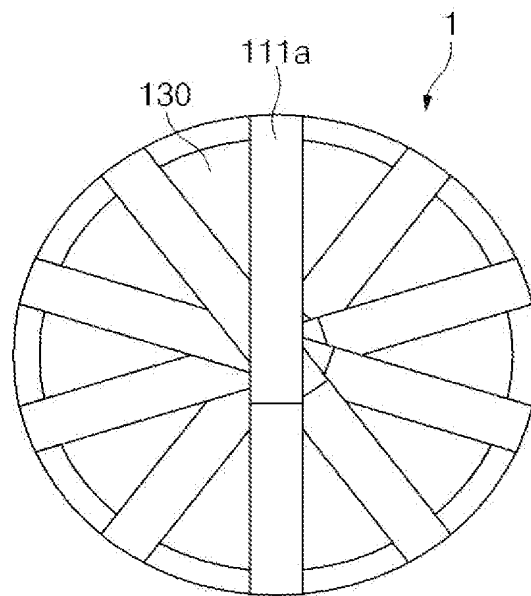


FIG. 8

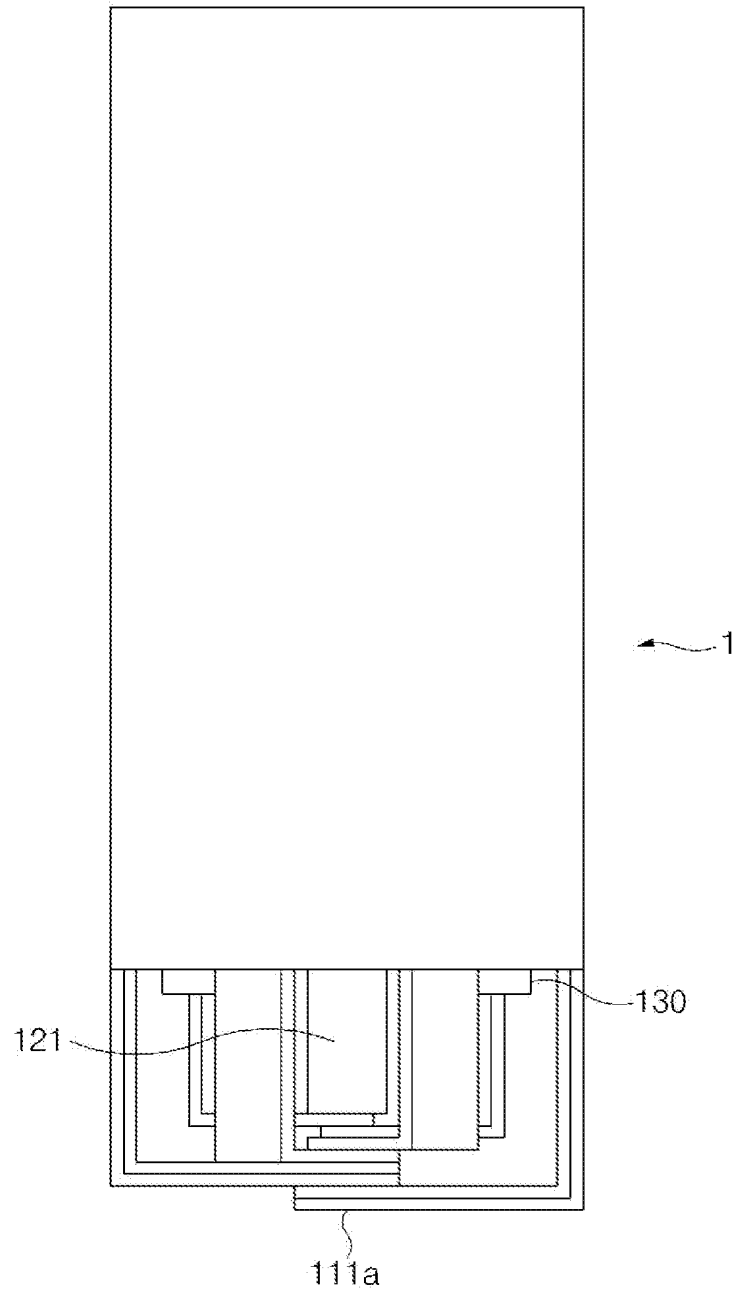


FIG.9

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		
	Filing Date		
	First Named Inventor	Jae Woong Kim	
	Art Unit	N/A	
	Examiner Name	Not Yet Assigned	
	Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII	

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	1	3761314		1973-09-25	CAILLEY J		
	2	4822377		1989-04-18	Wolff		
	3	6569565		2003-05-27	Ligeois et al.		
	4	11652232		2023-05-16	Kim et al.		
	5	11973177		2024-04-30	Kim et al.		
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	1	20010023038	A1	2001-09-20	Ligeois et al.		

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Application Number		
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First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII	

	2	20050003264	A1	2005-01-06	Oh et al.	
	3	20050142436	A1	2005-06-30	Arai et al.	
	4	20050277022	A1	2005-12-15	Kozuki, Kiyomi	
	5	20050287428	A1	2005-12-29	Cheon et al.	
	6	20050287432	A1	2005-12-29	Cheon et al.	
	7	20060024572	A1	2006-02-02	Lee, Sang-Won	
	8	20060127774	A1	2006-06-15	Kim et al.	
	9	20070196730	A1	2007-08-23	Kozuki, Kiyomi	
	10	20100081052	A1	2010-04-01	MORISHIMA et al.	
	11	20100190056	A	2010-07-29	Turner	
	12	20100310926	A1	2010-12-09	WANG et al.	

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First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII	

	13	20110067227	A1	2011-03-24	Sohn	
	14	20110293977	A1	2011-12-01	Kim et al.	
	15	20120009450	A1	2012-01-12	Chun	
	16	20120208055	A1	2012-08-16	Ahn	
	17	20130108903	A1	2013-05-02	Martin et al.	
	18	20130252054	A1	2013-09-26	BARONE et al.	
	19	20130344364	A1	2013-12-26	Nanaumi et al.	
	20	20140050956	A1	2014-02-20	HUANG	
	21	20150295270	A1	2015-10-15	Chun, Kwan-Sic	
	22	20160351966	A1	2016-12-01	Batson et al.	
	23	20180145304	A1	2018-05-24	KOGETSU et al.	

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number			
	Filing Date			
	First Named Inventor	Jae Woong Kim		
	Art Unit	N/A		
	Examiner Name	Not Yet Assigned		
	Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII		

	24	20190379028	A1	2019-12-12	Lim et al.	
	25	20240243335	A1	2024-07-18	Kim et al.	

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	1	101714624	CN	A	2010-05-26	Guangzhou Fullriver Battery New Technology Co., Lt	Machine generated translation attached.	☒
	2	102024936	CN	A	2011-04-20	Samsung Sdi Co., Ltd	U.S. Publication No. 20110067227 is the English language counterpart which was previously cited.	☐
	3	201450072	CN	U	2010-05-05	Dongguan AmpereX Electronics Technology Co., Ltd	U.S. Publication No. 20100310926 is the English language counterpart which was previously cited.	☐
	4	203250828	CN	U	2013-10-23	A123 Systems, Inc	U.S. Publication No. 20130108903 is the English lanuage counterpart which was previously cited.	☐
	5	204577523	CN	U	2015-08-19	Contemporary AmpereX Technology Ltd	Machine generated translation attached.	☒
	6	1134820	EP	A1	2001-09-19	Cit Alcatel	U.S. Publication No. 20010023038 A1 and U. S. Patent No. 6,569,565 is an English language counterpart.	☒
	7	2000-077078	JP	A	2000-03-14	Matsushita Electric Ind Co Ltd	Machine generated translation attached.	☒

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

Application Number		
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First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII	

8	2000-228182	JP	A	2000-08-15	Toyota Central Res & Dev	Machine generated translation attached.	☒
9	2003338276	JP	A	2003-11-28	Matsushita Electric Ind Co Ltd	Machine Generated Translation attached.	☒
10	2005276459	JP	A	2005-10-06	Sanyo Electric Co	English translation of Abstract only.	☒
11	2006128106	JP	A	2006-05-18	Samsung Sdi Co Ltd	U.S. Publication No. 20060127774 is an English language counterpart	☒
12	2006-324049	JP	A	2006-11-30	Sony Corp	Machine generated translation attached.	☒
13	2007227137	JP	A	2007-09-06	Matsushita Electric Ind Co Ltd	US Publication 2007196730 is an English Language counterpart.	☒
14	2008066040	JP	A	2008-03-21	Sony Corp	Machine Generated Translation attached.	☒
15	2011065981	JP	A	2011-03-31	Samsung Sdi Co Ltd	US Publication 2011067227 is an English Language counterpart.	☒
16	2016001575	JP	A	2016-01-07	Denso Corp	Machine generated translation is attached.	☒
17	2020509534	JP	A	2020-03-26	LG CHEM Ltd.	US Publication 2019379028 is an English Language counterpart.	☒
18	4401634	JP	B2	2010-01-20	Matsushita Electric Ind Co Ltd	Machine generated translation attached.	☒

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Application Number		
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First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII	

19	5117133	JP	B2	2013-01-09	Hitachi Vehicle Energy Ltd	Machine generated translation attached.	☒
20	5790576	JP	B2	2015-10-07	Toyota Industries Corp	Machine generated translation is attached.	☒
21	6045286	JP	B2	2016-12-14	Hitachi Vehicle Energy Ltd	Machine generated translation attached.	☒
22	H0935701	JP	A	1997-02-07	Denso Corp	Machine generated translation attached.	☒
23	H11339756	JP	A	1999-12-10	Toyota Central Res & Dev	Machine generated translation attached.	☒
24	100515833	KR	B1	2005-09-21	Samsung Sdi Co., Ltd	U.S. Publication No. 20050003264 A1 is an English language counterpart.	☒
25	100599710	KR	B1	2006-07-12	Samsung Sdi Co., Ltd	U.S. Publication No. 20060024572 A1 is an English translation attached.	☒
26	100599749	KR	B1	2006-07-12	Samsung Sdi Co., Ltd	U.S. Publication No. 20050287428 A1 is an English translation attached.	☒
27	20000051740	KR	A	2000-08-16	Samsung Sdi Co., Ltd	Machine generated translation is attached.	☒
28	20050121904	KR	A	2005-12-28	Samsung Sdi Co., Ltd	U.S. Publication No. 20050287428 A1 is an English language counterpart.	☒
29	20050121914	KR	A	2005-12-28	Samsung Sdi Co., Ltd	U.S. Publication No. 20050287432 A1 is an English translation attached.	☒

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII	

	30	20060022358	KR	A	2006-03-10	Samsung Sdi Co., Ltd	Machine generated translation is attached.	☒
	31	20100071941	KR	A	2010-06-29	Lg Chemical Ltd	U.S. Publication No. 20110293977 A1 is an English language counterpart.	☒
	32	20120006389	KR	A	2012-01-18	Samsung Sdi Co., Ltd	U.S. Publication No. 20120009450 A1 is an English language counterpart.	☒
	33	20120092367	KR	A	2012-08-21	Samsung Sdi Co., Ltd	U.S. Publication No. 20120208055 A1 is an English language counterpart.	☒
	34	20130073329	KR	A	2013-07-03	Datechnology Co., Ltd	Machine generated translation is attached.	☒
	35	20150117135	KR	A	2015-10-19	Samsung Sdi Co Ltd	U.S. Publication No. 20150295270 A1 is an English language counterpart.	☒
	36	2016-0009406	KR	A	2016-01-26	Lg Chemical Ltd	Machine generated translation attached.	☒
	37	20160113612	KR	A	2016-09-30	A123 Systems LLC	U.S. Publication No. 20160351966 A1 is an English language counterpart.	☒
	38	20170006437	KR	A	2017-01-18	Lg Chemical Ltd	Machine generated translation is attached.	☒
	39	2012086690	WO	A1	2012-06-28	Hitachi Vehicle Energy, Ltd	US Publication 2013344364 is an English Language counterpart.	☒
	40	2014188501	WO	A1	2014-11-27	Hitachi Automotive Systems, Ltd	Machine generated translation attached.	☒

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII	

If you wish to add additional Foreign Patent Document citation information, please click the Add button **Add**

NON-PATENT LITERATURE DOCUMENTS

Remove

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	Chiinese Search Report for Application No. 201880025247, dated August 24, 2021, 2 pages.	☐
	2	Extended European Seach Report including Written Opinion for Application No. 24187569.9 dated 10/11/24. 9 pages.	☐
	3	Extended European Seach Report including Written Opinion for Application No. 24187669.7 dated 10/18/24. 9 pages.	☐
	4	Extended European Seach Report including Written Opinion for Application No. EP18784869.2 dated January 21, 2020, 5 pages.	☐
	5	International Search Report from Application No. PCT/KR2018/004366 mailed July 20, 2018, pages 1 - 2	☒

If you wish to add additional non-patent literature document citation information, please click the Add button **Add**

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered and whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		
	Filing Date		
	First Named Inventor	Jae Woong Kim	
	Art Unit	N/A	
	Examiner Name	Not Yet Assigned	
	Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII	

TIMING STATEMENT			
Please see 37 CFR 1.97(e) to make the appropriate selection(s):			
☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).			
OR			
☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).			
☐ See attached timing statement under 37 CFR 1.97(e).			
☒ A timing statement under 37 CFR 1.97(e) is not submitted herewith.			
TIMING FEE			
☐ The fee set forth in 37 CFR 1.17(p) has been submitted herewith.			
COPIES			
☒ An identification of an earlier application pursuant to 37 CFR 1.98(d)(1) is attached.			
SIZE FEE ASSERTION			
Please see 37 CFR 1.17(v) and the IDS Auto-Load Instructions for completing this form to make the appropriate selection of an assertion under 37 CFR 1.98. For the information disclosure statement (IDS) submitted herewith, the applicant or patent owner certifies the following with respect to the cumulative number of applicant-provided or patent-owner provided items of information submitted to date including those in the accompanying IDS (select only one):			
☐ No IDS size fee is required under 37 CFR 1.17(v) at this time.			
☒ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(1).			
☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(2).			
☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(3).			
SIGNATURE			
A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 11.18. Please see 37 CFR 1.4(d) for the form of the signature.			
Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2025-08-28
Name/Print	Joseph H. Karlin	Registration Number	62,471

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

Application Number	
Filing Date	
First Named Inventor	Jae Woong Kim
Art Unit	N/A
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: August 28, 2025

Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGES 3.3F-1654 CCCCIIII
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: Not Yet Assigned

Confirmation No.: N/A

Filed: Concurrently Herewith

Art Unit: N/A

For: Secondary Battery and Method for Manufacturing
the Same

Examiner: Not Yet Assigned

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Dear Commissioner:

It is respectfully requested that the references listed on the enclosed form be made of record and considered with respect to the above-referenced U.S. patent application. A copy of each foreign patent document and each non-patent literature document was of record in Application Nos. 19/207,758, 18/620,269, 18/618,609, 18/132,719 and 16/498,751, the benefit of which is claimed under § 120. Submission of the present Information Disclosure Statement should not be taken as an admission that the cited references are legally available prior art or that the same are pertinent or material or even analogous art to the claimed invention.

The required fees are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments to our Deposit Account No. 12-1095.

Dated: August 28, 2025

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin
Registration No.: 62,471
LERNER DAVID LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant

LD-459A

**DECLARATION AND ASSIGNMENT FOR
UTILITY OR DESIGN APPLICATIONS****Attorney Docket Number:****Title of Invention**

SECONDARY BATTERY AND METHOD FOR MANUFACTURING THE SAME

As the below named inventor(s) (hereinafter designated as the undersigned), I/We hereby declare that:

This declaration
is directed to:

The attached application, or



United States application or PCT international application number _____

filed on _____

The above-identified application was made or authorized to be made by me/us.

I/We believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I/We hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

WHEREAS, the undersigned as a sole inventor (if only one name is listed below) or as a joint inventor (if plural names are listed below) has invented certain new and useful improvements in the application identified above.

AND, WHEREAS **LG CHEM, LTD.**, (hereinafter referred to as assignee), a corporation of Korea whose mailing address is 128, Yeoui-daero, Yeongdeungpo-gu, Seoul 07336, Republic of Korea is desirous of acquiring the entire right, title and interest in the same.

NOW, THEREFORE, in consideration of the sum of one dollar, the receipt whereof is acknowledged, and other good and valuable consideration, the undersigned hereby sell(s), assign(s) and set(s) over to said assignee my/our entire right, title and interest for the United States of America and all other countries in and to said invention and the aforesaid utility patent application and all original, divisional, continuation, substitute or reissue applications and patents applied for or granted therefor in the United States of America and all other countries, for said invention, including without limitation all applications and patents for said invention claiming priority or benefit of the aforesaid utility application pursuant to any law or treaty, and including all claims for damages by reason of past infringement with the right to sue for and collect the same for its own use and the use of its successors, assigns and legal representatives, and including the right to claim such priority or benefit and the Commissioner of Patents and Trademarks is hereby authorized and requested to issue all patents on said improvements or resulting therefrom to said assignee herein, as assignee of the entire interest therein; and the undersigned for me/us and my/our legal representatives, heirs and assigns do hereby agree and covenant without further remuneration, to execute and deliver all original, divisional, continuation, reissue and other applications for Letters Patent on said improvements and all assignments thereof to said assignee or its assigns, to communicate to said assignee or its representatives all facts known to the undersigned respecting said improvements, whenever requested, to testify in any interferences or other legal proceedings in which any of said applications or patents may become involved, to sign all lawful papers, make all rightful oaths, and to do generally everything necessary to aid assignee, its successors, assigns and nominees to obtain patent protection for said improvements in all countries, the expenses incident to said applications to be borne and paid by said assignee.

And I do hereby authorize my attorneys to insert on this deed the filing date and application number of said application when known.

FULL NAME OF INVENTOR(S)Inventor one: **KIM, Jae Woong**Date: September 25, 2019Signature: Additional inventors or a legal representative are being named on 1 additional form(s) attached hereto.**PLEASE SEE THE WARNINGS ON THE LAST PAGE OF THIS FORM.**

DECLARATION AND ASSIGNMENT FOR UTILITY OR DESIGN APPLICATIONS	ADDITIONAL INVENTOR(S) Supplemental Sheet
Inventor two: <u>KIM, Hyoung Kwon</u> Signature: <u>[Signature]</u>	Date: <u>September 25, 2019</u>
Inventor three: <u>KIM, Sang Uck</u> Signature: <u>[Signature]</u>	Date: <u>September 25, 2019</u>
Inventor four: _____ Signature: _____	Date: _____
Inventor five: _____ Signature: _____	Date: _____
Inventor six: _____ Signature: _____	Date: _____
Inventor seven: _____ Signature: _____	Date: _____
Inventor eight: _____ Signature: _____	Date: _____
Inventor nine: _____ Signature: _____	Date: _____
Inventor ten: _____ Signature: _____	Date: _____
WARNING: According to 37 C.F.R. 1.63 (c), "[a] person may not execute an oath or declaration unless the person has reviewed and understands the contents of the application, including the claims and is aware of the duty to disclose to the Office all information known to be material to patentability as defined in 1.56." Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available.	



Placeholder Sheet for Supplemental Content

Application Number: 19313233

Document Date: 08/28/2025

The presence of this form indicates that the following document type was received in electronic format on the date identified above. This document is stored separately from the typical case contents which are stored in image format.

Since this was an electronic submission, there is no physical artifact folder, no artifact folder is recorded and no paper documents or physical media exist.

- Drawing

At the time of document entry (noted above):

- USPTO employees may access this supplemental content via Docket Application Viewer (DAV) or Supplemental Content Viewer (SCV).
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United States Patent and Trademark Office

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Document Code:WFEE

User :Elizabeth Hailu

Sale Accounting Date:09/08/2025

Sale Item Reference Number

19313233

Effective Date

08/28/2025

Document Number	Fee Code	Fee Code Description	Amount Paid	Payment Method
I202598C55287811	1055	FILING AN APPLICATION OR PRESENTATION OF BENEFIT CLAIM MORE	\$2,700.00	Deposit Account



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
19/313,233	08/28/2025		2000	LGES 3.3F-1654 CCCCHII	18	1

CONFIRMATION NO. 7385

86765

LG

Lerner David LLP

20 Commerce Drive

Cranford, NJ 07016

FILING RECEIPT



0000000094194706

Date Mailed: 09/10/2025

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Jae Woong Kim, Daejeon, KOREA, REPUBLIC OF;
Hyoung Kwon Kim, Daejeon, KOREA, REPUBLIC OF;
Sang Uck Kim, Daejeon, KOREA, REPUBLIC OF;

Applicant(s)

LG Energy Solution, Ltd., Seoul, KOREA, REPUBLIC OF;

Assignment For Published Patent Application

LG Energy Solution, Ltd., Seoul, KOREA, REPUBLIC OF

Power of Attorney: The patent practitioners associated with Customer Number 86765

Domestic Applications for which benefit is claimed

This application is a CON of 19/207,758 05/14/2025
which is a CON of 18/620,269 03/28/2024 PAT 12,334,493
which is a CON of 18/618,609 03/27/2024 PAT 12,341,147
which is a CON of 18/132,719 04/10/2023 PAT 11,973,177
which is a CON of 16/498,751 09/27/2019 PAT 11,652,232
which is a 371 of PCT/KR2018/004366 04/13/2018

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

REPUBLIC OF KOREA 10-2018-0043191 04/13/2018 No Access Code Provided

REPUBLIC OF KOREA 10-2017-0048645 04/14/2017 No Access Code Provided

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 09/04/2025

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 19/313,233**

Projected Publication Date: 12/18/2025

Non-Publication Request: No

Early Publication Request: No

Title

Secondary Battery and Method for Manufacturing the Same

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative,

this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

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technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
19/313,233

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	350
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	770
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	880
TOTAL CLAIMS (37 CFR 1.16(i))	18 minus 20 =	*			OR	x =	0
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*				x =	0
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	2000

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus **	=	x =		OR	x =	
Independent (37 CFR 1.16(h))	*	Minus ***	=	x =		OR	x =	
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus **	=	x =		OR	x =	
Independent (37 CFR 1.16(h))	*	Minus ***	=	x =		OR	x =	
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.

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Application	Document	Mailroom Date	Attorney Docket No.
19313233	APP.FILE.REC	09/10/2025	LGES 3.3F-1654 CCCCIIICII
19313233	WELCOME.LET	09/10/2025	LGES 3.3F-1654 CCCCIIICII

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/313,233	08/28/2025	Jae Woong Kim	LGES 3.12F-1654 CCCCIICII	7385
86765	7590	09/18/2025	EXAMINER	
LG			CENTRAL, DOCKET	
Lerner David LLP			ART UNIT	
20 Commerce Drive			PAPER NUMBER	
Cranford, NJ 07016			OPAP	
			NOTIFICATION DATE	DELIVERY MODE
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<i>Decision Granting Request for Prioritized Examination (Track I)</i>	Application No. 19/313,233	Applicant(s) Kim et al.	
	Examiner FIKIRTE A GEREMEW	Art Unit OMBL	AIA (FITF) Status Yes
1. THE REQUEST FILED <u>28 August 2025</u> IS GRANTED . The above-identified application has met the requirements for prioritized examination A. ☒ for an original nonprovisional application (Track I). B. ☐ for an application undergoing continued examination (RCE). 2. The above-identified application will undergo prioritized examination. The application will be accorded special status throughout its entire course of prosecution until one of the following occurs: A. filing a <u>petition for extension of time</u> to extend the time period for filing a reply; B. filing an <u>amendment to amend the application to contain more than four independent claims, more than thirty total claims</u>, or a multiple dependent claim; C. filing a <u>request for continued examination</u> ; D. filing a notice of appeal; E. filing a request for suspension of action; F. mailing of a notice of allowance; G. mailing of a final Office action; H. completion of examination as defined in 37 CFR 41.102; or I. abandonment of the application. Telephone inquiries with regard to this decision should be directed to FIKIRTE GEREMEW at (703) 756-1930. In his/her absence, calls may be directed to Petition Help Desk at (571) 272-3282.			
/FIKIRTE A GEREMEW/ PROGRAM SUPPORT SPECIALIST, OMBL			

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FROM: noreply@uspto.gov
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19313233	TRACK1.GRANT	09/18/2025	LGES 3.12F-1654 CCCCIIICII

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/313,233	08/28/2025	Jae Woong Kim	LGES 3.12F-1654 CCCCIICII	7385
86765	7590	12/09/2025	EXAMINER	
LG Lerner David LLP 20 Commerce Drive Cranford, NJ 07016			EGGERDING, ALIX ECHELMAYER	
			ART UNIT	PAPER NUMBER
			1729	
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Office Action Summary

Application No.

19/313,233

Applicant(s)

Kim et al.

Examiner

ALIX E EGGERDING

Art Unit

1729

AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1) ☒ Responsive to communication(s) filed on 8/28/25.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2a) ☐ This action is **FINAL**.

2b) ☒ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☒ Claim(s) 1-18 is/are pending in the application.

5a) Of the above claim(s) ____ is/are withdrawn from consideration.

6) ☐ Claim(s) ____ is/are allowed.

7) ☒ Claim(s) 1-18 is/are rejected.

8) ☐ Claim(s) ____ is/are objected to.

9) ☐ Claim(s) ____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10) ☐ The specification is objected to by the Examiner.

11) ☐ The drawing(s) filed on ____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12) ☒ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☒ All b) ☐ Some** c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☒ Certified copies of the priority documents have been received in Application No. 16/498751.

3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

3) ☐ Interview Summary (PTO-413)

Paper No(s)/Mail Date ____.

2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)

4) ☐ Other: ____.

Paper No(s)/Mail Date 8/28/25.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

2. Acknowledgment is made of applicant's claim for foreign priority under 35 U.S.C. 119 (a)-(d). The certified copies have been filed in parent Application No. 16/498,751, filed on 9/27/2019.

Information Disclosure Statement

3. The information disclosure statement filed 8/28/25 has been considered. Copies of foreign and non-patent literature documents are provided in parent applications.

Claim Interpretation

4. Claim 1 includes the limitation "the notching tab part only extending from a first end region of the coating part in the longitudinal direction located at a radially outer portion of the jelly-roll remote from the winding axis of the jelly-roll in a radial direction." While the examiner recognizes that the claims appear to be attempting the notching part and notching tabs to be provided in a single layer only at the very outer radius, this limitation does not sufficiently limit the claim in that manner, since "a radially outer portion" does includes any portions that are not inner, and does not specifically define the point at which the structure is no longer "radially inner" and becomes "radially outer." This interpretation is further supported by the fourth indented limitation to the plurality of notching tabs "are spaced apart from one another in the longitudinal direction of the electrode collector by the cut

portions but *overlap one another in a radial direction of the jelly-roll from*” (emphasis added by the examiner).

Additionally, claim 1 includes the limitation to “wherein the notching tab part has a plurality of notching tabs.” The term “has” is interpreted by the examiner to be inclusive or open-ended and does not exclude additional notching tabs per MPEP 2111.03 I. As such, the limitation will be interpreted as being directed to some notching tabs but not limiting the assembly to have no other notching tabs in the notching tab part.

5. Claim 1 requires that the notching tabs are “defined by cut portions where the notching tab part is cut” in the fourth indented limitation. Since claim 1 is a product claim, the method by which the notching tabs are formed is not given patentable weight since the process does not appear to provide a structural limitation to the claims. MPEP 2113

6. Claim 1 requires that the notching tabs are bent radially inward ... “to provide welding portions” in the last limitation. Since claim 1 is a product claim, the method by which the notching tabs are connected, and the intended use of the portions, is not given patentable weight since the process does not appear to provide a structural limitation to the claims. MPEP 2113

7. Claim 1 requires “bent parts of the plurality of notching tabs” that “but overlap one another in a radial direction of the jelly-roll form” and “overlap one another in in an axial direction of the jelly-roll form”; however, claims 5 and 14 require “the bent parts of the notching tabs ... and do not overlap one another in the axial direction.” This limitation is directly contradictory to claim 1 and therefore is not given patentable weight.

8. Claims 1, 3-6, and 12-15 include limitations to a/the “winding axis.” The examiner notes that the instant specification does not specifically use the term “winding axis” but finds that the person having ordinary skill in the art would recognize that the winding axis is the same as the central axis of the jelly-roll in a wound state. The limitations will be interpreted as such.

Double Patenting

9. The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The filing of a terminal disclaimer by itself is not a complete reply to a nonstatutory double patenting (NSDP) rejection. A complete reply requires that the terminal disclaimer be accompanied by a reply requesting reconsideration of the prior Office action. Even where the NSDP rejection is provisional the reply must be complete. See MPEP § 804, subsection I.B.1. For a reply to a non-final Office action,

Art Unit: 1729

see 37 CFR 1.111(a). For a reply to final Office action, see 37 CFR 1.113(c). A request for reconsideration while not provided for in 37 CFR 1.113(c) may be filed after final for consideration. See MPEP §§ 706.07(e) and 714.13.

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The actual filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/apply/applying-online/eterminal-disclaimer.

10. Claims 1-18 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1, 3-5, and 8 of U.S. Patent No. 12,334,493. Although the claims at issue are not identical, they are not patentably distinct from each other for the reasons discussed below.

Regarding claims 1 and 10, the annotated claims provided below explain the obviousness of the claims over the claims of '493:

instant claims

12,334,493

CLAIMS

i. An electrode assembly in a jelly-roll form, the electrode assembly having a separator disposed between a first electrode and a second electrode, the electrode assembly being wound about a winding axis along a longitudinal direction, one of the first or second electrodes comprising:

- an electrode collector; and
- an active material coated on a coating portion of the electrode collector,

wherein the electrode collector comprises a notching tab part on which the active material is not coated, the notching tab part extending from one side of the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part only extending from a first end region of the coating portion in the longitudinal direction located at a radially outer portion of the jelly-roll form remote from the winding axis of the jelly-roll form in a radial direction,

wherein the notching tab part has a plurality of notching tabs defined by cut portions, where the notching tab part is cut, see claim interpretation section above

the plurality of notching tabs are arranged along the radially outer portion of the jelly-roll form such that they are spaced apart from one another in the longitudinal direction of the electrode collector by the cut portions but overlap one another in a radial direction of the jelly-roll form, and

each of the plurality of notching tabs is bent radially inward to define bent parts that overlap one another in an axial direction of the jelly-roll form perpendicular to the radial direction and the circumferential direction to provide welding portions.

ii. A secondary battery comprising:

- a can member; and
- the electrode assembly of claim 1 accommodated in the can member,

wherein the bent parts are welded to the can member.

The invention claimed is:

1. A secondary battery comprising:
 - a can member; and
 - an electrode assembly having a positive electrode, a negative electrode and a separator disposed therebetween, which are wound together in a wound state in a jelly-roll shape,wherein the positive electrode or the negative electrode comprises:
 - an electrode collector;
 - a coating portion coated with an active material on the electrode collector; and
 - a non-coating portion on which the active material is not applied to the electrode collector, the non-coating portion being adjacent to the coating portion in a width direction perpendicular to a longitudinal direction of the electrode collector, the non-coating portion having a plurality of notching tabs formed therein such that each of the plurality of notching tabs extends away from the coating portion in the width direction,wherein, when the electrode assembly is in the wound state, the plurality of notching tabs are circumferentially distributed about a central axis of the electrode assembly,

wherein the plurality of notching tabs are bent in a radial direction towards the central axis so as to form an overlapping region in which the plurality of notching tabs overlap one another in the width direction of the electrode collector,

wherein adjacent ones of the plurality of notching tabs are circumferentially spaced apart from one another in a peripheral region, the peripheral region being located further from the central axis in the radial direction than the overlapping region, and

wherein at least some of the plurality of notching tabs are directly bonded to others of the plurality of notching tabs of the overlapping region so as to define an electrical connection between the plurality of notching tabs and the can member.

2. The secondary battery of claim 1, wherein the overlapping region is positioned along the central axis of the electrode assembly.
3. The secondary battery of claim 1, wherein the plurality of notching tabs is formed on a radially outer portion of an end of the electrode assembly, the radially outer portion being remote from the central axis.
4. The secondary battery of claim 1, wherein the plurality of notching tabs are bonded to the can member by welding.
5. The secondary battery of claim 1, further comprising an insulation member disposed between bent parts of the plurality of notching tabs and an end of the electrode assembly.

Claims 2 and 11 are obvious over claim 5 of '493.

Claims 3-6 and 11-15 are obvious over claim 1 of '493 in light of the claim interpretations discussed above.

Claims 7-8 and 16-17 are obvious over claim 8 of '493.

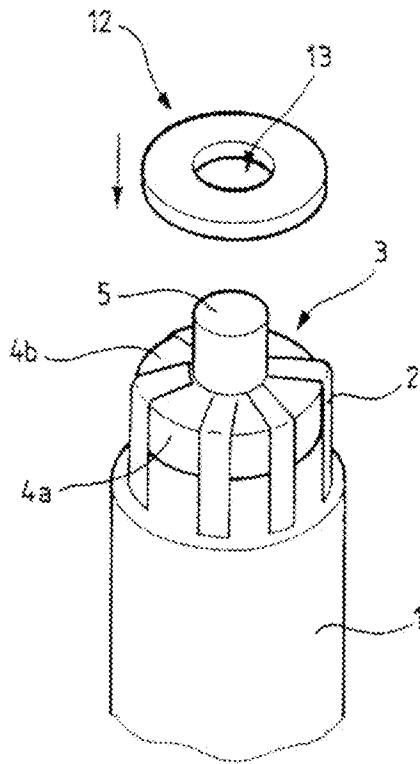
Claims 9 and 18 are rejected over the 8th indented limitation of claim 1 of '493.

Conclusion

11. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. Ligeouis et al. (US 2001/0023038) teaches a plurality of bent notches (2), wherein all of the

notches are spaced apart from one another at a radially outer portion of the jelly-roll form (1), or in other words do not overlap each other in the radial direction, as seen in Figure 7:

FIG. 7



12. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ALIX ECHELMEYER EGGERDING whose telephone number is (571)272-1101. The examiner can normally be reached 8:30am - 4:30pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

Art Unit: 1729

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Ruddock can be reached at 571-272-1481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/ALIX E EGGERDING/

Primary Examiner, Art Unit 1729

Notice of References Cited	Application/Control No. 19/313,233		Applicant(s)/Patent Under Reexamination Kim et al.	
	Examiner ALIX E EGGERDING		Art Unit 1729	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Name	CPC Classification	US Classification
*	A	US-12334493-B2	2025-06-17	Kim; Jae Woong	H01M10/0422	1/1
*	B	US-20010023038-A1	2001-09-20	Ligeois, Dominique	H01M50/528	429/178
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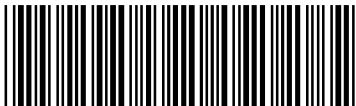
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<i>Search Notes</i> 	Application/Control No. 19/313,233	Applicant(s)/Patent Under Reexamination Kim et al.
	Examiner ALIX E EGGERDING	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M50/531,538; H01M10/0422	11/24/2025	AE

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Class	Subclass	Date	Examiner

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Inventor, assignee, similarity search; global dossier	11/24/2025	AE
Reviewed and updated search in parent cases	11/24/2025	AE
Searched PE2E SEARCH, subclasses searched with text keywords	11/24/2025	AE

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729	
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		
	Filing Date		
	First Named Inventor	Jae Woong Kim	
	Art Unit		N/A
	Examiner Name	Not Yet Assigned	
	Attorney Docket Number		LGES 3.3F-1654 CCCCIIICII

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	2	4822377		1989-04-18	Wolff		
	3	6569565		2003-05-27	Ligeois et al.		
	4	11652232		2023-05-16	Kim et al.		
	5	11973177		2024-04-30	Kim et al.		
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Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit		N/A
Examiner Name	Not Yet Assigned	
Attorney Docket Number		LGES 3.3F-1654 CCCCIIICII

	2	20050003264	A1	2005-01-06	Oh et al.	
	3	20050142436	A1	2005-06-30	Arai et al.	
	4	20050277022	A1	2005-12-15	Kozuki, Kiyomi	
	5	20050287428	A1	2005-12-29	Cheon et al.	
	6	20050287432	A1	2005-12-29	Cheon et al.	
	7	20060024572	A1	2006-02-02	Lee, Sang-Won	
	8	20060127774	A1	2006-06-15	Kim et al.	
	9	20070196730	A1	2007-08-23	Kozuki, Kiyomi	
	10	20100081052	A1	2010-04-01	MORISHIMA et al.	
	11	20100190056	A	2010-07-29	Turner	
	12	20100310926	A1	2010-12-09	WANG et al.	

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

Application Number		
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First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII	

	13	20110067227	A1	2011-03-24	Sohn	
	14	20110293977	A1	2011-12-01	Kim et al.	
	15	20120009450	A1	2012-01-12	Chun	
	16	20120208055	A1	2012-08-16	Ahn	
	17	20130108903	A1	2013-05-02	Martin et al.	
	18	20130252054	A1	2013-09-26	BARONE et al.	
	19	20130344364	A1	2013-12-26	Nanaumi et al.	
	20	20140050956	A1	2014-02-20	HUANG	
	21	20150295270	A1	2015-10-15	Chun, Kwan-Sic	
	22	20160351966	A1	2016-12-01	Batson et al.	
	23	20180145304	A1	2018-05-24	KOGETSU et al.	

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number			
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	First Named Inventor	Jae Woong Kim		
	Art Unit	N/A		
	Examiner Name	Not Yet Assigned		
	Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII		

	24	20190379028	A1	2019-12-12	Lim et al.	
	25	20240243335	A1	2024-07-18	Kim et al.	

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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1	101714624	CN	A	2010-05-26	Guangzhou Fullriver Battery New Technology Co., Lt	Machine generated translation attached.	☒
	2	102024936	CN	A	2011-04-20	Samsung Sdi Co., Ltd	U.S. Publication No. 20110067227 is the English language counterpart which was previously cited.	☐
	3	201450072	CN	U	2010-05-05	Dongguan Amperex Electronics Technology Co., Ltd	U.S. Publication No. 20100310926 is the English language counterpart which was previously cited.	☐
	4	203250828	CN	U	2013-10-23	A123 Systems, Inc	U.S. Publication No. 20130108903 is the English lanuage counterpart which was previously cited.	☐
	5	204577523	CN	U	2015-08-19	Contemporary Amperex Technology Ltd	Machine generated translation attached.	☒
	6	1134820	EP	A1	2001-09-19	Cit Alcatel	U.S. Publication No. 20010023038 A1 and U. S. Patent No. 6,569,565 is an English language counterpart.	☒
	7	2000-077078	JP	A	2000-03-14	Matsushita Electric Ind Co Ltd	Machine generated translation attached.	☒

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII	

8	2000-228182	JP	A	2000-08-15	Toyota Central Res & Dev	Machine generated translation attached.	☒
9	2003338276	JP	A	2003-11-28	Matsushita Electric Ind Co Ltd	Machine Generated Translation attached.	☒
10	2005276459	JP	A	2005-10-06	Sanyo Electric Co	English translation of Abstract only.	☒
11	2006128106	JP	A	2006-05-18	Samsung Sdi Co Ltd	U.S. Publication No. 20060127774 is an English language counterpart	☒
12	2006-324049	JP	A	2006-11-30	Sony Corp	Machine generated translation attached.	☒
13	2007227137	JP	A	2007-09-06	Matsushita Electric Ind Co Ltd	US Publication 2007196730 is an English Language counterpart.	☒
14	2008066040	JP	A	2008-03-21	Sony Corp	Machine Generated Translation attached.	☒
15	2011065981	JP	A	2011-03-31	Samsung Sdi Co Ltd	US Publication 2011067227 is an English Language counterpart.	☒
16	2016001575	JP	A	2016-01-07	Denso Corp	Machine generated translation is attached.	☒
17	2020509534	JP	A	2020-03-26	LG CHEM Ltd.	US Publication 2019379028 is an English Language counterpart.	☒
18	4401634	JP	B2	2010-01-20	Matsushita Electric Ind Co Ltd	Machine generated translation attached.	☒

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

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Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII	

19	5117133	JP	B2	2013-01-09	Hitachi Vehicle Energy Ltd	Machine generated translation attached.	☒
20	5790576	JP	B2	2015-10-07	Toyota Industries Corp	Machine generated translation is attached.	☒
21	6045286	JP	B2	2016-12-14	Hitachi Vehicle Energy Ltd	Machine generated translation attached.	☒
22	H0935701	JP	A	1997-02-07	Denso Corp	Machine generated translation attached.	☒
23	H11339756	JP	A	1999-12-10	Toyota Central Res & Dev	Machine generated translation attached.	☒
24	100515833	KR	B1	2005-09-21	Samsung Sdi Co., Ltd	U.S. Publication No. 20050003264 A1 is an English language counterpart.	☒
25	100599710	KR	B1	2006-07-12	Samsung Sdi Co., Ltd	U.S. Publication No. 20060024572 A1 is an English translation attached.	☒
26	100599749	KR	B1	2006-07-12	Samsung Sdi Co., Ltd	U.S. Publication No. 20050287428 A1 is an English translation attached.	☒
27	20000051740	KR	A	2000-08-16	Samsung Sdi Co., Ltd	Machine generated translation is attached.	☒
28	20050121904	KR	A	2005-12-28	Samsung Sdi Co., Ltd	U.S. Publication No. 20050287428 A1 is an English language counterpart.	☒
29	20050121914	KR	A	2005-12-28	Samsung Sdi Co., Ltd	U.S. Publication No. 20050287432 A1 is an English translation attached.	☒

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

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First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII	

30	20060022358	KR	A	2006-03-10	Samsung Sdi Co., Ltd	Machine generated translation is attached.	☒
31	20100071941	KR	A	2010-06-29	Lg Chemical Ltd	U.S. Publication No. 20110293977 A1 is an English language counterpart.	☒
32	20120006389	KR	A	2012-01-18	Samsung Sdi Co., Ltd	U.S. Publication No. 20120009450 A1 is an English language counterpart.	☒
33	20120092367	KR	A	2012-08-21	Samsung Sdi Co., Ltd	U.S. Publication No. 20120208055 A1 is an English language counterpart.	☒
34	20130073329	KR	A	2013-07-03	Datechnology Co., Ltd	Machine generated translation is attached.	☒
35	20150117135	KR	A	2015-10-19	Samsung Sdi Co Ltd	U.S. Publication No. 20150295270 A1 is an English language counterpart.	☒
36	2016-0009406	KR	A	2016-01-26	Lg Chemical Ltd	Machine generated translation attached.	☒
37	20160113612	KR	A	2016-09-30	A123 Systems LLC	U.S. Publication No. 20160351966 A1 is an English language counterpart.	☒
38	20170006437	KR	A	2017-01-18	Lg Chemical Ltd	Machine generated translation is attached.	☒
39	2012086690	WO	A1	2012-06-28	Hitachi Vehicle Energy, Ltd	US Publication 2013344364 is an English Language counterpart.	☒
40	2014188501	WO	A1	2014-11-27	Hitachi Automotive Systems, Ltd	Machine generated translation attached.	☒

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Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
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	1	Chiinese Search Report for Application No. 201880025247, dated August 24, 2021, 2 pages.	☐
	2	Extended European Seach Report including Written Opinion for Application No. 24187569.9 dated 10/11/24. 9 pages.	☐
	3	Extended European Seach Report including Written Opinion for Application No. 24187669.7 dated 10/18/24. 9 pages.	☐
	4	Extended European Seach Report including Written Opinion for Application No. EP18784869.2 dated January 21, 2020, 5 pages.	☐
	5	International Search Report from Application No. PCT/KR2018/004366 mailed July 20, 2018, pages 1 - 2	☒

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Application Number			
Filing Date			
First Named Inventor	Jae Woong Kim		
Art Unit	N/A		
Examiner Name	Not Yet Assigned		
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII		

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☐ The fee set forth in 37 CFR 1.17(p) has been submitted herewith.

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Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2025-08-28
Name/Print	Joseph H. Karlin	Registration Number	62,471

**INFORMATION DISCLOSURE
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Application Number	
Filing Date	
First Named Inventor	Jae Woong Kim
Art Unit	N/A
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGES 3.3F-1654 CCCCIIICII

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Bibliographic Data

Application No: 19/313,233

Foreign Priority claimed: ☒ Yes ☐ No

35 USC 119 (a-d) conditions met: ☒ Yes ☐ No ☐ Met After Allowance

Verified and Acknowledged: /ALIX E EGGERDING/

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Initials

Title: Secondary Battery and Method for Manufacturing the Same

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
08/28/2025	429	1729	LGES 3.12F-1654 CCCCIICII
RULE			

APPLICANTS

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Hyoung Kwon Kim, Daejeon, KOREA, REPUBLIC OF

Sang Uck Kim, Daejeon, KOREA, REPUBLIC OF

CONTINUING DATA

This application is a CON of 19207758 05/14/2025

19207758 is a CON of 18620269 03/28/2024 PAT 12334493

18620269 is a CON of 18618609 03/27/2024 PAT 12341147

18618609 is a CON of 18132719 04/10/2023 PAT 11973177

18132719 is a CON of 16498751 09/27/2019 PAT 11652232

16498751 is a 371 of PCT/KR2018/004366 04/13/2018

FOREIGN APPLICATIONS

KOREA, REPUBLIC OF KR10-2017-0048645 04/14/2017

KOREA, REPUBLIC OF KR10-2018-0043191 04/13/2018

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Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L2	14	("20010023038" "20050003264" "20150295270" "20110067227" "20120009450" "20160351966" "20110293977" "20100310926" "6569565" "20070196730" "20050287428" "20130344364" "20190379028" "20120208055").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/17 02:34 PM
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L23	66	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) with (plural\$4 several) near4 (tab notch lead terminal)	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/18 01:43 PM
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		"20060124973" "20060172190" "20060210857" "20070117011" "20090109099" "3823040" "4049888" "4383013" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237" "6071638" "6080506" "6083639" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "6780303" "6875540" "6884541" "7033697" "7927732" "8084158" "RE38518").PN. OR ("8236441").URPN.					
L26	1	"20090029240".pn.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 02:29 PM
L27	9	("2009/0029240").URP N.	(USPAT)	OR	ON	ON	2021/05/18 02:30 PM
L28	1	"20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/18 02:31 PM
L29	100	("4053687" "4237199" "4283843" "4539273" "5631102").PN. OR ("5849431").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 02:37 PM
L30	1	"20200295341".pn.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 03:13 PM
L32	38	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead terminal)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/27 11:35 AM
L36	3	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead terminal)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/27 03:37 PM
L37	66	(h01m50/538.cpc. h01m10/0422.cpc.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 03:40 PM

L38	0	h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:00 PM
L39	26	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not L37	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:00 PM
L40	27	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal not L37	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:04 PM
L41	4	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L37 not L40	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:15 PM
L42	8	"20050287428".pn. "20090148766".pn. "20090297941".pn. "20110076547".pn. "20110076547".pn. "20160197379".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:18 PM

L43	26	"7923139".pn. "7955736".pn. "8450006".pn. (h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L37 not L40	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:20 PM
L44	1	(15/573571).APP.	(US-PGPUB; USOCR)	OR	ON	ON	2021/05/27 04:35 PM
L45	2	("20130316209" "20150050535").PN. OR ("10566598").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/27 04:35 PM
L46	1	(13/984779).APP.	(USPAT; USOCR)	OR	ON	ON	2021/05/27 04:38 PM
L47	15	("20060240319" "20070269711" "20080241646" "20090136835" "20100124694" "20100129719" "20100310910" "6019802" "6465134" "6589694" "7261792" "7642015" "7935445" "9577231" "9655393").PN. OR ("10658633").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/27 04:38 PM
L48	66	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:49 AM
L49	27	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not L48	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:49 AM

L50	26	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L48 not L49	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:49 AM
L51	140	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L48 not L49 not L50	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:49 AM
L52	21	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L48 not L49 not L50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit)	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:59 AM
L53	274	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) not L48 not L49 not L50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not L52	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 10:19 AM
L54	74	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near4 (side end) not L48 not L49 not L50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not L52	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 10:22 AM
L55	2	(h01m50/\$.cpc.) and	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28

		(mea membrane near3 electrode near3 assembly) with (terminal lead tab connect\$4) near4 (side end) not L48 not L49 not L50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not L52 not L54					10:40 AM
L56	1	"20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 02:56 PM
L57	4	"20070196730".pn. "20100081052".pn. "20090029240".pn. "20110067227".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 03:04 PM
L58	0	"20100081052".pn. and second near3 embodiment with (roll\$4 coil\$4)	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 03:07 PM
L59	0	"20100081052".pn. and second near3 embodiment same2 (roll\$4 coil\$4)	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 03:08 PM
L60	1	"20180145304".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 05:13 PM
L61	2	"20100081052".pn. "20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:17 PM
L62	68	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:41 PM
L63	28	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not L62	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:41 PM
L64	26	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) near4 (edge top length) same (terminal lead tab connect\$4) near3	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:41 PM

L65	22	(attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L62 not L63 (h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L62 not L63 not L64 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit)	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:41 PM
L66	77	(h01m50/\$.cpc. h01m50/\$.ipc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near4 (side end) not L62 not L63 not L64 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not L65	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/09/17 03:41 PM
L67	1	"20110067227".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:50 PM
L68	68	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:58 PM
L69	40	(h01m50/\$.cpc. h01m\$.ipc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal not L68	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/09/17 03:58 PM
L70	3	"20100081052".pn. "20130108903".pn. "20180145304".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/01/18 12:49 PM
L71	1	"20200295341".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/08 09:51 AM

L72	59	("20010038945" "20030035993" "20030091893" "20030175587" "20030194601" "20030211388" "20040023108" "20040121230" "20040157115" "20040157120" "20040214076" "20040237290" "20040265683" "20050255378" "20060124973" "20060172190" "20060210857" "20070117011" "20090109099" "3823040" "4049888" "4383013" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237" "6071638" "6080506" "6083639" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "6780303" "6875540" "6884541" "7033697" "7927732" "8084158" "RE38518").PN. OR ("8236441").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/02/08 09:56 AM
L73	1	"20130252054".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 10:11 AM
L74	28	(h01m50/\$.cpc. h01m50/\$.ipc.) and insulat\$4 near4 (member plate collar) near4 (end top bottom) with (electrode negative positive cathode anode) near4 (tab lead) and cell with (wind\$4 wound\$4 roll\$4 jelly near4 roll\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/03/03 11:30 AM
L75	96	(h01m50/\$.cpc. h01m50/\$.ipc.) and insulat\$4 near4 (member plate collar) with (electrode negative positive cathode anode) near4 (tab lead) with (cell unit mea membrane near3 assembly) and cell with	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/03/03 11:33 AM

L76	19	(wind\$4 wound\$4 roll\$4 jelly near4 roll\$4) not L74	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 11:40 AM
L77	42	("4271241" "4822377").PN. OR ("5171647").URPN. (h01m50/\$.cpc. h01m50/\$.ipc.) and insulat\$4 near4 (member plate collar pad slab disc layer sheet panel) with (electrode negative positive cathode anode) near4 (tab lead) with (cell unit mea membrane near3 assembly) and cell with (wind\$4 wound\$4 roll\$4 jelly near4 roll\$4) not L74 not L75	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/03/03 11:43 AM
L78	32	("3262819" "3415690" "3861960" "3986083" "3994749" "4271241" "4523376").PN. OR ("4822377").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 11:48 AM
L79	3	"4822377".pn. "20110067227".pn. "20130252054".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 11:55 AM
L80	1	(13/861455).APP.	(USPAT; USOCR)	OR	ON	ON	2022/03/03 01:21 PM
L81	11	("20020192548" "20030235736" "20050277019" "20060110658" "20060147793" "20070154797" "20090169990" "20090317707" "20100124694" "5580679" "5662566").PN. OR ("9300004").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 01:21 PM
L82	11	("20020192548" "20030235736" "20050277019" "20060110658" "20060147793" "20070154797" "20090169990" "20090317707" "20100124694" "5580679" "5662566").PN. OR ("9300004").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 01:22 PM

L83	3	("2013/0252054").URP N.	(USPAT)	OR	ON	ON	2022/03/03 01:22 PM
L84	17	("20040029008" "20060123609" "20060194108" "20090239133" "20100124694" "20100281685" "20130252054" "5250784" "6337758" "6534952" "6746796" "6967828" "6995971" "7875087" "8974955").PN. OR ("9373865").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 01:23 PM
L85	3	"4822377".pn. "20130108903".pn. "20130252054".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:25 PM
L86	3	"4822377".pn. "20110067227".pn. "20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:25 PM
L87	1	"20110067227".pn. and tab near3 group with (plural\$4 number additional)	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:33 PM
L88	1	"20130108903".pn. and electrode near3 tab with (across plural\$4)	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:37 PM
L89	1	"20200295341".pn. and outer near4 portion with end with electrode	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:41 PM
L90	1	"20130252054".pn. and number near4 (tab winding "602" "604")	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:47 PM
L91	3	"20110067227".pn. "20130252054".pn. "4822377".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 04:44 PM
L92	48	((US-20130108903-A1 OR US-20150072204- A1 OR US- 20180026295-A1 OR US-20100081052-A1 OR US-20070117011- A1 OR US- 20030211388-A1 OR US-20010038945-A1 OR US-20090029240- A1 OR US- 20200343520-A1 OR US-20110151295-A1 OR US-20210005936- A1 OR US- 20180145304-A1 OR US-20080026293-A1 OR US-20100124694-	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2022/08/25 04:45 PM

		A1 OR US- 20160351966-A1 OR US-20110067227-A1 OR US-20070196730- A1 OR US- 20010023038-A1 OR US-20160351940-A1 OR US-20130252054- A1 OR US- 20100310926-A1 OR US-20190027789-A1 OR US-20170125764- A1 OR US- 20110008670-A1 OR US-20190379028-A1 OR US-20050225926- A1 OR US- 20060024572-A1 OR US-20200295341- A1).did. AND PGPB.dbnm.) OR ((US- 8236441-B2 OR US- 6521374-B1 OR US- 6432574-B1 OR US- 8871373-B2 OR US- 5849431-A OR US- 9640833-B2 OR US- 6071638-A OR US- 7935445-B2 OR US- 5171647-A OR US- 4822377-A OR US- 8310809-B2 OR US- 8628877-B2 OR US- 6456484-B1).did. AND USPT.dbnm.) OR ((WO-2017047278-A1 OR WO-2012051402- A1 OR EP-2325927-A1 OR JP-3754220- B2).did. AND FPRS.dbnm.) OR ((JP- 2009289714-A OR JP- 2013187077-A).did. AND JPAB.dbnm.) OR ((US-3761314-A).did. AND USOC.dbnm.)					
L93	12	("20150037638" "20110 067227" "20140147714 " "20070269685" "2012 0009450" "2005022117 8" "20110081574" "201 10262779" "201603519 40" "20100190056" "20 160218343" "20130252 054").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 04:49 PM
L94	15	("2007/0269685").urpn. AND (PGPB USPT	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 04:52 PM

L95	36	USOC).dbnm. (Artificial Intelligence) More like doc: US-20070269685-A1 Text: (US-8084158-B2 OR EP-2181482-A1 OR WO-2009015282-A1 OR US-20090029240- A1 OR US-8236441-B2 OR EP-2181482-B1 OR KR-20100058501-A OR JP-2010534916-A OR JP-5378377-B2 OR KR- 101566561-B1 OR EP- 2429010-B1 OR EP- 2429010-A1 OR US- 7927732-B2 OR JP- 2009507345-A OR KR- 101304125-B1 OR US- 4322484-A OR WO- 2007028152-A2 OR EP-1946393-A2 OR US-20070117011-A1 OR KR-20080050452- A).did.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2022/08/25 04:54 PM
L96	53	("3823040" "4383013" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237" "6071638" "6080506" "6083639" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "RE38518" "6780303" "6875540" "6884541" "7033697" "7927732" "20010038945" "20030035993" "20030091893" "20030175587" "20030194601" "20030211388" "20040023108" "20040121230" "20040157115" "20040157120" "20040214076" "20040237290" "20040265683" "20050255378" "20060124973" "20060172190" "20070117011" "20090109099").pn. OR	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 04:54 PM

L97	33	("8084158").urpn. AND (PGPB USPT USOC).dbnm. (h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 04:59 PM
L98	1	"20190348703".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 05:01 PM
L99	1	"11152638".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 05:02 PM
L100	11	("9299967" "20020068217" "20090208825" "20110262779" "20120009450" "20140186670" "20140199571" "20170084901" "20170092926" "20170250437" "20170346064").pn. OR ("11152638").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 05:02 PM
L101	1	"9012053".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 08:16 PM
L102	19	("2291110" "3507703" "3607417" "4816352" "7060387" "7276313" "20020061435" "20040161662" "20040202924" "20050142436" "20060240319" "20100178559").pn. OR ("9012053").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 08:16 PM
L103	13	("2012/0009450").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 08:23 PM
L104	87	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:31 PM

L105	668	notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4 plate cover) (h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4 plate cover) NOT L97	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:32 PM
L106	0	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) near4 (plate cover layer) NOT L97	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:32 PM
L107	131	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) near4 (plate cover layer)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:32 PM
L108	0	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L97	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 08:32 PM
L109	231	(h01m50/\$.cpc. h01m10/\$.cpc.	(US-PGPUB; USPAT; FPRS; EPO; JPO;	OR	ON	ON	2022/08/25 08:37 PM

		h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107	DERWENT)				
L110	174	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:37 PM
L111	1	("2014/0050956").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 08:42 PM
L112	15	("20020061435" "20060127751" "20090239133" "20100190056" "20110067227" "20110104541" "20110151295" "20120279053" "20140050956" "20160164133" "20170250437" "20170365839" "20180026295" "20180040918" "20190379028").pn. OR ("10700384").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 08:43 PM
L113	43	(Artificial Intelligence) More like doc: US-20140050956-A1 Text: (KR-20150030537-A OR TW-201407859-A OR JP-3187576-U OR CN-114050324-A OR KR-20180048419-A OR KR-102103900-B1 OR US-10700384-B2 OR EP-3514876-A1 OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK,	OR	ON	ON	2022/08/25 08:46 PM

		US-20190027789-A1 OR CN-103594736-A OR CN-108496269-B OR TW-201244225-A OR US-20040191625-A1 OR CN-102760905-A OR EP-3136494-A1 OR KR-20150122340-A OR CN-101714651-B OR CN-108496269-A OR CN-101714651-A OR CN-202308131-U).did.	TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
L114	0	20150037638.pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 08:48 PM
L115	1	"20150037638".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 08:48 PM
L116	53	((US-20130108903-A1 OR US-20150072204-A1 OR US-20180026295-A1 OR US-20100081052-A1 OR US-20070117011-A1 OR US-20030211388-A1 OR US-20010038945-A1 OR US-20090029240-A1 OR US-20200343520-A1 OR US-20110151295-A1 OR US-20210005936-A1 OR US-20180145304-A1 OR US-20080026293-A1 OR US-20100124694-A1 OR US-20160351966-A1 OR US-20110067227-A1 OR US-20070196730-A1 OR US-20010023038-A1 OR US-20160351940-A1 OR US-20130252054-A1 OR US-20100310926-A1 OR US-20190027789-A1 OR US-20170125764-A1 OR US-20110008670-A1 OR US-20190379028-A1 OR US-20050225926-A1 OR US-20060024572-A1 OR US-20150037638-A1 OR US-20190348703-A1 OR US-20120009450-A1 OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2022/08/25 08:50 PM

		US-20140050956-A1 OR US-20200295341-A1).did. AND PGPB.dbnm.) OR ((US- 8236441-B2 OR US- 6521374-B1 OR US- 6432574-B1 OR US- 8871373-B2 OR US- 5849431-A OR US- 9640833-B2 OR US- 6071638-A OR US- 7935445-B2 OR US- 5171647-A OR US- 4822377-A OR US- 8310809-B2 OR US- 8628877-B2 OR US- 6456484-B1).did. AND USPT.dbnm.) OR ((WO-2017047278-A1 OR WO-2012051402-A1 OR EP-2325927-A1 OR JP-3754220-B2 OR CN-114128033-A).did. AND FPRS.dbnm.) OR ((JP-2009289714-A OR JP-2013187077-A).did. AND JPAB.dbnm.) OR ((US-3761314-A).did. AND USOC.dbnm.)					
L117	2	"20120009450".pn. "20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 09:02 PM
L119	1	"3234421".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/30 02:52 PM
L120	6	waymouth-j\$.inv. AND koury-f\$.inv.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/30 02:53 PM
L121	2	"20140050956".pn. "20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/09/02 03:30 PM
L122	12	("20150037638" "20110 067227" "20140147714 "20070269685" "2012 0009450" "2005022117 8" "20110081574" "201 10262779" "201603519 40" "20100190056" "20 160218343" "20130252 054").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2022/09/06 01:20 PM
L123	6	"20140050956".pn. "20130108903".pn. "20120009450".pn. "20110067227".pn. "20130252054".pn. "4822377".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/09/07 02:36 PM
L124	2	("20060127774" "20100 190056").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 12:17 PM
L125	183	(h01m50/\$.cpc.	(US-PGPUB; USPAT;	OR	ON	ON	2022/12/16

		h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4)	FPRS; EPO; JPO; DERWENT)				12:40 PM
L126	197	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl\$5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:01 PM
L127	38	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl\$5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4) AND (tan lead terminal) WITH (overlap\$4 overl\$4 shingl\$4 superimpos\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:03 PM
L128	57	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl\$5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4) AND (tab lead terminal) WITH (overlap\$4 overl\$4 shingl\$4 superimpos\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:03 PM

L129	1	"20160351966".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 01:09 PM
L130	100	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4 plate cover)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:42 PM
L131	105	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl\$5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4 plate cover)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:43 PM
L132	0	(h01m50/538,536.cpc. h01m10/1422,0431.cpc.) AND (wound roll\$4 cylind\$5) SAME tab WITH ben\$4 same2 tab WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) near4 (equal\$4 axis around about)	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:07 PM
L133	0	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) SAME tab WITH ben\$4 same2 tab WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) near4 (equal\$4 axis around about)	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:07 PM
L134	0	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) SAME tab WITH ben\$4 same2 tab WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) .clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:08 PM

L135	86	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) SAME tab WITH ben\$4 .clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:08 PM
L136	73423	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) tab WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) .clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:08 PM
L137	168	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (tab electrode) WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) .clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:08 PM
L138	27	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (tab electrode) WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) same3 (tab electrode) WITH (overlap\$4 laminat\$4).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:09 PM
L139	1101	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (tab electrode) WITH (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:11 PM
L140	104	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) WITH (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:11 PM
L141	56	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) near3 (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:11 PM

L142	1	"11652232".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/03 01:28 PM
L143	1	"20230246222".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/03 01:33 PM
L144	4	"20110067227".pn. "20120009450".pn. "20130108903".pn. "20130252054".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/03 02:46 PM
L145	63	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) near3 (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/03 02:54 PM
L146	2	L142 L143	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/03 03:34 PM
L147	22	("4822377" "6569565" "20010023038" "20050003264" "20050287428" "20060127774" "20070196730" "20100081052" "20100190056" "20100310926" "20110067227" "20110293977" "20120009450" "20120208055" "20130108903" "20130252054" "20130344364" "20140050956" "20150295270" "20160351966" "20180145304" "20190379028").pn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2023/11/03 04:02 PM
L148	32	(Artificial Intelligence) More like doc: <u>US-11652232-B2</u> Text: (US-2020295341-A1 OR US-2023246222-A1 OR EP-3611781-A1 OR WO-2018190691-A1 OR KR-20180116156-A OR KR-102126970-B1 OR JP-7341585-B2 OR JP-2020505717-A OR JP-2021177490-A OR US-11349182-B2 OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO;	OR	ON	ON	2023/11/03 04:02 PM

		CN-110476273-B OR US-2019379028-A1 OR EP-3567666-B1 OR CN-110476273-A OR KR-101137372-B1 OR KR-20140140405-A OR KR-101584298-B1 OR JP-5194072-B2 OR KR- 102226916-B1 OR CN- 116544526-A).did.	JPO; DERWENT; IBM_TDB)				
L149	2	(h01m50/538,536.cpc. h01m10/1422,0431.cpc.) AND (wound roll\$4 cylind\$5) SAME tab near3 notch\$4 WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) near4 (equal\$4 axis around about)	(US-PGPUB; USPAT)	OR	ON	ON	2024/02/16 04:14 PM
L150	2	(h01m50/538,536.cpc. h01m10/1422,0431.cpc.) AND (wound roll\$4 cylind\$5) SAME tab near3 notch\$4 WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) near4 (equal\$4 axis around about).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2024/02/16 04:14 PM
L151	2	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) SAME tab near3 notch\$4 WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) near4 (equal\$4 axis around about).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2024/02/16 04:20 PM
L152	67	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) near3 (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2024/02/16 04:20 PM
L153	4	"3761314".pn. "20050287428".pn. "20060024572".pn. "20050287432".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/03/06 08:44 AM
L154	54	((US-20130108903-A1 OR US-20150072204- A1 OR US-	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY,	OR	ON	ON	2024/04/17 10:11 AM

		20180026295-A1 OR US-20100081052-A1 OR US-20070117011- A1 OR US- 20030211388-A1 OR US-20010038945-A1 OR US-20090029240- A1 OR US- 20160351966-A1 OR US-20110151295-A1 OR US-20210005936- A1 OR US- 20180145304-A1 OR US-20080026293-A1 OR US-20100124694- A1 OR US- 20190379028-A1 OR US-20110067227-A1 OR US-20070196730- A1 OR US- 20010023038-A1 OR US-20160351940-A1 OR US-20130252054- A1 OR US- 20100310926-A1 OR US-20190027789-A1 OR US-20170125764- A1 OR US- 20110008670-A1 OR US-20050225926-A1 OR US-20060024572- A1 OR US- 20200343520-A1 OR US-20150037638-A1 OR US-20190348703- A1 OR US- 20120009450-A1 OR US-20140050956-A1 OR US-20200295341- A1 OR US- 20230246222-A1).did. AND PGPB.dbnm.) OR ((US-8236441-B2 OR US-6521374-B1 OR US-6432574-B1 OR US-8871373-B2 OR US-5849431-A OR US- 9640833-B2 OR US- 6071638-A OR US- 7935445-B2 OR US- 5171647-A OR US- 4822377-A OR US- 8310809-B2 OR US- 8628877-B2 OR US- 6456484-B1 OR US- 11652232-B2).did. AND USPT.dbnm.) OR ((US- 3761314-A).did. AND	CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
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		USOC.dbnm.) OR ((WO-2012051402-A1 OR WO-2017047278- A1 OR EP-2325927-A1 OR JP-3754220- B2).did. AND FPRS.dbnm.) OR ((JP- 2009289714-A OR JP- 2013187077-A).did. AND JPAB.dbnm.)					
L155	2	"11652232".pn. "11973177".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:49 PM
L156	1	"20240243335".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:51 PM
L157	1	"20240243336".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:52 PM
L158	4	L155 L156 L157	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:53 PM
L159	0	L158 AND second near3 non\$2coating	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:53 PM
L160	0	L158 AND second WITH non\$2coating	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:54 PM
L161	4	L158 AND non\$2coating	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:54 PM
L162	4	L158 AND bent near4 electrical\$4 near4 connect\$4	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:56 PM
L163	1	L158 AND spac\$4 near2 apart	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:58 PM
L164	4	L158 AND (spac\$4 apart)	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:58 PM
L165	57	((US-20130108903-A1 OR US-20150072204- A1 OR US- 20180026295-A1 OR US-20100081052-A1 OR US-20070117011- A1 OR US- 20030211388-A1 OR US-20010038945-A1 OR US-20090029240- A1 OR US- 20160351966-A1 OR US-20110151295-A1 OR US-20210005936- A1 OR US- 20180145304-A1 OR US-20080026293-A1 OR US-20100124694- A1 OR US- 20190379028-A1 OR US-20110067227-A1 OR US-20070196730- A1 OR US-	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/22 09:59 PM

		20010023038-A1 OR US-20160351940-A1 OR US-20130252054- A1 OR US- 20100310926-A1 OR US-20190027789-A1 OR US-20170125764- A1 OR US- 20110008670-A1 OR US-20050225926-A1 OR US-20060024572- A1 OR US- 20200343520-A1 OR US-20150037638-A1 OR US-20190348703- A1 OR US- 20120009450-A1 OR US-20140050956-A1 OR US-20200295341- A1 OR US- 20230246222-A1 OR US-20240243335-A1 OR US-20240243336- A1).did. AND PGPB.dbnm.) OR ((US- 8236441-B2 OR US- 6521374-B1 OR US- 6432574-B1 OR US- 8871373-B2 OR US- 5849431-A OR US- 9640833-B2 OR US- 6071638-A OR US- 7935445-B2 OR US- 5171647-A OR US- 4822377-A OR US- 8310809-B2 OR US- 8628877-B2 OR US- 6456484-B1 OR US- 11652232-B2 OR US- 11973177-B2).did. AND USPT.dbnm.) OR ((US- 3761314-A).did. AND USOC.dbnm.) OR ((WO-2012051402-A1 OR WO-2017047278- A1 OR EP-2325927-A1 OR JP-3754220- B2).did. AND FPRS.dbnm.) OR ((JP- 2009289714-A OR JP- 2013187077-A).did. AND JPAB.dbnm.)					
L166	4	"3761314".pn. "20050287428".pn. "20060024572".pn. "20050287432".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 10:08 PM
L167	75	(h01m50/\$.cpc.	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22

		h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) near3 (overlap\$4 laminat\$4).clm. NOT L138					10:08 PM
L168	1	"11973177".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/10/08 09:39 AM
L169	2	"20240243335".pn. "20240243336".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/10/08 09:41 AM
L170	2	"11973177".pn. "11652232".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/10/08 09:58 AM
L171	24	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) AND (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv.) AND (electrode cathode anode) WITH (notch\$4) near4 (overlap\$4 plural\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2024/10/08 12:28 PM
L172	2	L170 AND overlap\$4.clm.	(US-PGPUB; USPAT)	OR	ON	ON	2024/10/08 02:44 PM
L173	1	"20210391579".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/10/09 02:50 PM
L174	3	"20240243335".pn. "20050277022".pn. "20050142436".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/12/06 10:43 AM
L175	27	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) AND (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv.) AND (electrode cathode anode) WITH (notch\$4) near4 (overlap\$4 plural\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2024/12/06 11:08 AM
L176	75	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) near3 (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2024/12/06 11:08 AM
L177	2	"20050142436".pn. "20050277022".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/02/06 09:36 AM
L178	50	(Artificial Intelligence) Similar to:	(US-PGPUB; USPAT; USOCR; FIT (AU, AP,	OR	ON	ON	2025/11/24 07:53 PM

		19/207,758 with 0 CPC Selections and 0 Text Selections Text: (EP-3567666-A1 OR US-20190379028-A1 OR EP-3567666-B1 OR CN-117457847-A OR WO-2023022560-A1 OR CN-101442139-B OR EP-4579867-A1 OR CN-101442139-A OR KR-20120002861-A OR US-20240136492-A1 OR KR-20230027600-A OR EP-4191737-B1 OR EP-2378595-A2 OR KR-20240010355-A OR KR-20130052410-A OR US-20110293977-A1 OR JP-2023541349-A OR JP-2025528822-A OR WO-2018080080-A1 OR WO-2019245144-A1 OR EP-3761419-A1 OR CN-119744469-A OR KR-20250091928-A OR KR-20190143304-A OR CN-111406338-A OR CN-105826514-A OR US-7666545-B2 OR EP-2136429-A1 OR EP-3761419-B1 OR EP-4044355-A1 OR JP-2024537650-A OR US-20200044225-A1 OR KR-20250106227-A OR US-20070231701-A1 OR CN-116072957-A OR JP-2011113970-A OR JP-7746530-B2 OR KR-102734115-B1 OR KR-20040059303-A OR KR-20150117135-A OR CN-104170126-A OR US-20060003221-A1 OR WO-2010071387-A2 OR KR-20140051885-A OR US-20210028494-A1 OR US-7700221-B2 OR KR-20160106526-A OR KR-20140141742-A OR KR-20240028850-A OR US-20140356674-A1).did.	AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN))				
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L179	50	(Artificial Intelligence) Similar to: 19/207,758 with 2 CPC Selections and 2 Text Selections Text: (US-20190379028-A1 OR US-20210028494- A1 OR EP-3567666-A1 OR CN-101442139-A OR US-7666545-B2 OR CN-101442139-B OR EP-3567666-B1 OR CN-1172400-C OR KR- 102785731-B1 OR EP- 4576294-A1 OR US- 20250210828-A1 OR EP-2378595-A2 OR US-20110293977-A1 OR KR-20250101825-A OR CN-117457847-A OR US-20020164524- A1 OR CN-104285319- B OR CN-116072957-A OR EP-1255310-A2 OR CN-1713418-A OR JP- 2011113970-A OR EP- 4262008-A1 OR US- 7700221-B2 OR US- 20060003221-A1 OR CN-101411003-B OR US-7214250-B2 OR US-20050246887-A1 OR EP-1255310-B1 OR KR-20250066216-A OR JP-7005642-B2 OR KR- 20130052410-A OR JP- 7604066-B2 OR CN- 111406338-A OR CN- 217239536-U OR JP- 2024167710-A OR JP- 2005302719-A OR DE- 69937078-T2 OR CN- 107112494-B OR US- 6884270-B2 OR US- 20250300309-A1 OR WO-2018080080-A1 OR CN-116365179-A OR US-20240421451- A1 OR CN-119009387- A OR JP-2025026968- A OR KR- 20250042130-A OR KR-20240168235-A OR KR-20150021030-A OR CN-116636058-A OR US-9350042-B2).did.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN))	OR	ON	ON	2025/11/24 07:53 PM
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L180	22	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) AND (uncoated (free remov\$4 without) near3 active near3 material) SAME (plural\$4 several) near4 (tab notch lead) SAME electrode WITH terminal	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2025/11/24 07:54 PM
L181	1071	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) AND (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv. lg.as.) AND (electrode cathode anode) WITH (tab notch) near4 (overlap\$4 plural\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2025/11/24 08:09 PM
L182	45	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) AND (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv. lg.as.) AND (electrode cathode anode) WITH (notching) near4 (overlap\$4 plural\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2025/11/24 08:09 PM
L183	30	("11652232" 11973177 " 20010023038" 2005 0003264" 2005014243 6" 20050277022" 200 50287428" 200502874 32" 20060024572" 20 060127774" 20070196 730" 20100081052" 2 0100190056" 2010031 0926" 20110067227" 20110293977" 201200 09450" 20120208055" "20130108903" 20130 252054" 20130344364 " 20140050956" 2015 0295270" 2016035196 6" 20180145304" 201 90379028" 202402433 35" 3761314" 482237 7" 6569565").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:10 PM
L184	5	"12334493".pn. "12341147".pn. "11973177".pn. "11652232".pn. "20250273726".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:13 PM

L185	2	L184 AND radial\$4 near3 ben\$4	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:13 PM
L186	5	L184 AND can	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:15 PM
L187	1	"20250273726".pn. AND longitudinal near3 direction	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:36 PM
L188	1	"20250273726".pn. AND longitudinal near3 direction WITH spac\$\$	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:37 PM
L189	1	"20250273726".pn. AND longitudinal near3 direction WITH spac\$3	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:37 PM
L190	4	L184 AND distributed	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:53 PM
L191	2	L184 AND radial\$4 near3 outer	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:06 PM
L192	22	("20110067227" "20110 067227" "20130252054 "4822377" "20100081 052" "20130108903" "2 0180145304" "2012000 9450" "20130108903" " 20140050956" "200701 96730" "20110067227" " 20130344364" "20190 379028" "20010023038 "20100310926" "6569 565" "20050003264" "2 0050287428" "2011029 3977" "20120009450" " 20120208055" "201502 95270" "20160351966" " 20160351966" "20060 127774" "20100190056 ").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:26 PM
L193	10	("2001/0023038").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/11/24 09:31 PM
L194	25	("4322484" "5227267" "5707758" "5849431" "6245457").pn. OR ("6569565").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/11/24 09:31 PM
L195	60	((WO-2012051402-A1 OR WO-2017047278- A1 OR EP-2325927-A1 OR JP-3754220- B2).did. AND FPRS.dbnm.) OR ((US- 3761314-A).did. AND USOC.dbnm.) OR ((US- 20190379028-A1 OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO,	OR	ON	ON	2025/11/24 09:34 PM

		US-20110067227-A1 OR US-20070196730-A1 OR US-20010023038-A1 OR US-20160351940-A1 OR US-20130252054-A1 OR US-20100310926-A1 OR US-20190027789-A1 OR US-20170125764-A1 OR US-20110008670-A1 OR US-20150072204-A1 OR US-20050225926-A1 OR US-20060024572-A1 OR US-20200343520-A1 OR US-20130108903-A1 OR US-20180026295-A1 OR US-20100081052-A1 OR US-20070117011-A1 OR US-20030211388-A1 OR US-20010038945-A1 OR US-20090029240-A1 OR US-20160351966-A1 OR US-20110151295-A1 OR US-20210005936-A1 OR US-20180145304-A1 OR US-20080026293-A1 OR US-20100124694-A1 OR US-20150037638-A1 OR US-20190348703-A1 OR US-20120009450-A1 OR US-20140050956-A1 OR US-20200295341-A1 OR US-20230246222-A1 OR US-20240243335-A1 OR US-20240243336-A1 OR US-20250273726-A1).did. AND PGPB.dbnm.) OR ((JP-2009289714-A OR JP-2013187077-A).did. AND JPAB.dbnm.) OR ((US-8310809-B2 OR US-8628877-B2 OR US-6456484-B1 OR US-5171647-A OR US-8236441-B2 OR US-6521374-B1 OR US-6432574-B1 OR US-	MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
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L196	55	8871373-B2 OR US-5849431-A OR US-9640833-B2 OR US-6071638-A OR US-7935445-B2 OR US-4822377-A OR US-11652232-B2 OR US-11973177-B2 OR US-12334493-B2 OR US-12341147-B2).did. AND USPT.dbnm.) L195 NOT L194	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/11/24 09:34 PM
L197	21	("2011/0067227").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/11/24 09:36 PM
L198	47	L195 NOT L192	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/11/24 09:38 PM
L199	122	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) AND tab WITH (only) WITH (outer outside circumference)	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:39 PM
L200	44	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) AND tab near3 (outer	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:40 PM

L201	0	outside) near4 (perimeter periphery circumference) (h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) AND tab near3 (outer outside) near4 (perimeter periphery circumference) NOT L200	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:41 PM
L202	42	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) AND tab near3 (outer outside) near4 (perimeter periphery circumference) NOT L199	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:41 PM
L203	96	(h01m50/\$.cpc. h01m10/\$.cpc.) AND tab near3 (outer outside) near4 (perimeter periphery circumference) NOT L199 NOT L202	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:45 PM
L204	1	"20010023038".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:52 PM
L205	5	L184 AND bent near2 part	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/25 10:30 AM
L206	1	"12334493".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/25 10:51 AM

PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N2	4	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) AND (uncoated (free remov\$4 without) near3 active near3 material) SAME (plural\$4 several) near4 (tab notch lead) SAME electrode WITH terminal.clm.	(US-PGPUB; USPAT)	OR	ON	ON	2024/12/06 11:10 AM
N3	5	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (uncoated (free remov\$4 without) near3 active near3 material) SAME (plural\$4 several) near4 (tab	(US-PGPUB; USPAT)	OR	ON	ON	2025/02/06 09:44 AM

N4	3	notch lead) SAME electrode WITH terminal.clm. (h01m50/\$.cpc. h01m10/\$.cpc.) AND (plural\$4 several) near4 (tab notch lead) WITH (fold\$4 bend\$4) WITH (bond\$4 weld\$4) SAME electrode WITH terminal.clm.	(US-PGPUB; USPAT)	OR	ON	ON	2025/02/06 09:44 AM
N5	5	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (uncoated (free remov\$4 without) near3 active near3 material) SAME (plural\$4 several) near4 (tab notch lead) SAME electrode WITH terminal.clm.	(US-PGPUB; USPAT)	OR	ON	ON	2025/02/06 09:50 AM

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Tue Dec 09 05:08:04 EST 2025

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Application	Document	Mailroom Date	Attorney Docket No.
19313233	CTNF	12/09/2025	LGES 3.12F-1654 CCCCIIICII
19313233	892	12/09/2025	LGES 3.12F-1654 CCCCIIICII
19313233	1449	12/09/2025	LGES 3.12F-1654 CCCCIIICII

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
19/313,233	08/28/2025	Jae Woong Kim	LGES 3.12F-1654 CCCCIIII CONFIRMATION NO. 7385

86765
LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

PUBLICATION NOTICE



OC00000099039603

Date Mailed: 12/18/2025

Title:Secondary Battery and Method for Manufacturing the Same

Publication No.US-2025-0385292-A1

Publication Date:12/18/2025

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be viewed using the USPTO's publicly available Searchable Databases via the Patent Public Search tool at www.uspto.gov. The direct link to access the Patent Public Search tool is currently <https://ppubs.uspto.gov/pubwebapp/static/pages/ppubsbasic.html>.

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FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
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Application	Document	Mailroom Date	Attorney Docket No.
19313233	NTC.PUB	12/18/2025	LGES 3.12F-1654 CCCCIIICII

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/313,233	08/28/2025	Jae Woong Kim	LGES 3.12F-1654	7385
86765	7590	02/20/2026	CCCCIICII	
LG			EXAMINER	
Lerner David LLP			EGGERDING, ALIX ECHELMAYER	
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Cranford, NJ 07016			ART UNIT	PAPER NUMBER
			1729	
			NOTIFICATION DATE	DELIVERY MODE
			02/20/2026	ELECTRONIC

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<i>Applicant-Initiated Interview Summary</i>	Application No. 19/313,233	Applicant(s) Kim et al.		
	Examiner ALIX E EGGERDING	Art Unit 1729	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
ALIX E EGGERDING	Primary Examiner	Telephonic
Joseph Karlin	Attorney of Record	

Date of Interview: 17 February 2026

Issues Discussed:

Proposed Amendment(s)

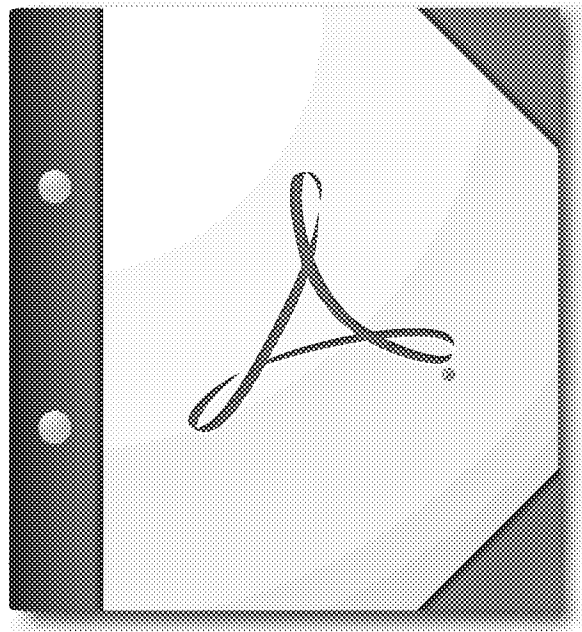
Discussed interpretations of claims 1, 5-6, and 14-15, specifically to the interpretations of the plurality of notching tabs and bent portions. No agreement was reached.

☒ Attachment

/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.



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TO: eOfficeAction@lernerdavid.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 86765

Fri Feb 20 05:05:29 EST 2026

Dear Patent Center Customer:

Correspondence Address:

LG

Lerner David LLP

20 Commerce Drive

Cranford, NEW JERSEY, 07016

UNITED STATES

This is a courtesy notification regarding the following USPTO patent application(s) associated with your Customer Number, 86765, that have new outgoing correspondence. This correspondence is now available for viewing in Patent Center.

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Application	Document	Mailroom Date	Attorney Docket No
19313233	EXIN	02/20/2026	LGES 3.12F-1654 CCCCIIICII
19313233	OA.APPENDIX	02/20/2026	LGES 3.12F-1654 CCCCIIICII

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Sincerely,

The Patent Center Team



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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/313,233	02/27/2026 02:51:37 PM Z ET	LGES 3.12F-1654 CCCCIIICII

Title of Invention

Secondary Battery and Method for Manufacturing the Same

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	7385	FILED BY	Kathleen Reynolds
PATENT CENTER #	74655586	FILING DATE	08/28/2025
CUSTOMER #	86765	FIRST NAMED INVENTOR	Jae Woong Kim
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	JAMES VEZERIS

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
Amendment Transmittal_1.pdf	1	Miscellaneous Incoming Letter	21 KB
Amendment_1.pdf	6	Amendment/Request for Reconsideration-After Non- Final Rejection	89 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
Amendment Transmittal_1.pdf	23B171B82E160DCDDBC761AE9A76FBCB84FAB36571C8C7E DD2EEEE734246146D97CC318A990078A89A1626CD338581A2 E33C9D7886A66E90834094D8ED2F9257

Amendment_1.pdf

C193D268F7E7A53FEA1FC89B7186470321982BB8CD0A76E8A
E383E1007E0A078D841F0636F5E861A4D99AC6C140517A3EC
C2EF4E34669F736915CFF19A5BD0F2

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

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If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

AMENDMENT TRANSMITTAL LETTER			Docket No. LGES 3.12F-1654 CCCCIIICII
Application No. 19/313,233-Conf. #7385	Filing Date August 28, 2025	Examiner A. E. Eggerding	Art Unit 1729

Applicant(s): LG Energy Solution, Ltd.

Invention: Secondary Battery and Method for Manufacturing the Same

TO THE COMMISSIONER FOR PATENTS

Transmitted herewith is an amendment in the above-identified application.
The fee has been calculated and is transmitted as shown below.

CLAIMS AS AMENDED					
	Claims Remaining After Amendment	Highest Number Previously Paid	Number Extra Claims Present	Rate	
Total Claims	18	- 20 =	0	x	
Independent Claims	1	- 3 =	0	x	
Multiple Dependent Claims (check if applicable) ☐					
Other fee (please specify):					
TOTAL ADDITIONAL FEE FOR THIS AMENDMENT:					0.00

☒ Large Entity
 ☐ Small Entity

☒ No additional fee is required for this amendment.

☐ Please charge Deposit Account No. _____ in the amount of \$ _____.

☐ A check in the amount of \$ _____ to cover the filing fee is enclosed.

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☒ The Director is hereby authorized to charge and credit Deposit Account No. 12-1095 as described below. A duplicate copy of this sheet is enclosed.

☒ Credit any overpayment.

☒ Charge any additional filing or application processing fees required under 37 C.F.R. § 1.16 and 1.17.

/Joseph H. Karlin/
 Joseph H. Karlin (Atty/Agent Reg. No.: 62,471)
 LERNER DAVID LLP
 20 Commerce Dr.
 Cranford, New Jersey 07016
 (908) 654-5000

Dated: February 27, 2026

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).	
Dated: <u>February 27, 2026</u>	Electronic Signature for Joseph H. Karlin: <u>/Joseph H. Karlin/</u>



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ELECTRONIC PAYMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/313,233	02/27/2026 03:13:01 PM Z ET	LGES 3.12F-1654 CCCCIIICII

Title of Invention

Secondary Battery and Method for Manufacturing the Same

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	7385	FILED BY	Joseph Karlin
PATENT CENTER #	74655657	AUTHORIZED BY	-
CUSTOMER #	86765	FILING DATE	08/28/2025
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Jae Woong Kim

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
CARD / 1060	E20262QF13236032	Joseph Karlin

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1814	STATUTORY DISCLAIMER, INCLUDING TERMINAL DISCLAIMER	183.00	1	183.00
TOTAL AMOUNT:				\$183.00

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371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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TERMINAL DISCLAIMER TO OBVIATE A DOUBLE PATENTING REJECTION OVER A PRIOR PATENT

APPLICATION #	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET #
19313233	08/28/2025	Jae Woong Kim	LGES 3.12F-1654 CCCCIIICII

Title of Invention

Secondary Battery and Method for Manufacturing the Same



Filing of terminal disclaimer does not obviate requirement for response under 37 CFR 1.111 to outstanding Office Action



This electronic Terminal Disclaimer is not being used for a Joint Research Agreement.

Owner	Percent interest
LG Energy Solution, Ltd.	100%

The owner(s) of percent interest listed above in the instant application hereby disclaims, except as provided below, the terminal part of the statutory term of any patent granted on the instant application which would extend beyond the expiration date of the full statutory term of any patent granted on pending reference Application Number(s)

Application #	Filing Date
---------------	-------------

as the term of any patent granted on said reference application may be shortened by any terminal disclaimer filed

prior to the grant of any patent on the pending reference application. The owner hereby agrees that any patent so granted on the instant application shall be enforceable only for and during such period that it and any patent granted on the reference application are commonly owned. This agreement runs with any patent granted on the instant application and is binding upon the grantee, its successors or assigns.

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The owner(s) of percent interest listed above in the instant application hereby disclaims, except as provided below, the terminal part of the statutory term of any patent granted on the instant application which would extend beyond the expiration date of the full statutory term of prior patent number(s)

Patent #

12334493

as the term of said prior patent is presently shortened by any terminal disclaimer. The owner hereby agrees that any patent so granted on the instant application shall be enforceable only for and during such period that it and the prior patent are commonly owned. This agreement runs with any patent granted on the instant application and is binding upon the grantee, its successors or assigns.

In making the above disclaimer, the owner does not disclaim the terminal part of the term of any patent granted on the instant application that would extend to the expiration date of the full statutory term of the prior patent, "as the term of said prior patent is presently shortened by any terminal disclaimer," in the event that said prior patent later:

- expires for failure to pay a maintenance fee;
- is held unenforceable;
- is found invalid by a court of competent jurisdiction;
- is statutorily disclaimed in whole or terminally disclaimed under 37 CFR 1.321;
- has all claims canceled by a reexamination certificate;
- is reissued; or
- is in any manner terminated prior to the expiration of its full statutory term as presently shortened by any terminal disclaimer.



Terminal disclaimer fee under 37 CFR 1.20(d) included with Electronic Terminal Disclaimer request.

Applicant claims the following entity status:

Regular Undiscounted

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

I certify, in accordance with 37 CFR 1.4(d)(4) that I am: An attorney or agent registered to practice before the Patent and Trademark Office who is of record in this application

Signature	Name	Registration #
/Joseph H. Karlin/	Joseph Karlin	62471



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APPROVAL LETTER

**APPLICATION #
19/313,233**

**FILING DATE
08/28/2025**

**APPLICANT/PATENT UNDER REEXAMINATION
Jae Woong Kim**

Title of Invention

Secondary Battery and Method for Manufacturing the Same

Electronic terminal disclaimer filed on 02/27/2026

☒ **Approved**

This patent is subject to a Terminal Disclaimer

Approved / Disapproved by: Electronic Terminal Disclaimer automatically approved



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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/313,233	02/27/2026 03:13:01 PM Z ET	LGES 3.12F-1654 CCCCIIICII

Title of Invention

Secondary Battery and Method for Manufacturing the Same

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	7385	FILED BY	Joseph Karlin
PATENT CENTER #	74655657	FILING DATE	08/28/2025
CUSTOMER #	86765	FIRST NAMED INVENTOR	Jae Woong Kim
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
petition-request.pdf	3	Terminal Disclaimer-Filed (Electronic)	42 KB
grantLetter.pdf	1	Terminal Disclaimer-Electronic- Approved	18 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
petition-request.pdf	1100E3249756F5C6DF924A4A9FDB9C7D2A2831D762999D444 ED7B0B0EBD0CBFDC4296B73E1C67454D49FD82D3464B32F9 4DB9FDCA02A965FD89F0223443B0F4A
grantLetter.pdf	4A71E568C12FFE6AC2FCEF209A6B7C00B3856B0F124DE758

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41D698736F827AF62573AF02047B0B0

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New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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Dated: __ February 27, 2026
Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGES 3.12F-1654 CCCCIIII
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: 19/313,233

Confirmation No.: 7385

Filed: August 28, 2025

Art Unit: 1729

For: Secondary Battery and Method for Manufacturing
the Same

Examiner: A. E. Eggerding

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT

Dear Commissioner:

In response to the official action mailed December 9, 2025, applicant submits the following amendments and remarks.

IN THE CLAIMS

1. (Original) An electrode assembly in a jelly-roll form, the electrode assembly having a separator disposed between a first electrode and a second electrode, the electrode assembly being wound about a winding axis along a longitudinal direction, one of the first or second electrodes comprising:

an electrode collector; and

an active material coated on a coating portion of the electrode collector,

wherein the electrode collector comprises a notching tab part on which the active material is not coated, the notching tab part extending from one side of the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part only extending from a first end region of the coating portion in the longitudinal direction located at a radially outer portion of the jelly-roll form remote from the winding axis of the jelly-roll form in a radial direction,

wherein the notching tab part has a plurality of notching tabs defined by cut portions where the notching tab part is cut,

the plurality of notching tabs are arranged along the radially outer portion of the jelly-roll form such that they are spaced apart from one another in the longitudinal direction of the electrode collector by the cut portions but overlap one another in a radial direction of the jelly-roll form, and

each of the plurality of notching tabs is bent radially inward to define bent parts that overlap one another in an axial direction of the jelly-roll form perpendicular to the radial direction and the circumferential direction to provide welding portions.

2. (Original) The electrode assembly of claim 1, further comprising an insulating member disposed radially inward of the plurality of notching tabs and axially inward of the bent parts of the plurality of notching tabs.

3. (Original) The electrode assembly of claim 1, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap one another in the radial direction are together distributed about the winding axis of the jelly-roll form.

4. (Original) The electrode assembly of claim 1, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap one another in the radial direction are spaced apart from the winding axis of the jelly-roll form.

5. (Currently Amended) The electrode assembly of claim 1, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction ~~and do not overlap one another in the radial direction~~ do not overlap one another in ~~the axial direction~~ a circumferential direction of the jelly-roll form.

6. (Currently Amended) The electrode assembly of claim 1, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction ~~and do not overlap one another in the radial direction~~ overlap one another in ~~the axial direction at the winding axis of~~ a circumferential direction of the jelly-roll form.

7. (Original) The electrode assembly of claim 1, wherein the electrode collector further comprises a non-coating portion disposed at a second end region of the coating portion opposite the first end region in the longitudinal direction, the non-coating portion being uncoated with the active material.

8. (Original) The electrode assembly of claim 7, wherein the second end region of the coating portion is located at an inner core of the jelly-roll form, and an electrode tab is attached to the non-coating portion, the electrode tab extending further in the width direction beyond the coating portion.

9. (Original) The electrode assembly of claim 8, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap one another in the radial direction overlap one another and the electrode tab in the axial direction at a central portion of the jelly-roll form.

10. (Original) A secondary battery comprising:
a can member; and
the electrode assembly of claim 1 accommodated in the can member,
wherein the bent parts are welded to the can member.

11. (Original) The secondary battery of claim 10, further comprising an insulating member disposed radially inward of the plurality of notching tabs and axially inward of the bent parts of the plurality of notching tabs.

12. (Currently Amended) The secondary battery of claim 10, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction ~~and do not overlap one another in the radial direction~~ are together distributed about the winding axis of the jelly-roll form.

13. (Currently Amended) The secondary battery of claim 10, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction ~~and do not overlap one another in the radial direction~~ are spaced apart from the winding axis of the jelly-roll form.

14. (Currently Amended) The secondary battery of claim 10, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction ~~and do not overlap one another in the radial direction~~ do not overlap one another in ~~the axial direction~~ a circumferential direction of the jell-roll form.

15. (Currently Amended) The secondary battery of claim 10, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction ~~and do not overlap one another in the radial direction~~ overlap one another in ~~the axial direction at the winding axis~~ a circumferential direction of the jelly-roll form.

16. (Original) The secondary battery of claim 10, wherein the electrode collector further comprises a non-coating portion disposed at a second end region of the coating portion opposite the first end region in the longitudinal direction, the non-coating portion being uncoated with the active material.

17. (Original) The secondary battery of claim 16, wherein the second end region of the coating portion is located at an inner core of the jelly-roll form, and an electrode tab is attached to the non-coating portion, the electrode tab extending further in the width direction beyond the coating portion.

18. (Original) The secondary battery of claim 17, wherein the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction and do not overlap one another in the radial direction overlap one another and the electrode tab in the axial direction at a central portion of the jelly-roll form.

REMARKS

The present amendments and remarks are being filed in response to the non-final office action dated December 9, 2025. Claims 1-18 are pending. Claims 5-6 and 12-15 are amended herein. No new matter is added by the present amendments. Applicant respectfully requests that the Examiner reconsider the claims in view of the amendments and remarks herein.

Interview Summary

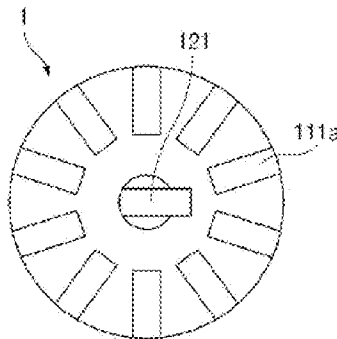
Applicants' representative thanks Examiner Eggerding for her time during the telephonic interview conducted on February 17, 2026, with the undersigned. Interpretations of claims 1, 5-6, and 14-15 and potential clarifying amendments to those claims were discussed. No agreement was reached regarding patentability.

Double Patenting Rejections

Claims 1-18 were rejected on the ground of nonstatutory double patenting as allegedly being unpatentable over claims 1, 3-5, and 8 of U.S. Patent No. 12,334,493 ("the '493 patent"). To address these rejections, a terminal disclaimer is being filed herewith with respect to the '493 patent. Accordingly, Applicant respectfully requests that these rejections should be withdrawn.

Claim Interpretation Amendments

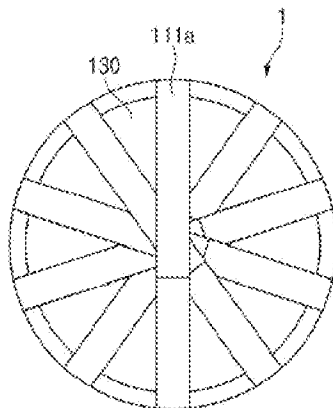
Claims 5-6 and 12-15 are amended herein to clarify the interpretation of those claims. Specifically, claim 5 is amended herein to recite that the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction do not overlap one other in *a circumferential direction, rather than an axial direction*, of the jelly-roll form as shown in Fig. 5, reproduced below.



Applicant respectfully submits that the overlapping of the plurality of notching tabs *in an axial direction* of the jelly-roll form does not contradict the overlapping of the bent parts of the

notching tabs in *a circumferential direction* of the jelly-roll form. That is, the axial overlapping and the circumferential overlapping are not mutually exclusive and may coexist.

Furthermore, claim 6 is amended herein to recite that the bent parts of the notching tabs that are spaced apart from one another in the longitudinal direction overlap one other in *a circumferential direction, rather than an axial direction*, of the jelly-roll form as shown in Fig. 8, reproduced below.



In addition, in claims 12-13, the redundant phrases of “and do not overlap one another in the radial direction” are deleted and claims 14-15 are similarly amended as in claims 5-6, respectively.

Conclusion

As it is believed that all of the rejections set forth in the Official Action have been fully met, favorable reconsideration and allowance are earnestly solicited. If, however, for any reason the Examiner does not believe that such action can be taken at this time, it is respectfully requested that the Examiner telephone applicant's attorney at (908) 654-5000 in order to overcome any additional objections which the Examiner might have.

Dated: February 27, 2026

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin
Registration No.: 62,471
LERNER DAVID LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 19/313,233		Filing Date 08/28/2025		☐ To be Mailed			
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO											
APPLICATION AS FILED - PART I											
		(Column 1)		(Column 2)							
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A					
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A					
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A					
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *				x \$200 =					
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *				x \$ 600 =					
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))											
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL					
APPLICATION AS AMENDED - PART II											
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT	02/27/2026	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 18	Minus	** 20	= 0			x \$ 200 =		0	
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0			x \$ 600 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=			x \$ 0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=			x \$ 0 =			
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.								SLIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".								/ALA H HUNTER/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".											
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.											

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

86765 7590 03/26/2026
LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

EXAMINER	
EGGERDING, ALIX ECHELMEYER	
ART UNIT	PAPER NUMBER
1729	

DATE MAILED: 03/26/2026

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/313,233	08/28/2025	Jae Woong Kim	LGES 3.12F-1654 CCCCIICH	7385

TITLE OF INVENTION: Secondary Battery and Method for Manufacturing the Same

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	06/26/2026

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 6 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

86765 7590 03/26/2026

LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/313,233	08/28/2025	Jae Woong Kim	LGES 3.12F-1654	7385
TITLE OF INVENTION: Secondary Battery and Method for Manufacturing the Same			CCCCIICII	

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	06/26/2026

EXAMINER	ART UNIT	CLASS-SUBCLASS
EGGERDING, ALIX ECHELMEYER	1729	429-094000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

6. Courtesy Ceremonial Copy

☐ Applicant requests a courtesy ceremonial copy of the electronic patent grant. Applicant understands that the electronic patent grant is the official statutory patent grant.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/313,233	08/28/2025	Jae Woong Kim	LGES 3.12F-1654	7385
86765	7590	03/26/2026	CCCCCH EXAMINER	
LG Lerner David LLP 20 Commerce Drive Cranford, NJ 07016			EGGERDING, ALIX ECHELMAYER	
			ART UNIT	PAPER NUMBER
			1729	
DATE MAILED: 03/26/2026				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).

<https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Notice of Allowability	Application No. 19/313,233	Applicant(s) Kim et al.	
	Examiner ALIX E EGGERDING	Art Unit 1729	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the amendment and terminal disclaimer filed 2/27/26.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-18. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov**.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some* c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☐ Examiner's Amendment/Comment |
| 2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date ____. | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material ____. | 7. ☐ Other ____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date ____. | |

/ALIX E EGGERDING/
Primary Examiner, Art Unit 1729

DETAILED ACTION

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendment

2. This Office Action is in response to the amendment filed 2/27/26. Claims 5-6 and 12-15 are amended. Claims 1-18 are allowed.

Allowable Subject Matter

3. The following is an examiner's statement of reasons for allowance: the prior art fails to teach or fairly suggest the claimed subject matter. Specifically, the prior art fails to teach or fairly suggest the claimed notching tab part configuration.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

4. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ALIX ECHELMEYER EGGERDING whose telephone number is (571)272-1101. The examiner can normally be reached 8:30am - 4:30pm.

Art Unit: 1729

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Ruddock can be reached at 571-272-1481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ALIX E EGGERDING/

Primary Examiner, Art Unit 1729

Issue Classification 	Application/Control No. 19/313,233	Applicant(s)/Patent Under Reexamination Kim et al.
	Examiner ALIX E EGGERDING	Art Unit 1729

CPC						
Symbol					Type	Version
H01M	/	10	/	0422	F	2013-01-01
H01M	/	10	/	0431	I	2013-01-01
H01M	/	50	/	538	I	2021-01-01
H01M	/	50	/	536	A	2021-01-01

CPC Combination Sets					
Symbol				Type	Set
	/		/		

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	18	
/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729	18 March 2026	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	1

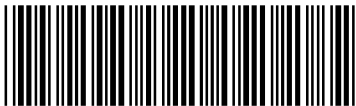
Issue Classification 	Application/Control No. 19/313,233	Applicant(s)/Patent Under Reexamination Kim et al.
	Examiner ALIX E EGGERDING	Art Unit 1729

INTERNATIONAL CLASSIFICATION			
CLAIMED			
H01M	/	10	/ 0422
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
429	94

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

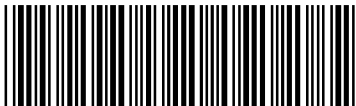
NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	18	
/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729	18 March 2026	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	1

Issue Classification 	Application/Control No. 19/313,233	Applicant(s)/Patent Under Reexamination Kim et al.
	Examiner ALIX E EGGERDING	Art Unit 1729

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☒ T.D. ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	10	10												
2	2	11	11												
3	3	12	12												
4	4	13	13												
5	5	14	14												
6	6	15	15												
7	7	16	16												
8	8	17	17												
9	9	18	18												

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 18	
/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729 (Primary Examiner) _____ (Date) 18 March 2026		O.G. Print Claim(s) 1	O.G. Print Figure 1

<i>Search Notes</i> 	Application/Control No. 19/313,233	Applicant(s)/Patent Under Reexamination Kim et al.
	Examiner ALIX E EGGERDING	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M50/531,538; H01M10/0422	11/24/2025	AE

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor, assignee, similarity search; global dossier	11/24/2025	AE
Reviewed and updated search in parent cases	11/24/2025	AE
Searched PE2E SEARCH, subclasses searched with text keywords	11/24/2025	AE
Updated above	03/18/2026	

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
H01M	50/531538; 10/0422	03/18/2026	AE

/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729	
---	--

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L2	14	("20010023038" "20050003264" "20150295270" "20110067227" "20120009450" "20160351966" "20110293977" "20100310926" "6569565" "20070196730" "20050287428" "20130344364" "20190379028" "20120208055").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/17 02:34 PM
L3	9	("20150037638" "20110067227" "20140147714" "20120009450" "20110081574" "20160351940" "20100190056" "20160218343" "20130252054").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/17 02:41 PM
L4	1	"20100310926".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/17 02:46 PM
L5	21124	(h01m10/\$.cpc. h01m50/\$.cpc.) and (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv.)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/17 02:56 PM
L6	260	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) and (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv.) and (electrode cathode anode) with (tab notch) near4 (overlap\$4 plural\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/17 02:57 PM
L7	144	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) and (electrode cathode anode) with (tab notch) near4 (overlap\$4 plural\$4) and (battery mea electrode near3 assembly) with (wound jelly near3 roll cylindrical)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/17 04:11 PM
L8	132	L7 not L6	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/17 04:12 PM
L9	8	("20050277019" "20090169990" "20090317707" "20030235736" "2006	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/17 04:15 PM

L10	0	0147793" 20020192548" 20060110658" 20070154797").PN. ("2016/0351966").URP N.	(USPAT)	OR	ON	ON	2021/05/17 04:31 PM
L11	1	(15/115220).APP.	(USPAT; USOCR)	OR	ON	ON	2021/05/17 04:31 PM
L12	10	("20050287428" "20060024572" "20070196730" "20120251854" "20120328924" "20130216879" "20140045041" "7364817" "7700222" "7955736").PN. OR ("10381688").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/17 04:31 PM
L13	6	("2007/0196730").URP N.	(USPAT)	OR	ON	ON	2021/05/17 04:33 PM
L14	10	("3490949" "4053687" "6376121" "6596434" "6818025" "7332246" "7364817").PN. OR ("7700222").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/17 04:33 PM
L15	14	("20020164524" "20030118912" "20050225926" "5736270" "6071638" "6222720" "6328769" "7364817" "7618742" "7670715" "7678501" "7700222").PN. OR ("8310809").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/17 04:34 PM
L16	11	("20020191370" "20030086239" "20060024572" "20060240322" "20060240323" "3237274" "5849431" "6456484" "6896993" "8310809").PN. OR ("8628877").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/17 04:35 PM
L17	13	("3761314" "5525441" "5635312" "5736270").PN. OR ("6328769").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 10:22 AM
L18	44	("3761314").URPN.	(USPAT)	OR	ON	ON	2021/05/18 10:23 AM
L19	5	("20050142436" "3761314").PN. OR ("7955736").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 10:24 AM
L20	3	("2005/0142436").URP N.	(USPAT)	OR	ON	ON	2021/05/18 10:34 AM
L21	1863	h01m50/538.cpc.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/18

L22	4637	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.)	(US-PGPUB; USPAT)	OR	ON	ON	11:26 AM 2021/05/18 01:20 PM
L23	66	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) with (plural\$4 several) near4 (tab notch lead terminal)	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/18 01:43 PM
L24	40	("20010038945" "20030091893" "20030175587" "20030194601" "20030211388" "20040023108" "20040121230" "20040157115" "20040157120" "20040214076" "20040237290" "20040265683" "20050255378" "20060124973" "20060172190" "20070117011" "20090109099" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237" "6080506" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "6780303" "6875540" "6884541" "7033697" "RE38518").PN. OR ("7927732").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 02:24 PM
L25	49	("20010038945" "20030035993" "20030091893" "20030175587" "20030194601" "20030211388" "20040023108" "20040121230" "20040157115" "20040157120" "20040214076" "20040237290" "20040265683" "20050255378"	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 02:29 PM

		"20060124973" "20060172190" "20060210857" "20070117011" "20090109099" "3823040" "4049888" "4383013" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237" "6071638" "6080506" "6083639" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "6780303" "6875540" "6884541" "7033697" "7927732" "8084158" "RE38518").PN. OR ("8236441").URPN.					
L26	1	"20090029240".pn.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 02:29 PM
L27	9	("2009/0029240").URP N.	(USPAT)	OR	ON	ON	2021/05/18 02:30 PM
L28	1	"20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/18 02:31 PM
L29	100	("4053687" "4237199" "4283843" "4539273" "5631102").PN. OR ("5849431").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 02:37 PM
L30	1	"20200295341".pn.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 03:13 PM
L32	38	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead terminal)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/27 11:35 AM
L36	3	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead terminal)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/27 03:37 PM
L37	66	(h01m50/538.cpc. h01m10/0422.cpc.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 03:40 PM

L38	0	h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:00 PM
L39	26	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not L37	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:00 PM
L40	27	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal not L37	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:04 PM
L41	4	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L37 not L40	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:15 PM
L42	8	"20050287428".pn. "20090148766".pn. "20090297941".pn. "20110076547".pn. "20110076547".pn. "20160197379".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:18 PM

L43	26	"7923139".pn. "7955736".pn. "8450006".pn. (h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L37 not L40	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:20 PM
L44	1	(15/573571).APP.	(US-PGPUB; USOCR)	OR	ON	ON	2021/05/27 04:35 PM
L45	2	("20130316209" "20150050535").PN. OR ("10566598").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/27 04:35 PM
L46	1	(13/984779).APP.	(USPAT; USOCR)	OR	ON	ON	2021/05/27 04:38 PM
L47	15	("20060240319" "20070269711" "20080241646" "20090136835" "20100124694" "20100129719" "20100310910" "6019802" "6465134" "6589694" "7261792" "7642015" "7935445" "9577231" "9655393").PN. OR ("10658633").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/27 04:38 PM
L48	66	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:49 AM
L49	27	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not L48	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:49 AM

L50	26	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L48 not L49	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:49 AM
L51	140	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L48 not L49 not L50	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:49 AM
L52	21	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L48 not L49 not L50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit)	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:59 AM
L53	274	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) not L48 not L49 not L50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not L52	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 10:19 AM
L54	74	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near4 (side end) not L48 not L49 not L50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not L52	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 10:22 AM
L55	2	(h01m50/\$.cpc.) and	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28

		(mea membrane near3 electrode near3 assembly) with (terminal lead tab connect\$4) near4 (side end) not L48 not L49 not L50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not L52 not L54					10:40 AM
L56	1	"20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 02:56 PM
L57	4	"20070196730".pn. "20100081052".pn. "20090029240".pn. "20110067227".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 03:04 PM
L58	0	"20100081052".pn. and second near3 embodiment with (roll\$4 coil\$4)	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 03:07 PM
L59	0	"20100081052".pn. and second near3 embodiment same2 (roll\$4 coil\$4)	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 03:08 PM
L60	1	"20180145304".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 05:13 PM
L61	2	"20100081052".pn. "20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:17 PM
L62	68	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:41 PM
L63	28	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not L62	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:41 PM
L64	26	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) near4 (edge top length) same (terminal lead tab connect\$4) near3	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:41 PM

L65	22	(attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L62 not L63 (h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L62 not L63 not L64 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit)	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:41 PM
L66	77	(h01m50/\$.cpc. h01m50/\$.ipc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near4 (side end) not L62 not L63 not L64 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not L65	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/09/17 03:41 PM
L67	1	"20110067227".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:50 PM
L68	68	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:58 PM
L69	40	(h01m50/\$.cpc. h01m\$.ipc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal not L68	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/09/17 03:58 PM
L70	3	"20100081052".pn. "20130108903".pn. "20180145304".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/01/18 12:49 PM
L71	1	"20200295341".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/08 09:51 AM

L72	59	("20010038945" "20030035993" "20030091893" "20030175587" "20030194601" "20030211388" "20040023108" "20040121230" "20040157115" "20040157120" "20040214076" "20040237290" "20040265683" "20050255378" "20060124973" "20060172190" "20060210857" "20070117011" "20090109099" "3823040" "4049888" "4383013" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237" "6071638" "6080506" "6083639" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "6780303" "6875540" "6884541" "7033697" "7927732" "8084158" "RE38518").PN. OR ("8236441").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/02/08 09:56 AM
L73	1	"20130252054".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 10:11 AM
L74	28	(h01m50/\$.cpc. h01m50/\$.ipc.) and insulat\$4 near4 (member plate collar) near4 (end top bottom) with (electrode negative positive cathode anode) near4 (tab lead) and cell with (wind\$4 wound\$4 roll\$4 jelly near4 roll\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/03/03 11:30 AM
L75	96	(h01m50/\$.cpc. h01m50/\$.ipc.) and insulat\$4 near4 (member plate collar) with (electrode negative positive cathode anode) near4 (tab lead) with (cell unit mea membrane near3 assembly) and cell with	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/03/03 11:33 AM

L76	19	(wind\$4 wound\$4 roll\$4 jelly near4 roll\$4) not L74	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 11:40 AM
L77	42	("4271241" "4822377").PN. OR ("5171647").URPN. (h01m50/\$.cpc. h01m50/\$.ipc.) and insulat\$4 near4 (member plate collar pad slab disc layer sheet panel) with (electrode negative positive cathode anode) near4 (tab lead) with (cell unit mea membrane near3 assembly) and cell with (wind\$4 wound\$4 roll\$4 jelly near4 roll\$4) not L74 not L75	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/03/03 11:43 AM
L78	32	("3262819" "3415690" "3861960" "3986083" "3994749" "4271241" "4523376").PN. OR ("4822377").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 11:48 AM
L79	3	"4822377".pn. "20110067227".pn. "20130252054".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 11:55 AM
L80	1	(13/861455).APP.	(USPAT; USOCR)	OR	ON	ON	2022/03/03 01:21 PM
L81	11	("20020192548" "20030235736" "20050277019" "20060110658" "20060147793" "20070154797" "20090169990" "20090317707" "20100124694" "5580679" "5662566").PN. OR ("9300004").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 01:21 PM
L82	11	("20020192548" "20030235736" "20050277019" "20060110658" "20060147793" "20070154797" "20090169990" "20090317707" "20100124694" "5580679" "5662566").PN. OR ("9300004").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 01:22 PM

L83	3	("2013/0252054").URP N.	(USPAT)	OR	ON	ON	2022/03/03 01:22 PM
L84	17	("20040029008" "20060123609" "20060194108" "20090239133" "20100124694" "20100281685" "20130252054" "5250784" "6337758" "6534952" "6746796" "6967828" "6995971" "7875087" "8974955").PN. OR ("9373865").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 01:23 PM
L85	3	"4822377".pn. "20130108903".pn. "20130252054".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:25 PM
L86	3	"4822377".pn. "20110067227".pn. "20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:25 PM
L87	1	"20110067227".pn. and tab near3 group with (plural\$4 number additional)	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:33 PM
L88	1	"20130108903".pn. and electrode near3 tab with (across plural\$4)	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:37 PM
L89	1	"20200295341".pn. and outer near4 portion with end with electrode	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:41 PM
L90	1	"20130252054".pn. and number near4 (tab winding "602" "604")	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:47 PM
L91	3	"20110067227".pn. "20130252054".pn. "4822377".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 04:44 PM
L92	48	((US-20130108903-A1 OR US-20150072204- A1 OR US- 20180026295-A1 OR US-20100081052-A1 OR US-20070117011- A1 OR US- 20030211388-A1 OR US-20010038945-A1 OR US-20090029240- A1 OR US- 20200343520-A1 OR US-20110151295-A1 OR US-20210005936- A1 OR US- 20180145304-A1 OR US-20080026293-A1 OR US-20100124694-	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2022/08/25 04:45 PM

		A1 OR US- 20160351966-A1 OR US-20110067227-A1 OR US-20070196730- A1 OR US- 20010023038-A1 OR US-20160351940-A1 OR US-20130252054- A1 OR US- 20100310926-A1 OR US-20190027789-A1 OR US-20170125764- A1 OR US- 20110008670-A1 OR US-20190379028-A1 OR US-20050225926- A1 OR US- 20060024572-A1 OR US-20200295341- A1).did. AND PGPB.dbnm.) OR ((US- 8236441-B2 OR US- 6521374-B1 OR US- 6432574-B1 OR US- 8871373-B2 OR US- 5849431-A OR US- 9640833-B2 OR US- 6071638-A OR US- 7935445-B2 OR US- 5171647-A OR US- 4822377-A OR US- 8310809-B2 OR US- 8628877-B2 OR US- 6456484-B1).did. AND USPT.dbnm.) OR ((WO-2017047278-A1 OR WO-2012051402- A1 OR EP-2325927-A1 OR JP-3754220- B2).did. AND FPRS.dbnm.) OR ((JP- 2009289714-A OR JP- 2013187077-A).did. AND JPAB.dbnm.) OR ((US-3761314-A).did. AND USOC.dbnm.)					
L93	12	("20150037638" "20110 067227" "20140147714 "20070269685" "2012 0009450" "2005022117 8" "20110081574" "201 10262779" "201603519 40" "20100190056" "20 160218343" "20130252 054").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 04:49 PM
L94	15	("2007/0269685").urpn. AND (PGPB USPT	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 04:52 PM

L95	36	USOC).dbnm. (Artificial Intelligence) More like doc: US-20070269685-A1 Text: (US-8084158-B2 OR EP-2181482-A1 OR WO-2009015282-A1 OR US-20090029240- A1 OR US-8236441-B2 OR EP-2181482-B1 OR KR-20100058501-A OR JP-2010534916-A OR JP-5378377-B2 OR KR- 101566561-B1 OR EP- 2429010-B1 OR EP- 2429010-A1 OR US- 7927732-B2 OR JP- 2009507345-A OR KR- 101304125-B1 OR US- 4322484-A OR WO- 2007028152-A2 OR EP-1946393-A2 OR US-20070117011-A1 OR KR-20080050452- A).did.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2022/08/25 04:54 PM
L96	53	("3823040" "4383013" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237" "6071638" "6080506" "6083639" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "RE38518" "6780303" "6875540" "6884541" "7033697" "7927732" "20010038945" "20030035993" "20030091893" "20030175587" "20030194601" "20030211388" "20040023108" "20040121230" "20040157115" "20040157120" "20040214076" "20040237290" "20040265683" "20050255378" "20060124973" "20060172190" "20070117011" "20090109099").pn. OR	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 04:54 PM

L97	33	("8084158").urpn. AND (PGPB USPT USOC).dbnm. (h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 04:59 PM
L98	1	"20190348703".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 05:01 PM
L99	1	"11152638".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 05:02 PM
L100	11	("9299967" "20020068217" "20090208825" "20110262779" "20120009450" "20140186670" "20140199571" "20170084901" "20170092926" "20170250437" "20170346064").pn. OR ("11152638").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 05:02 PM
L101	1	"9012053".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 08:16 PM
L102	19	("2291110" "3507703" "3607417" "4816352" "7060387" "7276313" "20020061435" "20040161662" "20040202924" "20050142436" "20060240319" "20100178559").pn. OR ("9012053").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 08:16 PM
L103	13	("2012/0009450").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 08:23 PM
L104	87	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:31 PM

L105	668	notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4 plate cover) (h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4 plate cover) NOT L97	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:32 PM
L106	0	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) near4 (plate cover layer) NOT L97	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:32 PM
L107	131	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) near4 (plate cover layer)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:32 PM
L108	0	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L97	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 08:32 PM
L109	231	(h01m50/\$.cpc. h01m10/\$.cpc.	(US-PGPUB; USPAT; FPRS; EPO; JPO;	OR	ON	ON	2022/08/25 08:37 PM

		h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107	DERWENT)				
L110	174	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:37 PM
L111	1	("2014/0050956").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 08:42 PM
L112	15	("20020061435" "20060127751" "20090239133" "20100190056" "20110067227" "20110104541" "20110151295" "20120279053" "20140050956" "20160164133" "20170250437" "20170365839" "20180026295" "20180040918" "20190379028").pn. OR ("10700384").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 08:43 PM
L113	43	(Artificial Intelligence) More like doc: US-20140050956-A1 Text: (KR-20150030537-A OR TW-201407859-A OR JP-3187576-U OR CN-114050324-A OR KR-20180048419-A OR KR-102103900-B1 OR US-10700384-B2 OR EP-3514876-A1 OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK,	OR	ON	ON	2022/08/25 08:46 PM

		US-20190027789-A1 OR CN-103594736-A OR CN-108496269-B OR TW-201244225-A OR US-20040191625-A1 OR CN-102760905-A OR EP-3136494-A1 OR KR-20150122340-A OR CN-101714651-B OR CN-108496269-A OR CN-101714651-A OR CN-202308131-U).did.	TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
L114	0	20150037638.pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 08:48 PM
L115	1	"20150037638".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 08:48 PM
L116	53	((US-20130108903-A1 OR US-20150072204-A1 OR US-20180026295-A1 OR US-20100081052-A1 OR US-20070117011-A1 OR US-20030211388-A1 OR US-20010038945-A1 OR US-20090029240-A1 OR US-20200343520-A1 OR US-20110151295-A1 OR US-20210005936-A1 OR US-20180145304-A1 OR US-20080026293-A1 OR US-20100124694-A1 OR US-20160351966-A1 OR US-20110067227-A1 OR US-20070196730-A1 OR US-20010023038-A1 OR US-20160351940-A1 OR US-20130252054-A1 OR US-20100310926-A1 OR US-20190027789-A1 OR US-20170125764-A1 OR US-20110008670-A1 OR US-20190379028-A1 OR US-20050225926-A1 OR US-20060024572-A1 OR US-20150037638-A1 OR US-20190348703-A1 OR US-20120009450-A1 OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2022/08/25 08:50 PM

		US-20140050956-A1 OR US-20200295341-A1).did. AND PGPB.dbnm.) OR ((US- 8236441-B2 OR US- 6521374-B1 OR US- 6432574-B1 OR US- 8871373-B2 OR US- 5849431-A OR US- 9640833-B2 OR US- 6071638-A OR US- 7935445-B2 OR US- 5171647-A OR US- 4822377-A OR US- 8310809-B2 OR US- 8628877-B2 OR US- 6456484-B1).did. AND USPT.dbnm.) OR ((WO-2017047278-A1 OR WO-2012051402-A1 OR EP-2325927-A1 OR JP-3754220-B2 OR CN-114128033-A).did. AND FPRS.dbnm.) OR ((JP-2009289714-A OR JP-2013187077-A).did. AND JPAB.dbnm.) OR ((US-3761314-A).did. AND USOC.dbnm.)					
L117	2	"20120009450".pn. "20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 09:02 PM
L119	1	"3234421".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/30 02:52 PM
L120	6	waymouth-j\$.inv. AND koury-f\$.inv.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/30 02:53 PM
L121	2	"20140050956".pn. "20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/09/02 03:30 PM
L122	12	("20150037638" "20110 067227" "20140147714 "20070269685" "2012 0009450" "2005022117 8" "20110081574" "201 10262779" "201603519 40" "20100190056" "20 160218343" "20130252 054").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2022/09/06 01:20 PM
L123	6	"20140050956".pn. "20130108903".pn. "20120009450".pn. "20110067227".pn. "20130252054".pn. "4822377".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/09/07 02:36 PM
L124	2	("20060127774" "20100 190056").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 12:17 PM
L125	183	(h01m50/\$.cpc.	(US-PGPUB; USPAT;	OR	ON	ON	2022/12/16

		h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4)	FPRS; EPO; JPO; DERWENT)				12:40 PM
L126	197	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl\$5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:01 PM
L127	38	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl\$5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4) AND (tan lead terminal) WITH (overlap\$4 overl\$4 shingl\$4 superimpos\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:03 PM
L128	57	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl\$5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4) AND (tab lead terminal) WITH (overlap\$4 overl\$4 shingl\$4 superimpos\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:03 PM

L129	1	"20160351966".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 01:09 PM
L130	100	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4 plate cover)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:42 PM
L131	105	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl\$5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4 plate cover)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:43 PM
L132	0	(h01m50/538,536.cpc. h01m10/1422,0431.cpc.) AND (wound roll\$4 cylind\$5) SAME tab WITH ben\$4 same2 tab WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) near4 (equal\$4 axis around about)	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:07 PM
L133	0	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) SAME tab WITH ben\$4 same2 tab WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) near4 (equal\$4 axis around about)	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:07 PM
L134	0	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) SAME tab WITH ben\$4 same2 tab WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) .clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:08 PM

L135	86	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) SAME tab WITH ben\$4 .clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:08 PM
L136	73423	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) tab WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) .clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:08 PM
L137	168	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (tab electrode) WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) .clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:08 PM
L138	27	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (tab electrode) WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) same3 (tab electrode) WITH (overlap\$4 laminat\$4).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:09 PM
L139	1101	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (tab electrode) WITH (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:11 PM
L140	104	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) WITH (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:11 PM
L141	56	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) near3 (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:11 PM

L142	1	"11652232".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/03 01:28 PM
L143	1	"20230246222".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/03 01:33 PM
L144	4	"20110067227".pn. "20120009450".pn. "20130108903".pn. "20130252054".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/03 02:46 PM
L145	63	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) near3 (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/03 02:54 PM
L146	2	L142 L143	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/03 03:34 PM
L147	22	("4822377" "6569565" "20010023038" "20050003264" "20050287428" "20060127774" "20070196730" "20100081052" "20100190056" "20100310926" "20110067227" "20110293977" "20120009450" "20120208055" "20130108903" "20130252054" "20130344364" "20140050956" "20150295270" "20160351966" "20180145304" "20190379028").pn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2023/11/03 04:02 PM
L148	32	(Artificial Intelligence) More like doc: US-11652232-B2 Text: (US-2020295341-A1 OR US-2023246222-A1 OR EP-3611781-A1 OR WO-2018190691-A1 OR KR-20180116156-A OR KR-102126970-B1 OR JP-7341585-B2 OR JP-2020505717-A OR JP-2021177490-A OR US-11349182-B2 OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO;	OR	ON	ON	2023/11/03 04:02 PM

		CN-110476273-B OR US-2019379028-A1 OR EP-3567666-B1 OR CN-110476273-A OR KR-101137372-B1 OR KR-20140140405-A OR KR-101584298-B1 OR JP-5194072-B2 OR KR- 102226916-B1 OR CN- 116544526-A).did.	JPO; DERWENT; IBM_TDB)				
L149	2	(h01m50/538,536.cpc. h01m10/1422,0431.cpc.) AND (wound roll\$4 cylind\$5) SAME tab near3 notch\$4 WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) near4 (equal\$4 axis around about)	(US-PGPUB; USPAT)	OR	ON	ON	2024/02/16 04:14 PM
L150	2	(h01m50/538,536.cpc. h01m10/1422,0431.cpc.) AND (wound roll\$4 cylind\$5) SAME tab near3 notch\$4 WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) near4 (equal\$4 axis around about).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2024/02/16 04:14 PM
L151	2	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) SAME tab near3 notch\$4 WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) near4 (equal\$4 axis around about).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2024/02/16 04:20 PM
L152	67	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) near3 (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2024/02/16 04:20 PM
L153	4	"3761314".pn. "20050287428".pn. "20060024572".pn. "20050287432".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/03/06 08:44 AM
L154	54	((US-20130108903-A1 OR US-20150072204- A1 OR US-	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY,	OR	ON	ON	2024/04/17 10:11 AM

		20180026295-A1 OR US-20100081052-A1 OR US-20070117011- A1 OR US- 20030211388-A1 OR US-20010038945-A1 OR US-20090029240- A1 OR US- 20160351966-A1 OR US-20110151295-A1 OR US-20210005936- A1 OR US- 20180145304-A1 OR US-20080026293-A1 OR US-20100124694- A1 OR US- 20190379028-A1 OR US-20110067227-A1 OR US-20070196730- A1 OR US- 20010023038-A1 OR US-20160351940-A1 OR US-20130252054- A1 OR US- 20100310926-A1 OR US-20190027789-A1 OR US-20170125764- A1 OR US- 20110008670-A1 OR US-20050225926-A1 OR US-20060024572- A1 OR US- 20200343520-A1 OR US-20150037638-A1 OR US-20190348703- A1 OR US- 20120009450-A1 OR US-20140050956-A1 OR US-20200295341- A1 OR US- 20230246222-A1).did. AND PGPB.dbnm.) OR ((US-8236441-B2 OR US-6521374-B1 OR US-6432574-B1 OR US-8871373-B2 OR US-5849431-A OR US- 9640833-B2 OR US- 6071638-A OR US- 7935445-B2 OR US- 5171647-A OR US- 4822377-A OR US- 8310809-B2 OR US- 8628877-B2 OR US- 6456484-B1 OR US- 11652232-B2).did. AND USPT.dbnm.) OR ((US- 3761314-A).did. AND	CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
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		USOC.dbnm.) OR ((WO-2012051402-A1 OR WO-2017047278- A1 OR EP-2325927-A1 OR JP-3754220- B2).did. AND FPRS.dbnm.) OR ((JP- 2009289714-A OR JP- 2013187077-A).did. AND JPAB.dbnm.)					
L155	2	"11652232".pn. "11973177".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:49 PM
L156	1	"20240243335".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:51 PM
L157	1	"20240243336".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:52 PM
L158	4	L155 L156 L157	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:53 PM
L159	0	L158 AND second near3 non\$2coating	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:53 PM
L160	0	L158 AND second WITH non\$2coating	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:54 PM
L161	4	L158 AND non\$2coating	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:54 PM
L162	4	L158 AND bent near4 electrical\$4 near4 connect\$4	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:56 PM
L163	1	L158 AND spac\$4 near2 apart	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:58 PM
L164	4	L158 AND (spac\$4 apart)	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 09:58 PM
L165	57	((US-20130108903-A1 OR US-20150072204- A1 OR US- 20180026295-A1 OR US-20100081052-A1 OR US-20070117011- A1 OR US- 20030211388-A1 OR US-20010038945-A1 OR US-20090029240- A1 OR US- 20160351966-A1 OR US-20110151295-A1 OR US-20210005936- A1 OR US- 20180145304-A1 OR US-20080026293-A1 OR US-20100124694- A1 OR US- 20190379028-A1 OR US-20110067227-A1 OR US-20070196730- A1 OR US-	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/22 09:59 PM

		20010023038-A1 OR US-20160351940-A1 OR US-20130252054- A1 OR US- 20100310926-A1 OR US-20190027789-A1 OR US-20170125764- A1 OR US- 20110008670-A1 OR US-20050225926-A1 OR US-20060024572- A1 OR US- 20200343520-A1 OR US-20150037638-A1 OR US-20190348703- A1 OR US- 20120009450-A1 OR US-20140050956-A1 OR US-20200295341- A1 OR US- 20230246222-A1 OR US-20240243335-A1 OR US-20240243336- A1).did. AND PGPB.dbnm.) OR ((US- 8236441-B2 OR US- 6521374-B1 OR US- 6432574-B1 OR US- 8871373-B2 OR US- 5849431-A OR US- 9640833-B2 OR US- 6071638-A OR US- 7935445-B2 OR US- 5171647-A OR US- 4822377-A OR US- 8310809-B2 OR US- 8628877-B2 OR US- 6456484-B1 OR US- 11652232-B2 OR US- 11973177-B2).did. AND USPT.dbnm.) OR ((US- 3761314-A).did. AND USOC.dbnm.) OR ((WO-2012051402-A1 OR WO-2017047278- A1 OR EP-2325927-A1 OR JP-3754220- B2).did. AND FPRS.dbnm.) OR ((JP- 2009289714-A OR JP- 2013187077-A).did. AND JPAB.dbnm.)					
L166	4	"3761314".pn. "20050287428".pn. "20060024572".pn. "20050287432".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22 10:08 PM
L167	75	(h01m50/\$.cpc.	(US-PGPUB; USPAT)	OR	ON	ON	2024/08/22

		h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) near3 (overlap\$4 laminat\$4).clm. NOT L138					10:08 PM
L168	1	"11973177".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/10/08 09:39 AM
L169	2	"20240243335".pn. "20240243336".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/10/08 09:41 AM
L170	2	"11973177".pn. "11652232".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/10/08 09:58 AM
L171	24	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) AND (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv.) AND (electrode cathode anode) WITH (notch\$4) near4 (overlap\$4 plural\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2024/10/08 12:28 PM
L172	2	L170 AND overlap\$4.clm.	(US-PGPUB; USPAT)	OR	ON	ON	2024/10/08 02:44 PM
L173	1	"20210391579".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/10/09 02:50 PM
L174	3	"20240243335".pn. "20050277022".pn. "20050142436".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/12/06 10:43 AM
L175	27	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) AND (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv.) AND (electrode cathode anode) WITH (notch\$4) near4 (overlap\$4 plural\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2024/12/06 11:08 AM
L176	75	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) near3 (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2024/12/06 11:08 AM
L177	2	"20050142436".pn. "20050277022".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/02/06 09:36 AM
L178	50	(Artificial Intelligence) Similar to:	(US-PGPUB; USPAT; USOCR; FIT (AU, AP,	OR	ON	ON	2025/11/24 07:53 PM

		19/207,758 with 0 CPC Selections and 0 Text Selections Text: (EP-3567666-A1 OR US-20190379028-A1 OR EP-3567666-B1 OR CN-117457847-A OR WO-2023022560-A1 OR CN-101442139-B OR EP-4579867-A1 OR CN-101442139-A OR KR-20120002861-A OR US-20240136492-A1 OR KR-20230027600-A OR EP-4191737-B1 OR EP-2378595-A2 OR KR-20240010355-A OR KR-20130052410-A OR US-20110293977-A1 OR JP-2023541349-A OR JP-2025528822-A OR WO-2018080080-A1 OR WO-2019245144-A1 OR EP-3761419-A1 OR CN-119744469-A OR KR-20250091928-A OR KR-20190143304-A OR CN-111406338-A OR CN-105826514-A OR US-7666545-B2 OR EP-2136429-A1 OR EP-3761419-B1 OR EP-4044355-A1 OR JP-2024537650-A OR US-20200044225-A1 OR KR-20250106227-A OR US-20070231701-A1 OR CN-116072957-A OR JP-2011113970-A OR JP-7746530-B2 OR KR-102734115-B1 OR KR-20040059303-A OR KR-20150117135-A OR CN-104170126-A OR US-20060003221-A1 OR WO-2010071387-A2 OR KR-20140051885-A OR US-20210028494-A1 OR US-7700221-B2 OR KR-20160106526-A OR KR-20140141742-A OR KR-20240028850-A OR US-20140356674-A1).did.	AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN))				
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L179	50	(Artificial Intelligence) Similar to: 19/207,758 with 2 CPC Selections and 2 Text Selections Text: (US-20190379028-A1 OR US-20210028494- A1 OR EP-3567666-A1 OR CN-101442139-A OR US-7666545-B2 OR CN-101442139-B OR EP-3567666-B1 OR CN-1172400-C OR KR- 102785731-B1 OR EP- 4576294-A1 OR US- 20250210828-A1 OR EP-2378595-A2 OR US-20110293977-A1 OR KR-20250101825-A OR CN-117457847-A OR US-20020164524- A1 OR CN-104285319- B OR CN-116072957-A OR EP-1255310-A2 OR CN-1713418-A OR JP- 2011113970-A OR EP- 4262008-A1 OR US- 7700221-B2 OR US- 20060003221-A1 OR CN-101411003-B OR US-7214250-B2 OR US-20050246887-A1 OR EP-1255310-B1 OR KR-20250066216-A OR JP-7005642-B2 OR KR- 20130052410-A OR JP- 7604066-B2 OR CN- 111406338-A OR CN- 217239536-U OR JP- 2024167710-A OR JP- 2005302719-A OR DE- 69937078-T2 OR CN- 107112494-B OR US- 6884270-B2 OR US- 20250300309-A1 OR WO-2018080080-A1 OR CN-116365179-A OR US-20240421451- A1 OR CN-119009387- A OR JP-2025026968- A OR KR- 20250042130-A OR KR-20240168235-A OR KR-20150021030-A OR CN-116636058-A OR US-9350042-B2).did.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN))	OR	ON	ON	2025/11/24 07:53 PM
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L180	22	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) AND (uncoated (free remov\$4 without) near3 active near3 material) SAME (plural\$4 several) near4 (tab notch lead) SAME electrode WITH terminal	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2025/11/24 07:54 PM
L181	1071	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) AND (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv. lg.as.) AND (electrode cathode anode) WITH (tab notch) near4 (overlap\$4 plural\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2025/11/24 08:09 PM
L182	45	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) AND (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv. lg.as.) AND (electrode cathode anode) WITH (notching) near4 (overlap\$4 plural\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2025/11/24 08:09 PM
L183	30	("11652232" 11973177 " 20010023038" 2005 0003264" 2005014243 6" 20050277022" 200 50287428" 200502874 32" 20060024572" 20 060127774" 20070196 730" 20100081052" 2 0100190056" 2010031 0926" 20110067227" 20110293977" 201200 09450" 20120208055" "20130108903" 20130 252054" 20130344364 " 20140050956" 2015 0295270" 2016035196 6" 20180145304" 201 90379028" 202402433 35" 3761314" 482237 7" 6569565").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:10 PM
L184	5	"12334493".pn. "12341147".pn. "11973177".pn. "11652232".pn. "20250273726".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:13 PM

L185	2	L184 AND radial\$4 near3 ben\$4	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:13 PM
L186	5	L184 AND can	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:15 PM
L187	1	"20250273726".pn. AND longitudinal near3 direction	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:36 PM
L188	1	"20250273726".pn. AND longitudinal near3 direction WITH spac\$\$	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:37 PM
L189	1	"20250273726".pn. AND longitudinal near3 direction WITH spac\$3	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:37 PM
L190	4	L184 AND distributed	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 08:53 PM
L191	2	L184 AND radial\$4 near3 outer	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:06 PM
L192	22	("20110067227" "20110 067227" "20130252054 "4822377" "20100081 052" "20130108903" "2 0180145304" "2012000 9450" "20130108903" " 20140050956" "200701 96730" "20110067227" " 20130344364" "20190 379028" "20010023038 "20100310926" "6569 565" "20050003264" "2 0050287428" "2011029 3977" "20120009450" " 20120208055" "201502 95270" "20160351966" " 20160351966" "20060 127774" "20100190056 ").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:26 PM
L193	10	("2001/0023038").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/11/24 09:31 PM
L194	25	("4322484" "5227267" "5707758" "5849431" "6245457").pn. OR ("6569565").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/11/24 09:31 PM
L195	60	((WO-2012051402-A1 OR WO-2017047278- A1 OR EP-2325927-A1 OR JP-3754220- B2).did. AND FPRS.dbnm.) OR ((US- 3761314-A).did. AND USOC.dbnm.) OR ((US- 20190379028-A1 OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO,	OR	ON	ON	2025/11/24 09:34 PM

		US-20110067227-A1 OR US-20070196730-A1 OR US-20010023038-A1 OR US-20160351940-A1 OR US-20130252054-A1 OR US-20100310926-A1 OR US-20190027789-A1 OR US-20170125764-A1 OR US-20110008670-A1 OR US-20150072204-A1 OR US-20050225926-A1 OR US-20060024572-A1 OR US-20200343520-A1 OR US-20130108903-A1 OR US-20180026295-A1 OR US-20100081052-A1 OR US-20070117011-A1 OR US-20030211388-A1 OR US-20010038945-A1 OR US-20090029240-A1 OR US-20160351966-A1 OR US-20110151295-A1 OR US-20210005936-A1 OR US-20180145304-A1 OR US-20080026293-A1 OR US-20100124694-A1 OR US-20150037638-A1 OR US-20190348703-A1 OR US-20120009450-A1 OR US-20140050956-A1 OR US-20200295341-A1 OR US-20230246222-A1 OR US-20240243335-A1 OR US-20240243336-A1 OR US-20250273726-A1).did. AND PGPB.dbnm.) OR ((JP-2009289714-A OR JP-2013187077-A).did. AND JPAB.dbnm.) OR ((US-8310809-B2 OR US-8628877-B2 OR US-6456484-B1 OR US-5171647-A OR US-8236441-B2 OR US-6521374-B1 OR US-6432574-B1 OR US-	MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
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L196	55	8871373-B2 OR US-5849431-A OR US-9640833-B2 OR US-6071638-A OR US-7935445-B2 OR US-4822377-A OR US-11652232-B2 OR US-11973177-B2 OR US-12334493-B2 OR US-12341147-B2).did. AND USPT.dbnm.) L195 NOT L194	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/11/24 09:34 PM
L197	21	("2011/0067227").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/11/24 09:36 PM
L198	47	L195 NOT L192	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/11/24 09:38 PM
L199	122	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) AND tab WITH (only) WITH (outer outside circumference)	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:39 PM
L200	44	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) AND tab near3 (outer	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:40 PM

L201	0	outside) near4 (perimeter periphery circumference) (h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) AND tab near3 (outer outside) near4 (perimeter periphery circumference) NOT L200	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:41 PM
L202	42	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) AND tab near3 (outer outside) near4 (perimeter periphery circumference) NOT L199	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:41 PM
L203	96	(h01m50/\$.cpc. h01m10/\$.cpc.) AND tab near3 (outer outside) near4 (perimeter periphery circumference) NOT L199 NOT L202	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:45 PM
L204	1	"20010023038".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/24 09:52 PM
L205	5	L184 AND bent near2 part	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/25 10:30 AM
L206	1	"12334493".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/11/25 10:51 AM
L207	50	(Artificial Intelligence) Similar to: 19/313,233 with 0 CPC Selections and 0 Text Selections Text: (US-20190379028-A1 OR EP-3567666-B1 OR US-11349182-B2 OR EP-3567666-A1 OR CN-117457847-A OR EP-4191737-A1 OR WO-2023022560-A1 OR EP-2378595-A2 OR KR-20230027600-A OR EP-4191737-B1 OR WO-2019245144-A1 OR EP-4016683-B1 OR EP-4579867-A1 OR CN-101442139-A OR EP-2136429-B1 OR KR-20250091928-A OR KR-20190143304-A OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN))	OR	ON	ON	2026/03/18 12:23 PM

		JP-2015053266-A OR KR-20240010355-A OR KR-20120002861-A OR KR-101653320-B1 OR KR-102410911-B1 OR KR-101192138-B1 OR KR-101792572-B1 OR CN-111406338-B OR US-9905881-B2 OR KR-20150040437-A OR CN-110476273-A OR US-20220393251-A1 OR KR-20200065946-A OR KR-20200086932-A OR CN-112262491-A OR EP-3761419-A1 OR CN-119009387-A OR KR-102266592-B1 OR JP-2025528822-A OR KR-101615712-B1 OR KR-20130071462-A OR EP-4044355-A1 OR KR-102234292-B1 OR KR-20250106227-A OR JP-2019194998-A OR US-20090317707-A1 OR KR-20080047165-A OR KR-102709293-B1 OR EP-3761419-B1 OR CN-116235362-A OR CN-111406338-A OR KR-20140141742-A OR US-20060003221-A1).did.					
L208	50	(Artificial Intelligence) Similar to: 19/313,233 with 3 CPC Selections and 2 Text Selections Text: (EP-2733778-A2 OR US-20140147716-A1 OR KR-20250085695-A OR US-20250226557-A1 OR KR- 20150122340-A OR JP- 2017069190-A OR KR- 20150040437-A OR CN-119009387-A OR KR-101653320-B1 OR AU-2016222400-A1 OR JP-2023550097-A OR AU-2016222400-B2 OR EP-3151310-A1 OR US-20170092925-A1 OR KR-20170038672-A OR US-9929393-B2 OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN))	OR	ON	ON	2026/03/18 12:26 PM

		EP-4496113-A1 OR WO-2023106835-A1 OR US-20250300236-A1 OR CN-104769763-B OR WO-2024106976-A1 OR US-20250279554-A1 OR US-20240186659-A1 OR US-20250023088-A1 OR CA-3238461-A1 OR EP-4293813-A1 OR KR-101879941-B1 OR US-11127985-B2 OR US-20220399576-A1 OR US-12191460-B2 OR WO-2020256967-A1 OR JP-2015053266-A OR US-20240120555-A1 OR WO-2022158856-A1 OR JP-2008262898-A OR US-12374762-B2 OR JP-4490209-B2 OR KR-101716542-B1 OR KR-20150124590-A OR EP-4641747-A1 OR AU-2022446963-A1 OR WO-2023173271-A1 OR EP-4601062-A1 OR US-20220223922-A1 OR KR-102917980-B1 OR KR-20150032971-A OR US-20240063423-A1 OR US-20240113402-A1 OR JP-2019194998-A OR US-20160248071-A1).did.					
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PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N2	4	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) AND (uncoated (free remov\$4 without) near3 active near3 material) SAME (plural\$4 several) near4 (tab notch lead) SAME electrode WITH terminal.clm.	(US-PGPUB; USPAT)	OR	ON	ON	2024/12/06 11:10 AM
N3	5	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (uncoated (free	(US-PGPUB; USPAT)	OR	ON	ON	2025/02/06 09:44 AM

N4	3	remov\$4 without) near3 active near3 material) SAME (plural\$4 several) near4 (tab notch lead) SAME electrode WITH terminal.clm. (h01m50/\$.cpc. h01m10/\$.cpc.) AND (plural\$4 several) near4 (tab notch lead) WITH (fold\$4 bend\$4) WITH (bond\$4 weld\$4) SAME electrode WITH terminal.clm.	(US-PGPUB; USPAT)	OR	ON	ON	2025/02/06 09:44 AM
N5	5	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (uncoated (free remov\$4 without) near3 active near3 material) SAME (plural\$4 several) near4 (tab notch lead) SAME electrode WITH terminal.clm.	(US-PGPUB; USPAT)	OR	ON	ON	2025/02/06 09:50 AM
N6	5	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) AND (uncoated (free remov\$4 without) near3 active near3 material) SAME (plural\$4 several) near4 (tab notch lead) SAME electrode WITH terminal.clm.	(US-PGPUB; USPAT)	OR	ON	ON	2026/03/18 12:53 PM
N7	6	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (uncoated (free remov\$4 without) near3 active near3 material) SAME (plural\$4 several) near4 (tab notch lead) SAME electrode WITH terminal.clm.	(US-PGPUB; USPAT)	OR	ON	ON	2026/03/18 12:53 PM

TO: eOfficeAction@lernerdavid.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 86765

Thu Mar 26 12:02:46 EDT 2026

Dear Patent Center Customer:

Correspondence Address:

LG

Lerner David LLP

20 Commerce Drive

Cranford, NEW JERSEY, 07016

UNITED STATES

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Application	Document	Mailroom Date	Attorney Docket No.
19313233	NOA	03/26/2026	LGES 3.12F-1654 CCCCIIICII

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If you have any questions, please contact the [Patent Electronic Business Center](#) (EBC) at ebc@uspto.gov or 866-217-9197 Monday – Friday, 6 a.m. to midnight ET.

Please do not reply to this email as it was sent from an unmonitored mailbox.

Sincerely,

The Patent Center Team

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with the applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 6 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

LERNER DAVID LLP
20 Commerce Dr.
Cranford, New Jersey 07016

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

Joseph H. Karlin

(Typed or printed name)

/Joseph H. Karlin/

(Signature)

April 27, 2026

(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/313,233	08/28/2025	Jae Woong Kim	LGES 3.12F-1654 CCCCIICII	7385

TITLE OF INVENTION: Secondary Battery and Method for Manufacturing the Same

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1,290.00			\$1,290.00	06/26/2026

EXAMINER	ART UNIT	CLASS-SUBCLASS
A. E. Eggerding	1729	429-094000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Lerner David LLP

2

3

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

LG Energy Solution, Ltd.

Seoul, Republic of Korea

Please check the appropriate assignee category or categories (will not be printed on the patent):

☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees Submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via the USPTO patent electronic filing system

☐ Enclosed check

☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 12-1095

5. Change of Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

6. Courtesy Ceremonial Copy

☐ Applicant requests a courtesy ceremonial copy of the electronic patent grant. Applicant understands that the electronic patent grant is the official statutory patent grant.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Joseph H. Karlin/

Date April 27, 2026

Typed or printed name

Joseph H. Karlin

Registration No.

62,471



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC PAYMENT RECEIPT

APPLICATION # 19/313,233	RECEIPT DATE / TIME 04/27/2026 05:15:41 PM Z ET	ATTORNEY DOCKET # LGES 3.12F-1654 CCCCIIICII
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Title of Invention

Secondary Battery and Method for Manufacturing the Same

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	7385	FILED BY	Sharon McGee
PATENT CENTER #	75579199	AUTHORIZED BY	KEVIN KOCUN
CUSTOMER #	86765	FILING DATE	08/28/2025
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Jae Woong Kim

Payment Information

PAYMENT METHOD CARD / 1011	PAYMENT TRANSACTION ID E20264QH15585753	PAYMENT AUTHORIZED BY Sharon McGee
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FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1501	UTILITY ISSUE FEE	1290.00	1	1290.00
TOTAL AMOUNT:				\$1,290.00

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/313,233	04/27/2026 05:15:41 PM Z ET	LGES 3.12F-1654 CCCCIIICII

Title of Invention

Secondary Battery and Method for Manufacturing the Same

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CONFIRMATION #	7385	FILED BY	Sharon McGee
PATENT CENTER #	75579199	FILING DATE	08/28/2025
CUSTOMER #	86765	FIRST NAMED INVENTOR	Jae Woong Kim
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	KEVIN KOCUN

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
Issue Fee Transmittal_2.pdf	2	Issue Fee Payment (PTO-85B)	41 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
Issue Fee Transmittal_2.pdf	B1500AE41854BC1E8294AD824C3DADFA0239BDAC08B323CC D6AF70446D65A35E27C2EB5BA9B5F9A4E0BADE293EC2EDC D8BB475C8F3FBB08D94E7CE78A76EED91

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New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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United States Patent and Trademark Office
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www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/313,233	06/02/2026	12646735	LGES 3.12F-1654 CCCCIICII	7385
86765	7590	05/20/2026		

LG

Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

The USPTO will automatically mail a courtesy ceremonial copy of the electronic patent grant to the correspondence address of record if the issue fee was paid before March 9, 2026. If the issue fee was paid on or after March 9, 2026, a courtesy ceremonial copy will be mailed if the applicant opted-in to receiving it. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

INVENTOR(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional inventors):

Jae Woong Kim, Daejeon, KOREA, REPUBLIC OF;
Hyoung Kwon Kim, Daejeon, KOREA, REPUBLIC OF;
Sang Uck Kim, Daejeon, KOREA, REPUBLIC OF;

APPLICANT(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional applicants):

LG Energy Solution, Ltd., Seoul, KOREA, REPUBLIC OF;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit [SelectUSA.gov](https://selectusa.gov).

TO: eOfficeAction@lernerdavid.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 86765

Thu May 21 07:00:17 EDT 2026

Dear Patent Center Customer:

Correspondence Address:

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Lerner David LLP
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Cranford, NEW JERSEY, 07016
UNITED STATES

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Application	Document	Mailroom Date	Attorney Docket No
19313233	ISSUE.NTF	05/20/2026	LGES 3.12F-1654 CCCCIIICII

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Sincerely,

The Patent Center Team